The impact of intrinsic excitability heterogeneity on the macro-scale brain dynamics: Computational analysis on stability, resilience, and synchrony

MASc. Oral Examination

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Outline

1. Motivation (Chapter 1 and 2)

2. Proposed mathematical brain model (Chapter 3)

3. Key results (Chapter 4 and 5)

4. Conclusions and future work (Chapter 6)

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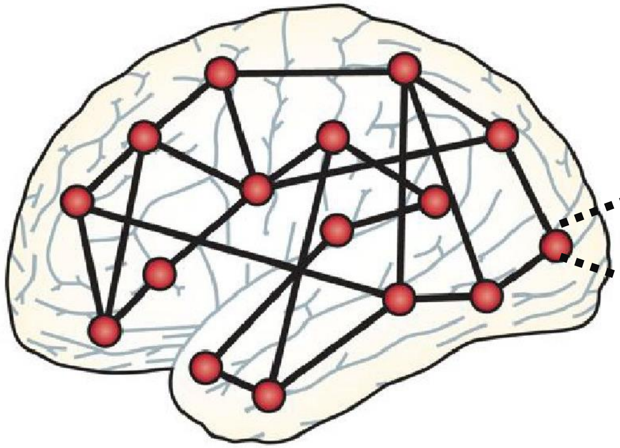
Computational neuroscience

Brain as a network based on neuro-anatomy

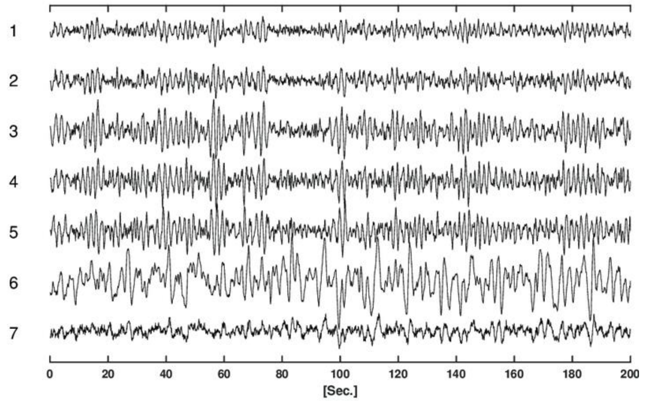
Node dynamics based on statistical physics models

Emerged brain activities

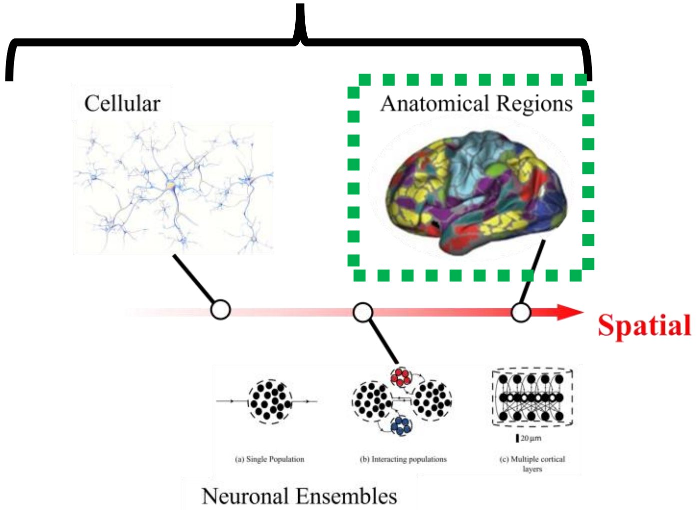


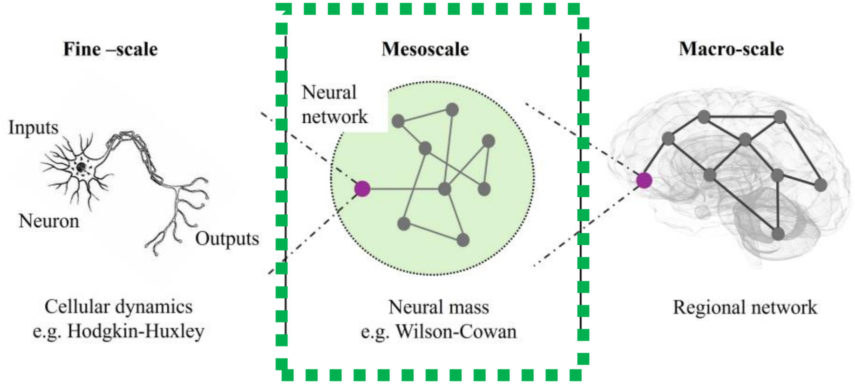
$$\dot{x}_j = (a_j - x_j^2 - y_j^2)x_j - \omega_j y_j + G \sum_i C_{ij}(x_i - x_j)$$
$$\dot{y}_j = (a_j - x_j^2 - y_j^2)y_j - \omega_j x_j + G \sum_i C_{ij}(y_i - y_j)$$

Simulation



Hypothesis testing



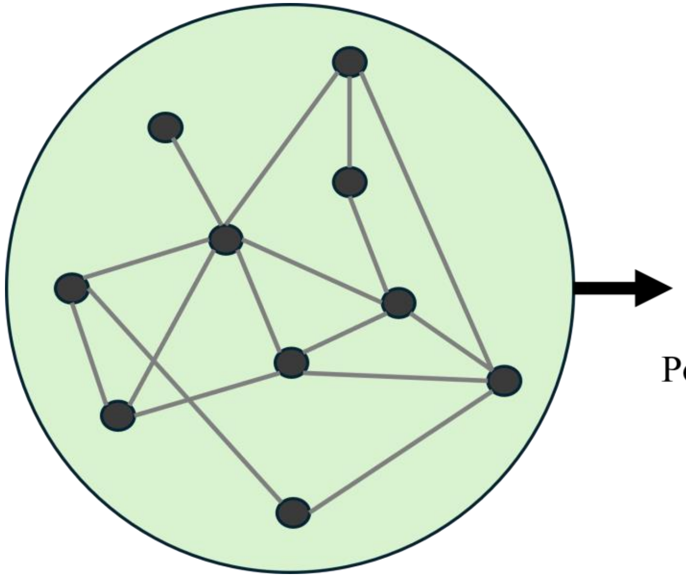


Scale matters!

Population dynamics assume homogeneity!

Why?

- If all neurons in a population are roughly the same, all neurons are exchangeable.
- Based on the parameters of a typical neuron in the population, we would be able to predict the activity of the population.



No. of active neurons within a time window of Δt

$$\boxed{A(t)} = \lim_{\Delta t \to 0} \frac{1}{\Delta t} \frac{\boxed{n_{\mathrm{act}}(t; t+\Delta t)}}{N} = \underbrace{\frac{1}{N} \sum_{j=1}^{N} \sum_{f} \delta(t - t_j^f)}_{\text{Mean firing of the neurons in the population}}$$

Population activity

Figure: The population activity becomes the simple average of all the neurons in the population which can be obtained considering a typical neuron and its spike activity (Dayan & Abbot, 2005).

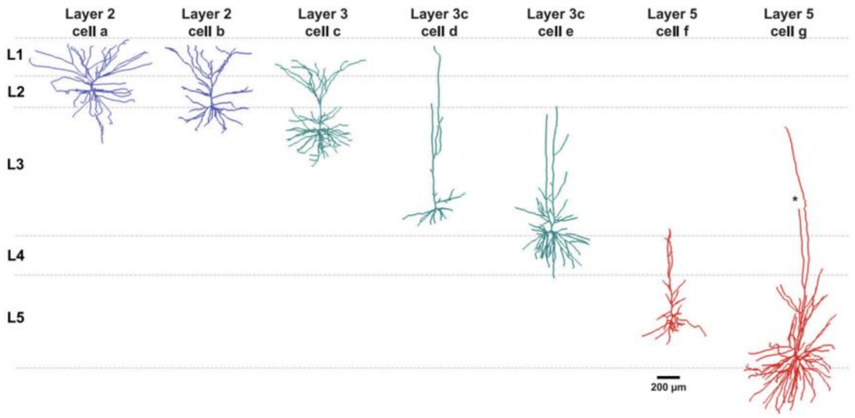
Physiologically, the neurons of the same type are not homogeneous!!

Strong heterogeneity deviates the population activity from an averaging.

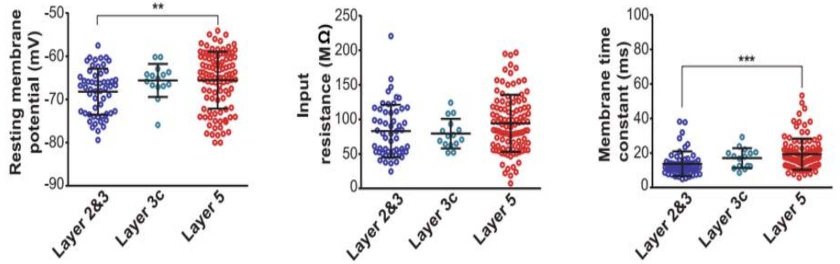
Within cell-type heterogeneity

Individual differences between cells of the same type (Zeng et al., 2022, Dueck et al., 2016)

- Distinct within cell-type heterogeneous properties (Chameh et al., 2021)
  - **Morphological heterogeneity** (Berg et al., 2021, Galakhova et al., 2022)
    - Diverse structural patterns of neurons, including size, dendritic lengths, axonal branching, and connectivity.
  - **Functional heterogeneity** (Berg et al., 2020, Chung et al., 2015)
    - Electrophysiological properties (firing patterns, membrane properties, and synaptic dynamics) and neurotransmitter profiles (glutamatergic, GABAergic, dopaminergic, and serotonergic).
  - **Transcriptomic heterogeneity** (Chameh et al., 2023, Berg et al., 2021)
    - Molecular signatures
- Other sources of heterogeneity
  - Synaptic heterogeneity (Chameh et al., 2023)
  - Intrinsic plasticity heterogeneity (Beck & Yaari, 2008)



(a) Morphological heterogeneity of pyramidal cells.



(b) Electrophyiological heterogeneity of pyramidal cells.

Figure: Morphological and electrophysiological diversity among pyramidal cells in the human neocortex. The figure is adapted from Chameh et al. (2021).

Computational studies of within cell-type heterogeneity

- **Improves information coding and learning**

  Mejias & Longtin (2004), Shamir & Sompolinsky (2006), Chelaru & Dragoi (2008), Padmanabhan & Urban (2010), Brette (2012), Tripathy et al. (2013), Perez-Nieve et al. (2014) …..

- **Renders neuronal networks resilient to a myriad of insults that could potentially cause instabilities in dynamics**

  Hutt et al. (2023), Hutt et al. (2024)

- **Stabilizes networks against synchronous activity that accompany seizures**

  Gast, Solla & Kennedy (2024), Rich et al. (2022), Mejias & Longtin (2014), Yim et al. (2013), Buchin et al. (2022)…..

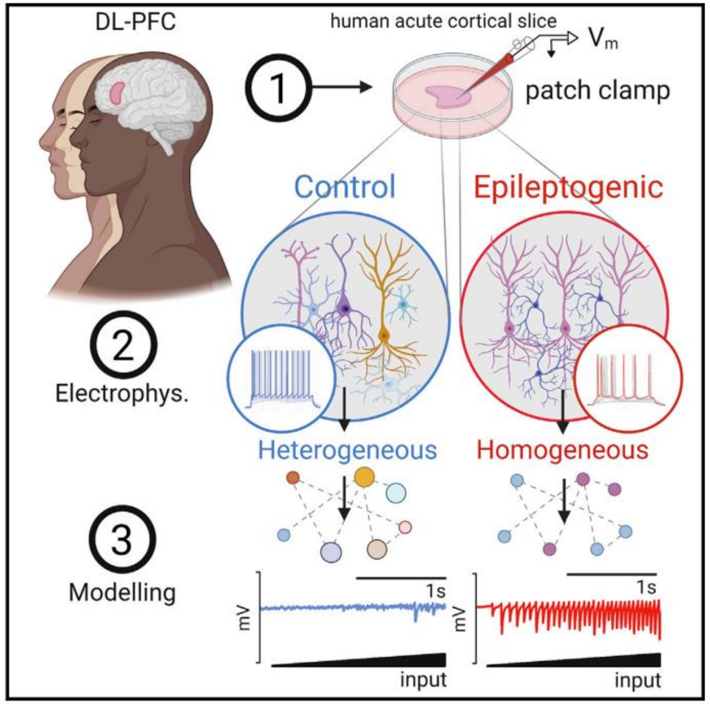


Figure: Loss of neuronal heterogeneity is associated with epileptogenic states (Rich et al., 2022).

Regional heterogeneity

Considers the diversity between the anatomic brain regions.

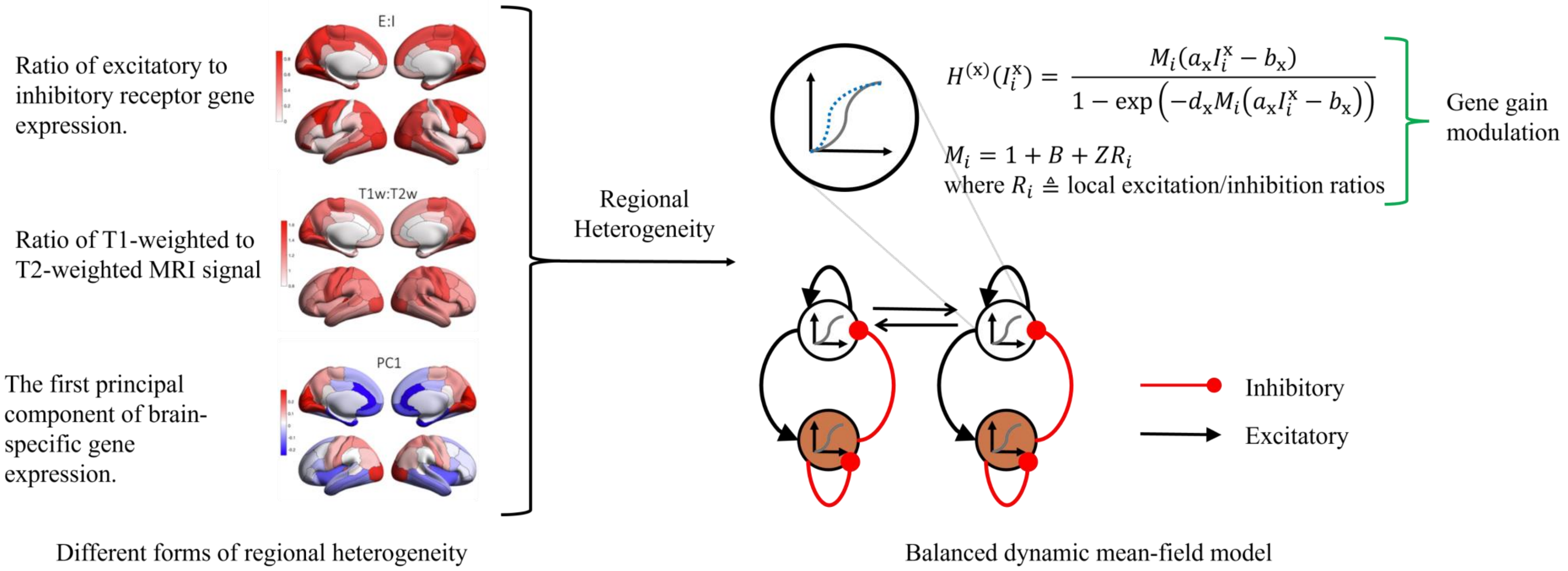


Figure: Whole-brain dynamic modeling based on observed regional heterogeneity. The figure is adapted from Deco et al. (2021).

- Gene expression profile variation (Negi & Guda, 2017, Fornito et al., 2019); Architectonic diversity (Demirtas et al., 2019); Distribution diversity of neurotransmitter receptor (Deco et al., 2021) ……..
- Explains the observed coarse-scale functional activities from fMRI, EEG and MEG (Deco et al., 2017, Perl et al., 2023)

What’s missing?

Main drawbacks in the current heterogeneity studies

- Studies with microcircuits
  - Better theoretical analysis (Hutt et al., 2023, Hutt et al., 2024).
  - Limited to **patches** within whole-brain.
  - **Clinical translation is challenging** as data is inaccessible at fine-scale resolution.

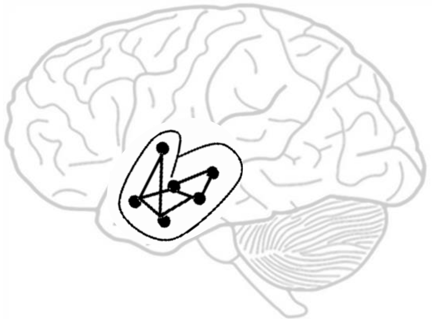


Figure: Microcircuits only represent a patch in the whole-brain

- Studies on regional heterogeneity
  - Establish anatomical regions have different properties (E/I neuron ratios) and explains observed functionality better.
  - However, there is **no connection to the local neuronal population diversity** in a region.
  - Only changes the mean-firing rate parameters of neural mass models at each anatomical region.

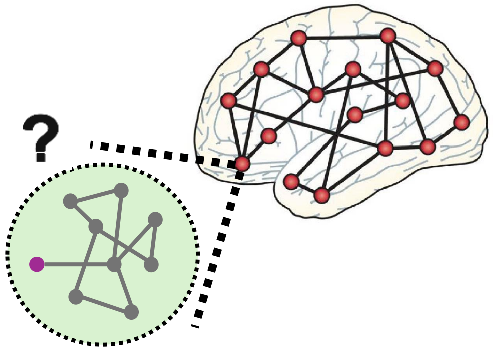


Figure: Regions are different but no analysis into local heterogeneity within a region.

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Dynamics of macro-scale node $n \in [N]$

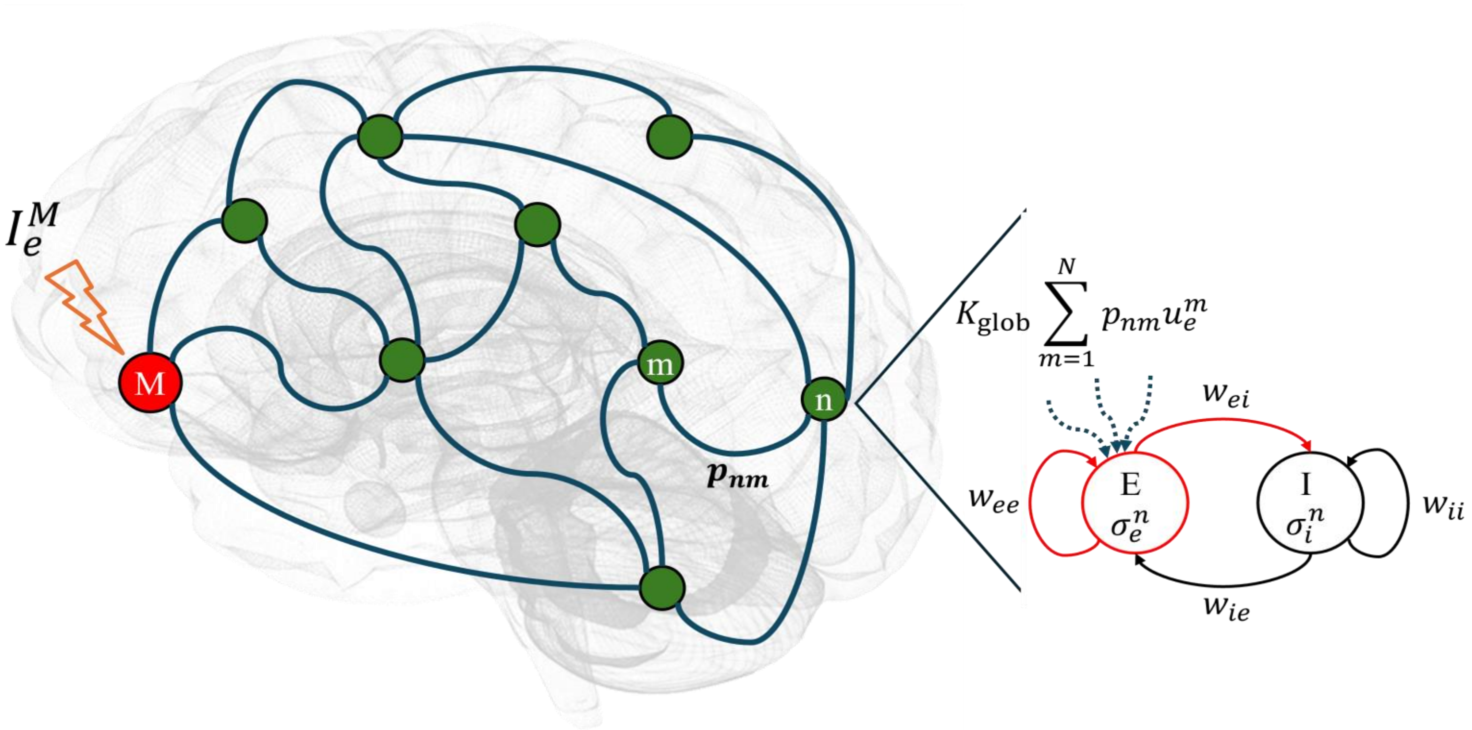

Figure: Macro-scale brain network setup with interacting excitatory and inhibitory sub-populations at each node.

Linear coupling between nodes

$$\tau_{\mathrm{e}} \dot{u}_{\mathrm{e}}^{n} = -u_{\mathrm{e}}^{n} + w_{\mathrm{ee}} F_{\beta}(u_{\mathrm{e}}^{n}, \sigma_{\mathrm{e}}^{n}) + w_{\mathrm{ie}} F_{\beta}(u_{\mathrm{i}}^{n}, \sigma_{\mathrm{i}}^{n}) + I_{\mathrm{e},0} + I_{\mathrm{e}}^{n} + \boxed{K_{\mathrm{glob}} \sum_{m=1}^{N} p_{nm} u_{\mathrm{e}}^{m}}$$

$$\tau_{\mathrm{i}} \dot{u}_{\mathrm{i}}^{n} = -u_{\mathrm{i}}^{n} + w_{\mathrm{e}i} F_{\beta}(u_{\mathrm{e}}^{n}, \sigma_{\mathrm{e}}^{n}) + w_{\mathrm{ii}} F_{\beta}(u_{\mathrm{i}}^{n}, \sigma_{\mathrm{i}}^{n}) + I_{\mathrm{i},0}$$

Key parameters

u_{x}^{n}	Neural dynamics of sub-population $\mathrm{x} \in \{\mathrm{e}, \mathrm{i}\}$
τ_{x}	Time constants of evolution of sub-populations $\mathrm{x} \in \{\mathrm{e}, \mathrm{i}\}$
σ_{x}^{n}	Excitability heterogeneity of sub-population $\mathrm{x} \in \{\mathrm{e}, \mathrm{i}\}$
w_{xy}	Intra-node connectivity weights between sub-populations $\mathrm{x}, \mathrm{y} \in \{\mathrm{e}, \mathrm{i}\}$
$I_{\mathrm{x},0}$	Bias voltages of sub-population $\mathrm{x} \in \{\mathrm{e}, \mathrm{i}\}$
I_{e}^{n}	Modulatory signal amplitude on excitatory sub-population
p_{nm}	Inter-node structural connectivity weights
K_{glob}	Global synaptic coupling parameter

Single neuron firing function

$$\boxed{F_\beta(x, s)} = \boxed{f_\beta(x)} * \boxed{\varphi(x; s)} = \frac{1}{\sqrt{2\pi s^2}} \int_{-\infty}^{\infty} \frac{1}{1 + \exp\left(-\beta(x + \theta)\right)} e^{-\left(\frac{\theta}{\sqrt{2}s}\right)^2} d\theta$$

Mean-field activation function

Zero-mean Gaussian distribution with $s^2 = (\sigma_{\mathrm{y}})^2$

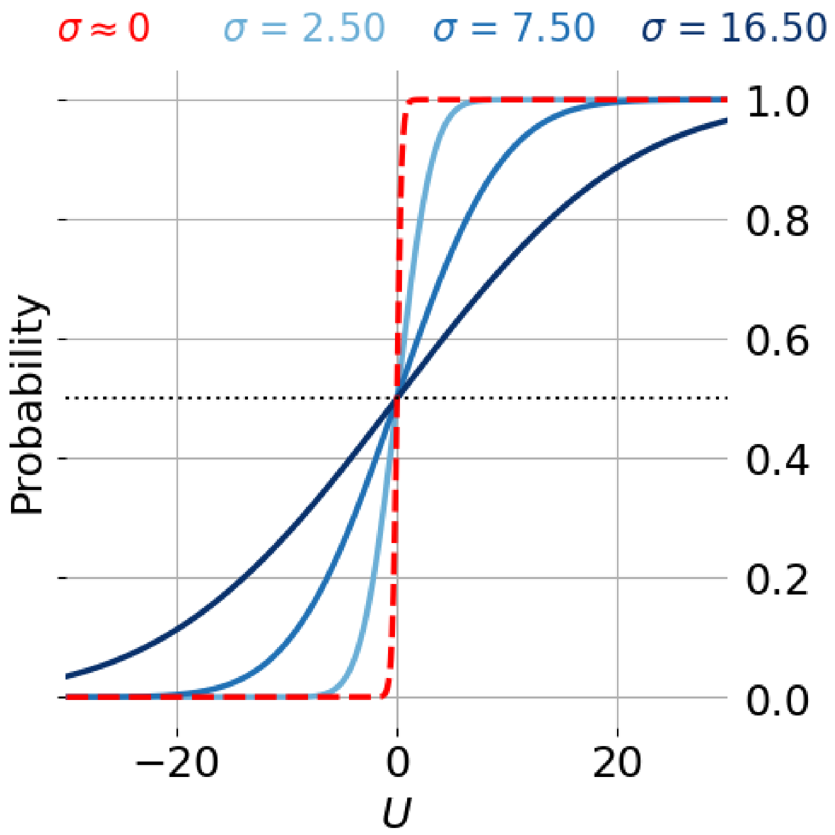

Figure: Change in the mean-field firing function $F_\beta(x, s)$ with excitability heterogeneity $s^2 = \sigma^2$.

Main modeling assumptions

- Properties of underlying microcircuits of each macro-scale node are **<u>unknown</u>**, **<u>except</u>** the overall diversity of neuronal excitability (σ_{x}^{n}) and the intra-population connectivity (w_{xy}).

- Only a subset of nodes i.e., $\mathcal{J}_M \subset$ [N] is modulated by external modulatory signals
  $I_{\mathrm{e}}^{n} \neq 0 \iff n \notin \mathcal{J}_M$.

- Only $\boldsymbol{\sigma}^n = [\sigma_{\mathrm{e}}^{n}, \sigma_{\mathrm{i}}^{n}]^T$ and K_{glob} are changed for each simulation setup. The constant parameters are selected following Scott et al. (2022).

Specific assumptions to our model/study.

- Time delays between the nodes are ignored (Perl et al., 2023).

Generic assumptions in whole-brain modeling.

Simulation setups – Two nodes vs. Multi-node

N = 2

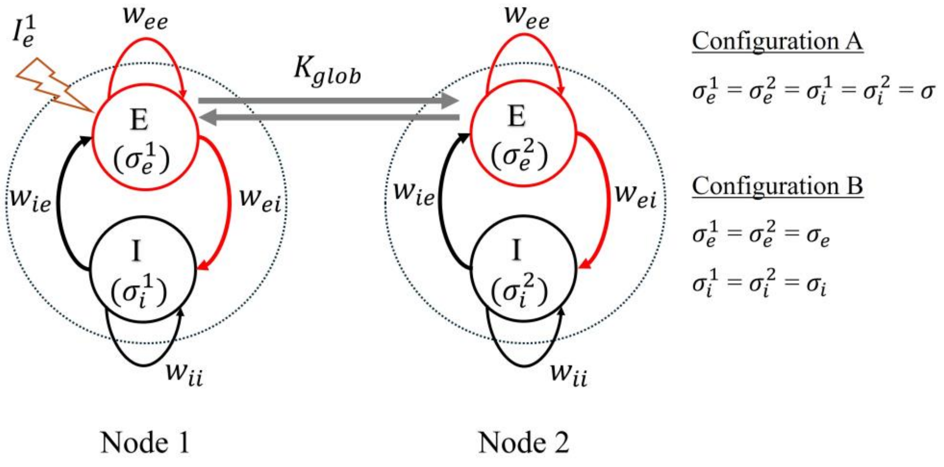

Figure: Two nodes macro-scale brain network. The E sub-population of node 1 is modulated by a constant external signal amplitude I_e^1 . The two configurations considered for assigning intrinsic excitability heterogeneity in the network are summarized in the figure.

N > 2 (N = 10)

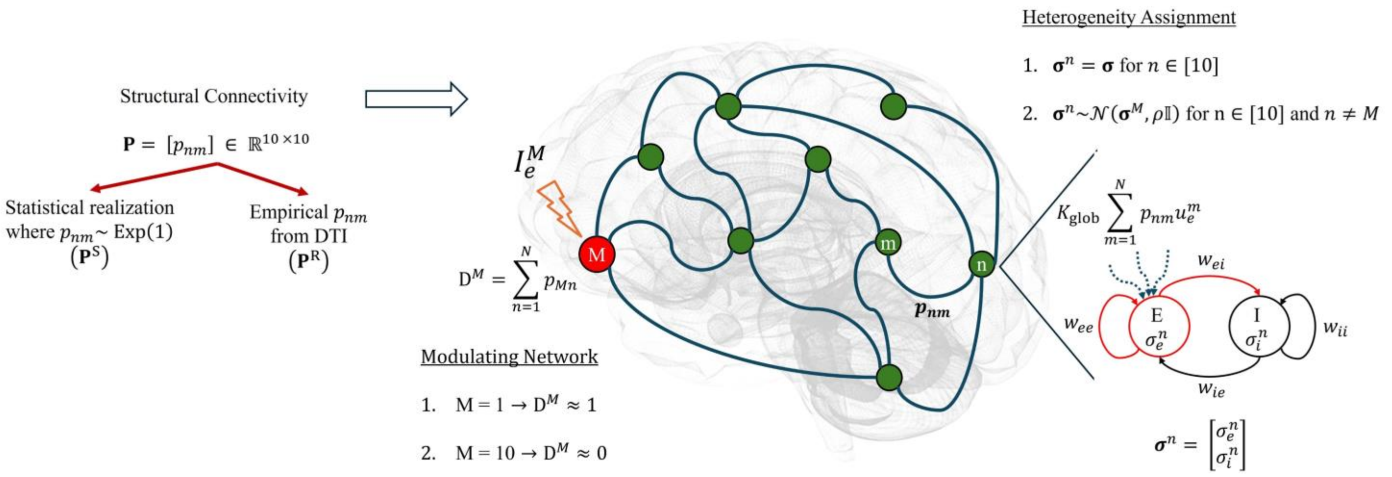

Figure: Multi-node macro-scale brain network modeling considerations. The multi-node brain network modeling involves: (i) defining structural connectivity **P**, (ii) selecting the modulated node M, and (iii) assigning heterogeneity to sub-populations of each node. Structural connectivity **P** is defined through 1) statistical realization via exponential distribution sampling, and 2) empirical data from [2]. Modulation scenarios include 1) the highest-degree and 2) the lowest-degree node modulation. Heterogeneity assignment methods are 1) equal heterogeneity vectors for all nodes, and 2) $\boldsymbol{\sigma}^n \sim \mathcal{N}(\boldsymbol{\sigma}^M, \rho\mathbb{I}_2)$ for n $\neq M$.

[2] Škoch, A., Rehák Bučková, B., Mareš, J., et al. (2022). Human brain structural connectivity matrices–ready for modelling. *Scientific Data*, *9*(1), 486.

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Quantifying stability/volatility of dynamics

[1] Lyapunov Stability: Estimated using temporally averaged Lyapunov exponents (ℓ_{e}^{n}).

$$\ell_{\mathrm{e}}^{n} = \frac{1}{T - t_s - 1} \sum_{k=1}^{T-t_s} \left|\log(u_{\mathrm{e}}^{n}[t_k] - u_{\mathrm{e}}^{n}[t_k - 1])\right|$$

$$\bar{\ell}_{\mathrm{e}} = \frac{1}{|\mathcal{I}_M|} \sum_{n \in \mathcal{I}_M} \ell_{\mathrm{e}}^{n}$$

Lyapunov stability of the modulated node(s).

$$\bar{\ell}'_{\mathrm{e}} = \frac{1}{N - |\mathcal{I}_M|} \sum_{n \notin \mathcal{I}_M} \ell_{\mathrm{e}}^{n}$$

Lyapunov stability of the unmodulated nodes.

[2] Coefficient of Variation (CV): Estimates the overall network volatility [3].

$$\mathrm{CV} = \left| \frac{1}{N} \sum_{n=1}^{N} \frac{\sigma_u^n}{\mu_u^n} \right|$$

σ_u^n: temporal variation in excitatory neural dynamics of the node $n \in [N]$

μ_u^n: mean of excitatory neural dynamics of the node $n \in [N]$

[3] P. J. Hellyer, B. Jachs, C. Clopath, and R. Leech, “Local inhibitory plasticity tunes macroscopic brain dynamics and allows the emergence of functional brain networks”, NeuroImage, 2016.

Intrinsic excitability heterogeneity reduces volatility and achieve stability.

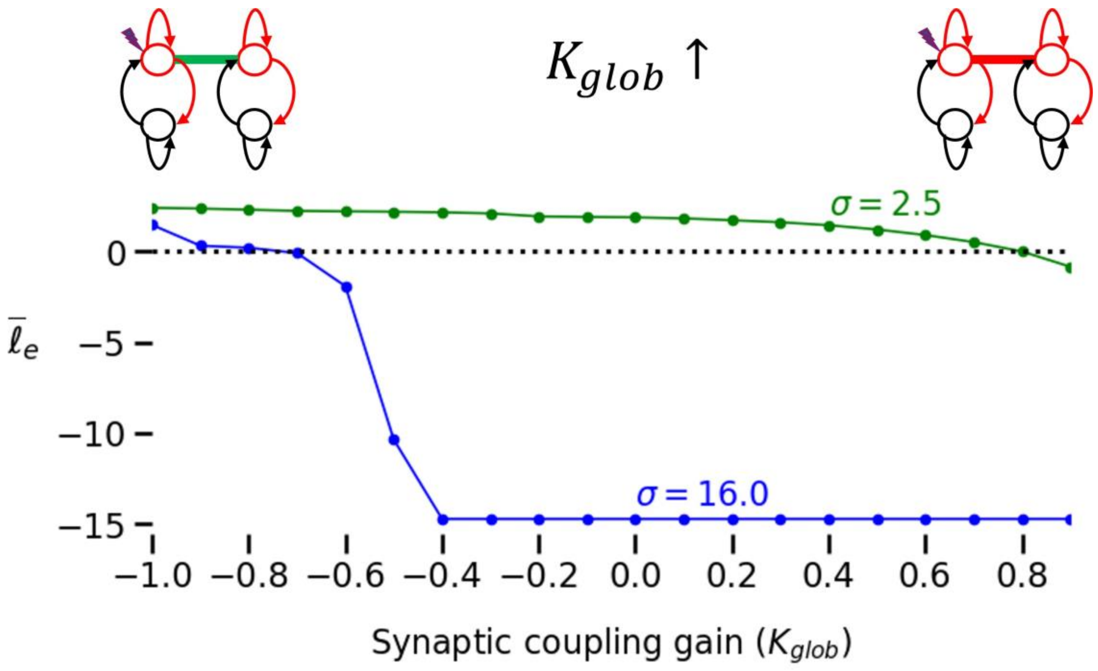


Figure: Excitability heterogeneity, in combination with global coupling, collectively achieves <u>Lyapunov stability</u> against external modulation at the macro-scale.

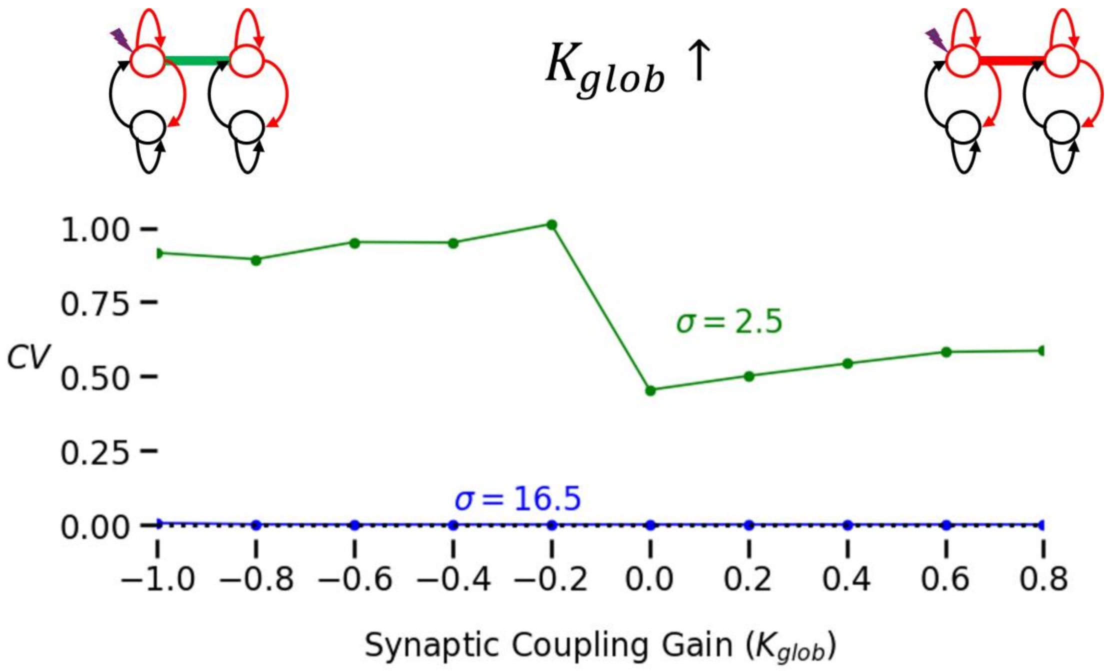


Figure: Excitability heterogeneity, in combination with global coupling, collectively quenches <u>the overall network volatility</u> at the macro-scale.

- The intrinsic excitability heterogeneity and synaptic coupling **<u>collectively</u>** reduces network volatility and improves the stability of the perturbed node.

Excitatory and inhibitory heterogeneity have differential effects on the stability and volatility.

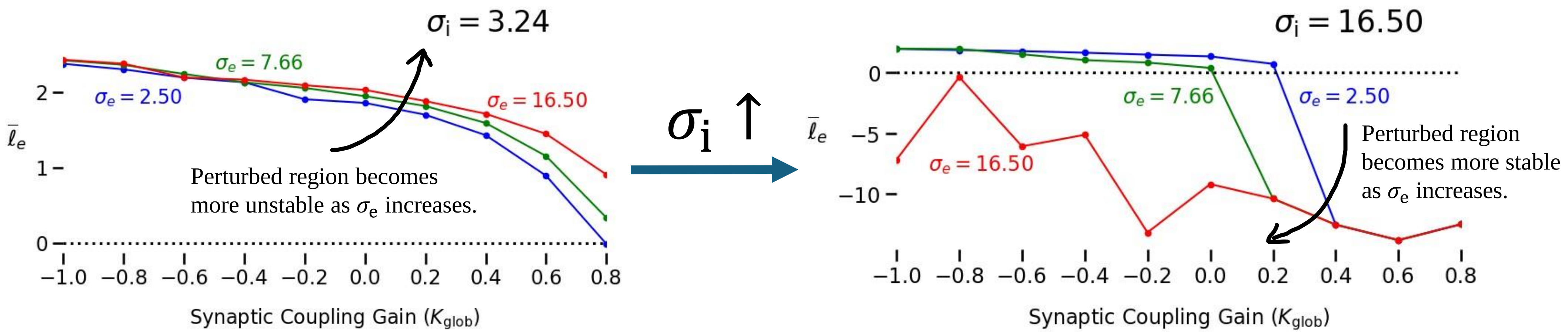


Figure: Excitatory and inhibitory heterogeneity exert distinct effects on the stability of macro-scale neural dynamics.

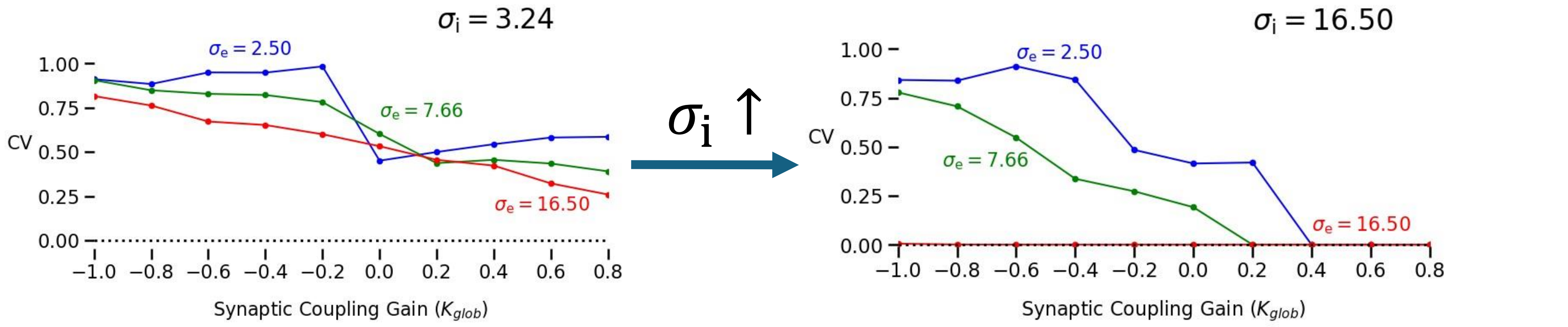
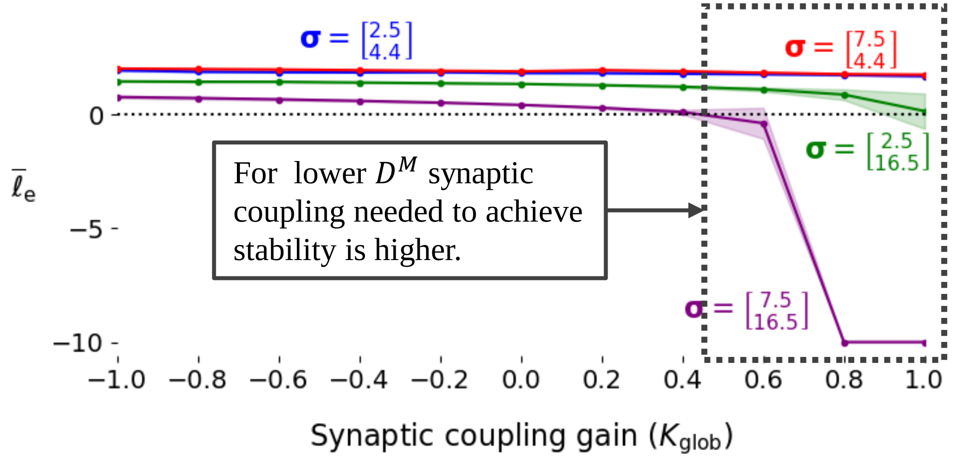


Figure: Inhibitory heterogeneity plays a more crucial role in quenching overall network volatility compared to excitatory heterogeneity.

- Inhibitory heterogeneity is **<u>more important</u>** than the excitatory heterogeneity!

Weighted connectivity of the perturbed node affects reducing volatility and achieving stability.

Lowest $D^M \approx 0$



Weighted connectivity of modulated node (D^M) increases

Highest $D^M \approx 1$

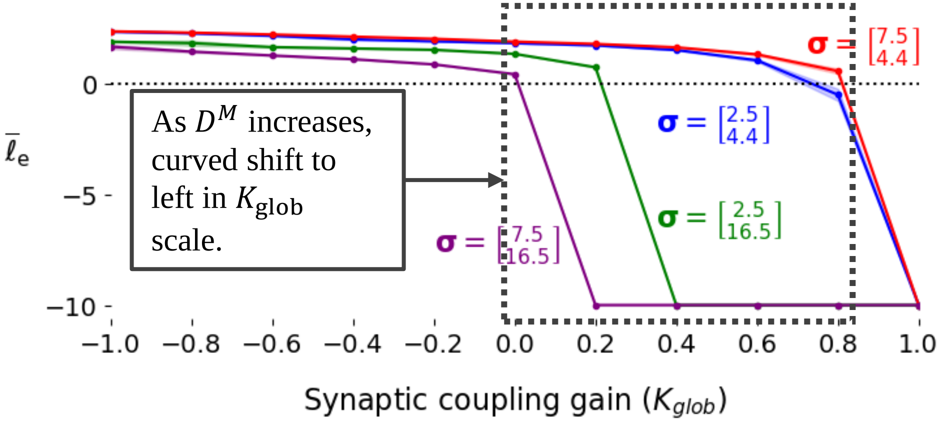
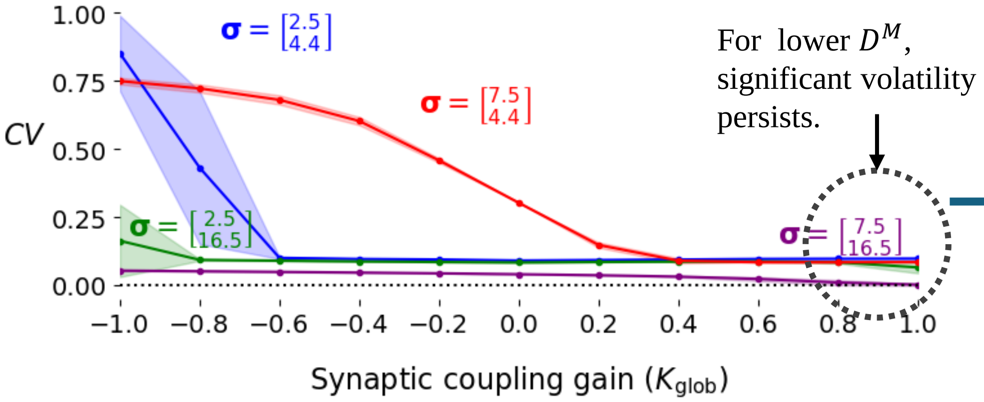


Figure: Change in Lyapunov stability of modulated node as the weighted connectivity changes.



Weighted connectivity of modulated node (D^M) increases

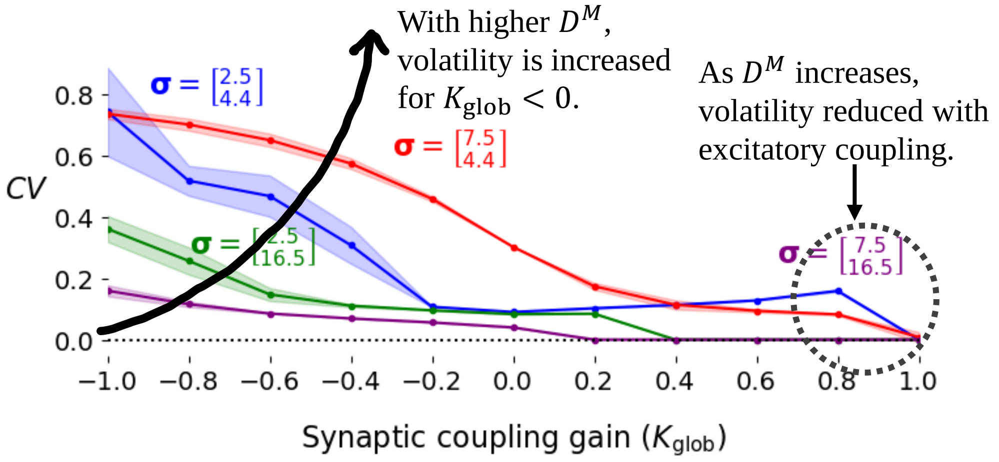


Figure: Change in overall network volatility as the weighted connectivity changes.

** Results corresponds to multi-node network setting with connectivity defined using empirically measured structural weights obtained from [2]

Change in the excitability heterogeneity distribution affects stability and overall network volatility

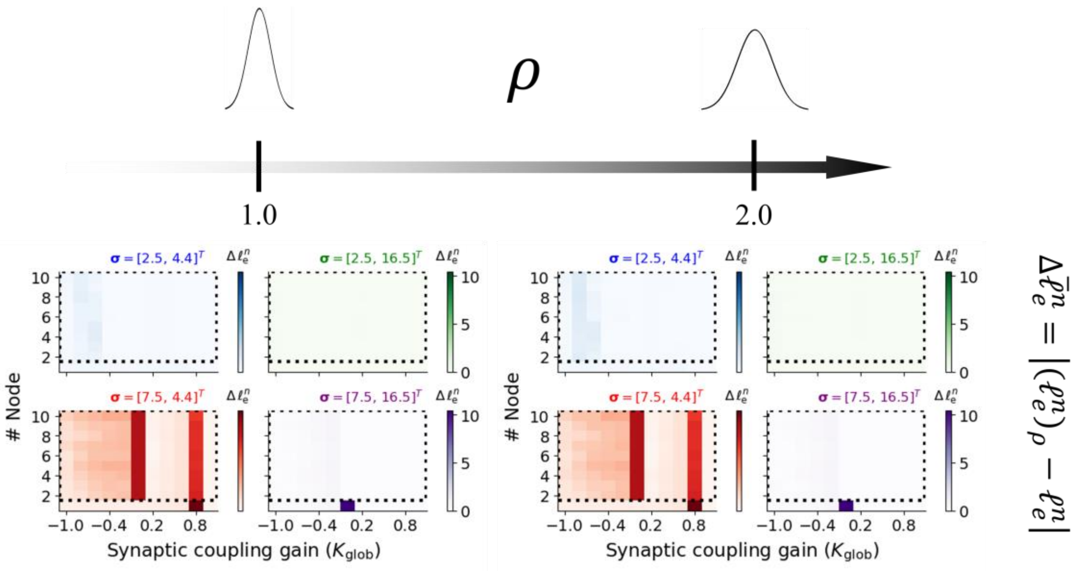


Figure: Change in Lyapunov exponents as the distribution of intrinsic excitability heterogeneity of unmodulated nodes changes.

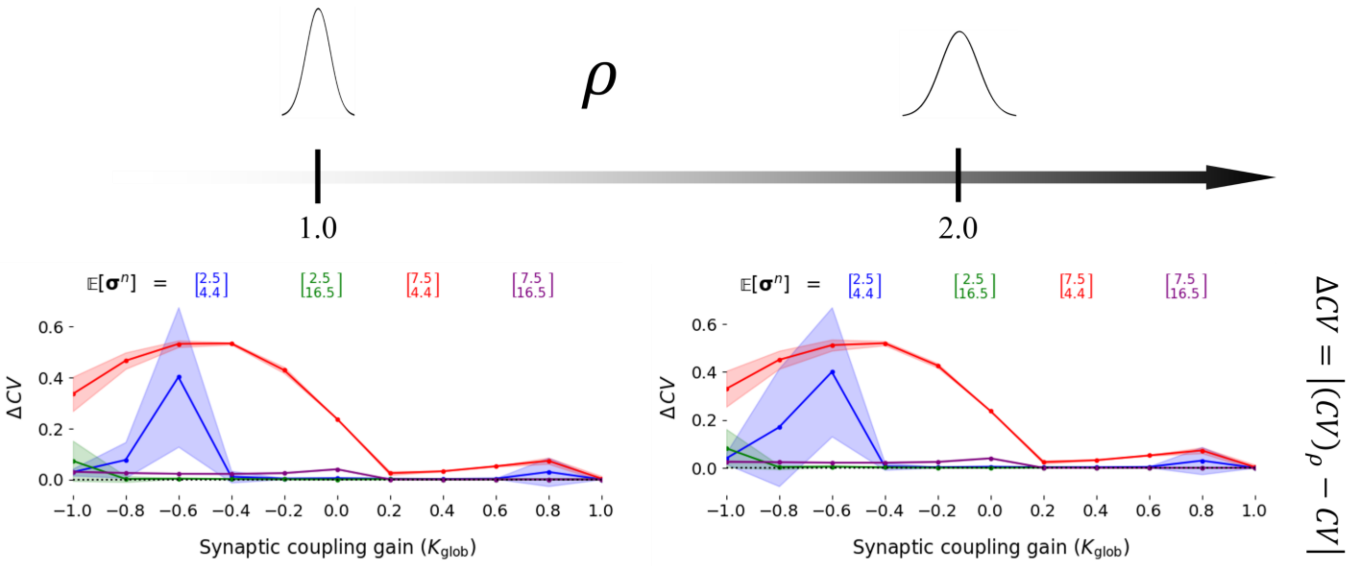


Figure: Change in Coefficient of Variation as the distribution of intrinsic excitability heterogeneity of unmodulated nodes changes.

Mean inhibitory heterogeneity is reduced → Volatility and stability changes as the inter-regional excitability heterogeneity changes.

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Quantifying system resilience and multi-stable region emergence

- Estimate equilibria in the dynamic field of the system (Algorithm 1 in the thesis).

$$\bar{\boldsymbol{u}}_{\mathrm{net}}^{l} \in \mathcal{U}_{I_{\mathrm{e}}^{M}} \text{ s.t. } \mathbf{0} = \mathbf{T}_{\mathrm{net}}\left(-\bar{\boldsymbol{u}}_{\mathrm{net}}^{l} + \boldsymbol{W}_{\mathrm{net}}\bar{\boldsymbol{F}}_{\mathrm{net}}\left(\bar{\boldsymbol{u}}_{\mathrm{net}}^{l}, \boldsymbol{\sigma}_{\mathrm{net}}\right) + (\mathbf{i}_{\mathbf{0}})_{\mathrm{net}} + (\mathbf{i}_{\boldsymbol{t}})_{\mathrm{net}} + K_{\mathrm{glob}}\mathbf{P}_{\mathrm{net}}\bar{\boldsymbol{u}}_{\mathrm{net}}^{l}\right)$$

- Estimate Jacobian matrix at each equilibria

$$\mathbf{J}\left(\bar{\boldsymbol{u}}_{\mathrm{net}}^{l}; \mathbf{P}, \boldsymbol{\sigma}_{\mathrm{net}}, K_{\mathrm{glob}}\right) = \mathbf{T}_{\mathrm{net}}\left(K_{\mathrm{glob}}\mathbf{P}_{\mathrm{net}} - \mathbb{I}_{2N} + \boldsymbol{W}_{\mathrm{net}}\bar{\boldsymbol{R}}_{\mathrm{net}}\left(\bar{\boldsymbol{u}}_{\mathrm{net}}^{l}, \boldsymbol{\sigma}_{\mathrm{net}}\right)\right) = \mathbf{T}_{\mathrm{net}}\boldsymbol{Q}^{\boldsymbol{l}}\left(\widetilde{\boldsymbol{\Lambda}}^{l} - \mathbb{I}_{2N}\right)\left(\boldsymbol{Q}^{\boldsymbol{l}}\right)^{\boldsymbol{H}}$$

- Estimate the eigen spectrum of the dynamic system at each equilibrium point $\bar{\boldsymbol{u}}_{\mathrm{net}}^{l} \in \mathcal{U}_{I_{\mathrm{e}}^{M}}$

$$\boldsymbol{\Lambda}^{l} = \mathbf{T}_{\mathrm{net}}\left(\widetilde{\boldsymbol{\Lambda}}^{l} - \mathbb{I}_{2N}\right)\mathbf{1}_{2N} = \left[\lambda^{l,1}, \lambda^{l,2}, \cdots, \lambda^{l,2N}\right]^{T} \in \mathbb{C}^{2N\times 1}$$

- Determine the local stability state based on the eigen spectrum (Table 3.2).

- Define a system to be resilient against the modulation I_{e}^{M} as:

$$\zeta^{l} = \max_{k\in[2N]}\left\{\Re\{\lambda^{k,l}\}\right\} < 0 \;\; \forall l \in \left|\mathcal{U}_{I_{\mathrm{e}}^{M}}\right|$$

Intrinsic excitability heterogeneity trivializes the network and improves resilience.

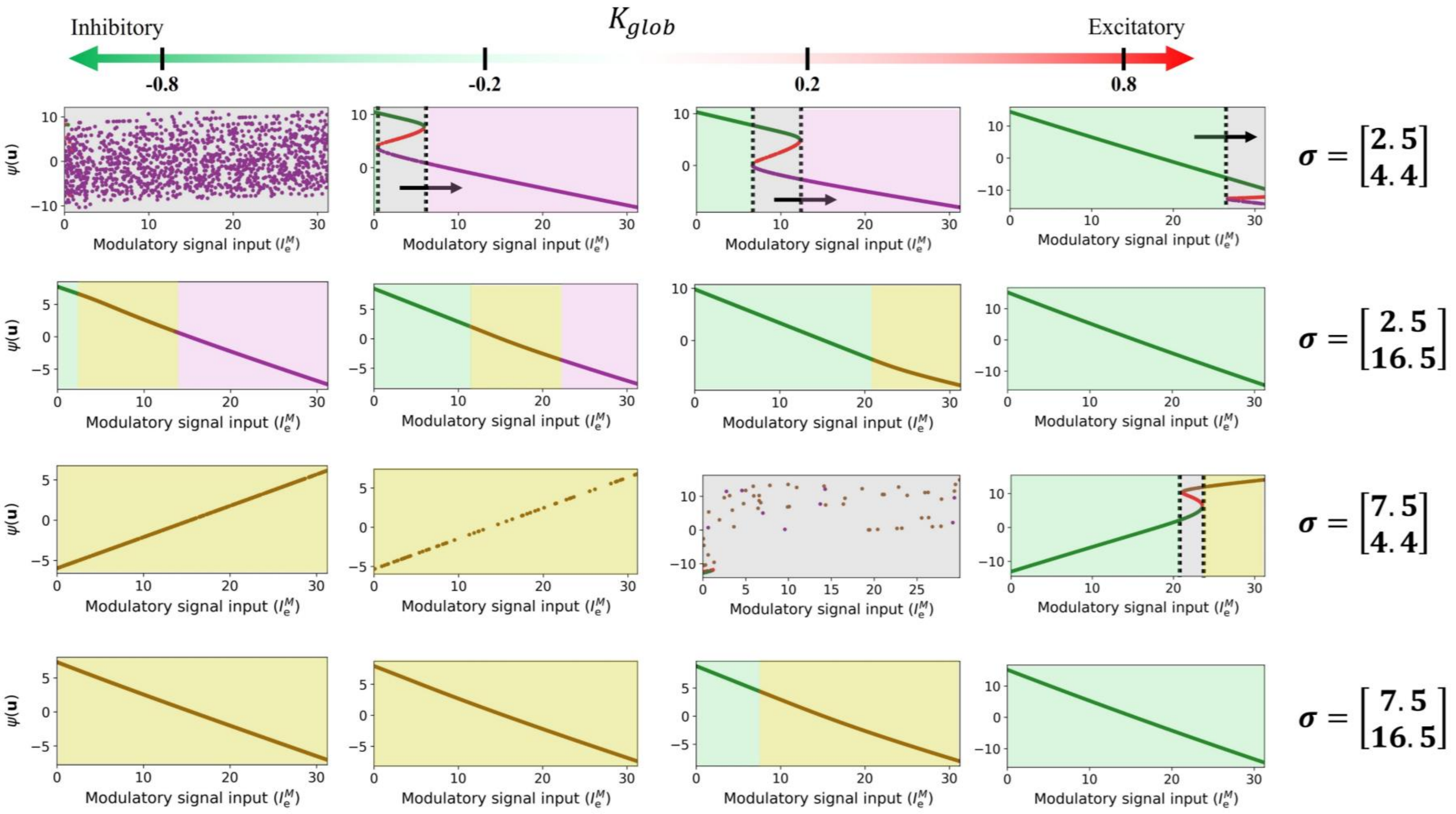

Figure: Excitability heterogeneity reduces the emergence of multiple equilibria (trivialize) at the macro-scale and achieves resilience against external perturbations.

- Intrinsic excitability heterogeneity **trivializes** the system as external perturbation increases.
- Intrinsic excitability heterogeneity and synaptic coupling **collectively improves the resilience**.
- Excitatory and inhibitory heterogeneity **differentially impact** network trivialization and resilience.

Weighted connectivity of perturbed node introduces subtle changes resilience.

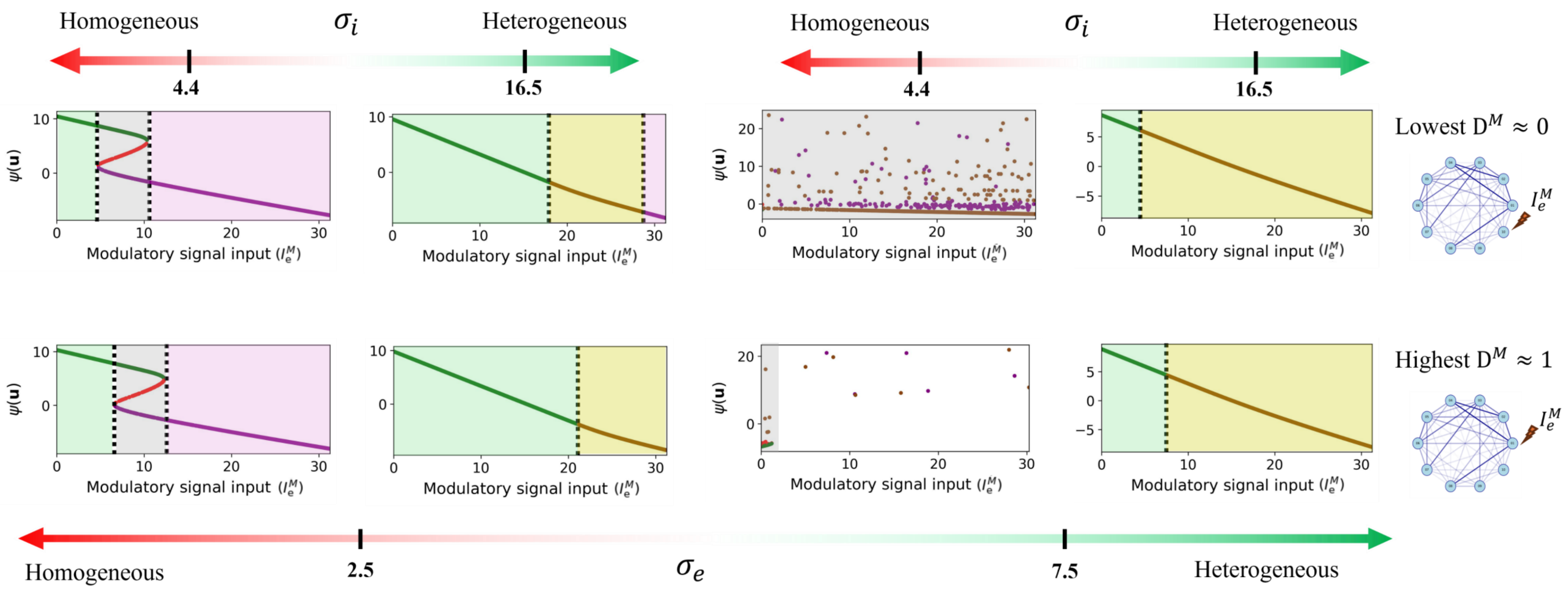

Figure: Trivialization and resilience subtly changes with the weighted connectivity of the perturbed node in the network.

- As D^M increases, system becomes **<u>more resilient</u>**.

Change in excitability heterogeneity distribution has no significant impact on the trivialization.

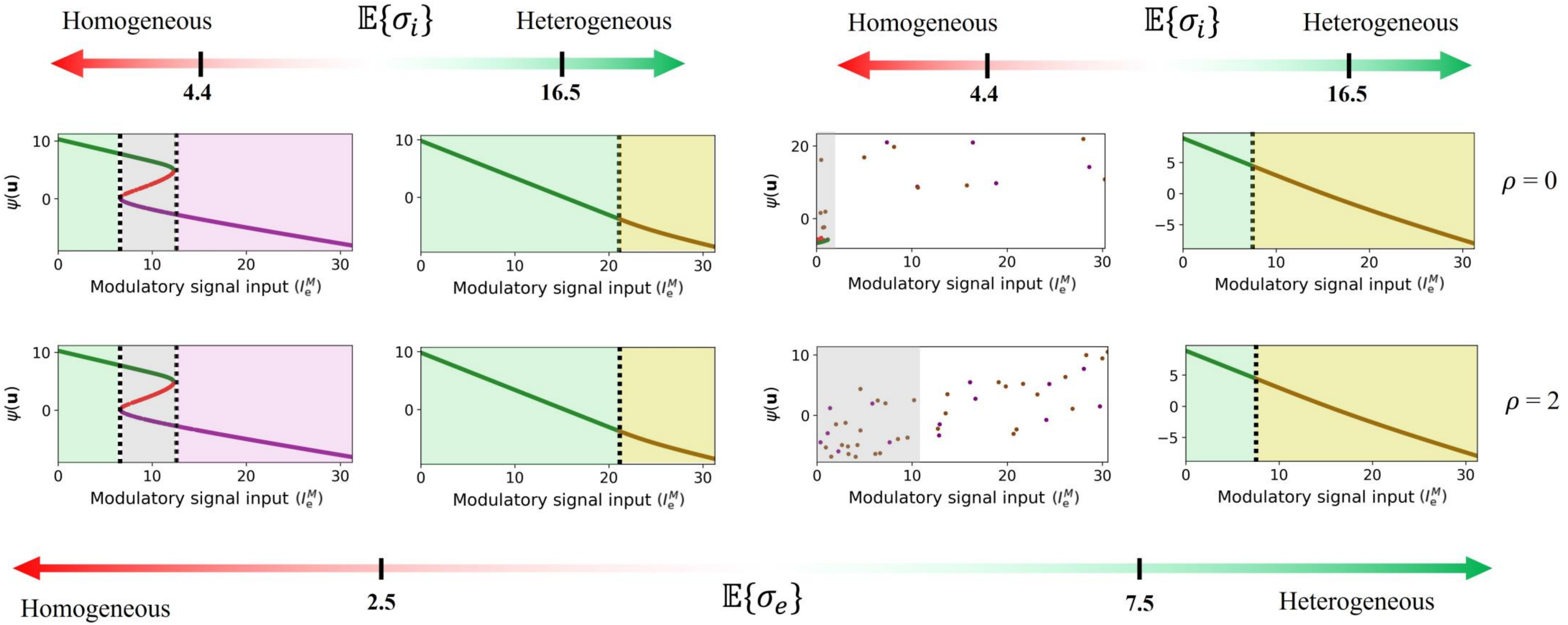

Figure: Comparison of equilibrium emergence and their states with mean heterogeneity vector $\mathbb{E}\{\sigma\} = \sigma^M$ and variance ρ.

- Inter-regional changes in the intrinsic excitability heterogeneity **<u>does not change</u>** trivialization and resilience of the system.

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Quantifying temporal and spectral synchronization among nodal dynamics.

Temporal-domain synchronization

1. Avalanche Event Synchronization [4]
2. Distance Correlation [5]
3. Kuramoto Order Parameter [6]

Spectral-domain synchronization

1. Global phase synchronization [3]
2. Phase Locking Value (PLV) [4]
3. SVD-based synchronization

Evaluated for three spectral bands:

a. Low-frequency oscillatory band (0.5 – 30 Hz)
b. Gamma-frequency band (40-85 Hz)
c. High-frequency oscillatory band (90-400 Hz)

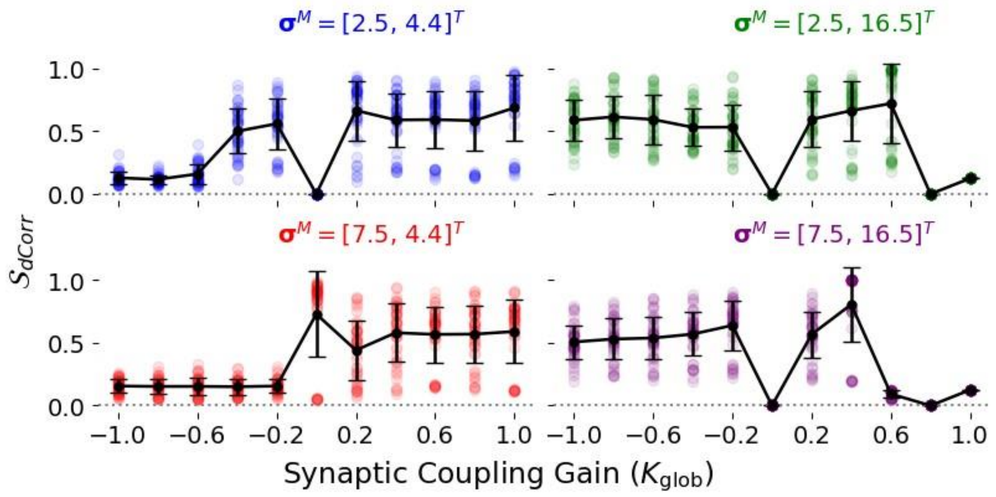
[4] T. Kreuz, F. Mormann, R. G. Andrzejak, A. Kraskov, K. Lehnertz, and P. Grassberger, “Measuring synchronization in coupled model systems: A comparison of different approaches,” Physica D: Nonlinear Phenomena, vol. 225, no. 1, pp. 29–42, 2007.

[5] D. Edelmann, T. F. M´ori, and G. J. Sz´ekely, “On relationships between the pearson and the distance correlation coefficients,” Statistics & probability letters, vol. 169, p. 108 960, 2021.

[6] G. Deco, V. Jirsa, A. R. McIntosh, O. Sporns, and R. K¨otter, “Key role of coupling, delay, and noise in resting brain fluctuations,” Proceedings of the National Academy of Sciences, vol. 106, no. 25, pp. 10 302–10 307, 2009.

Temporal synchrony with differential adjustments in excitatory and inhibitory heterogeneities .

Lowest $D^M \approx 0$



D^M increases

Highest $D^M \approx 1$

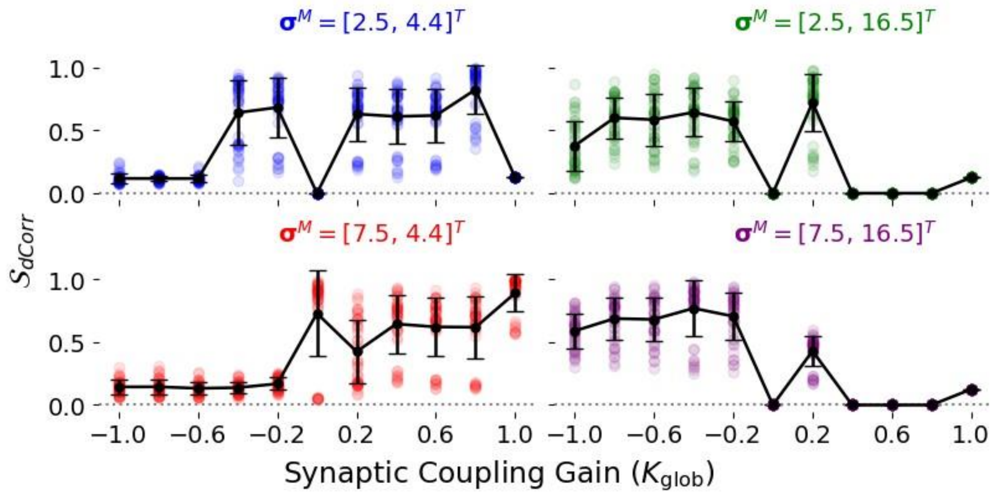


Figure: Change in temporal synchronization with differential excitatory and inhibitory heterogeneities as D^M changes.

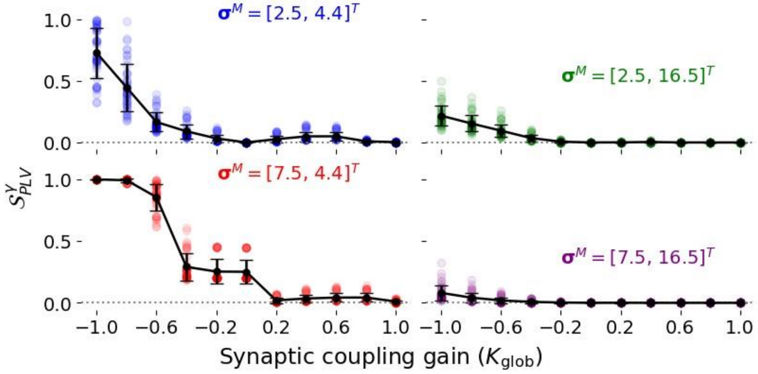
- **Inhibitory heterogeneity and synaptic coupling type** is associated with synchronous states in temporal domain.
- Variations in D^M **influence** the range of K_{glob} for which asynchronous dynamics exist.

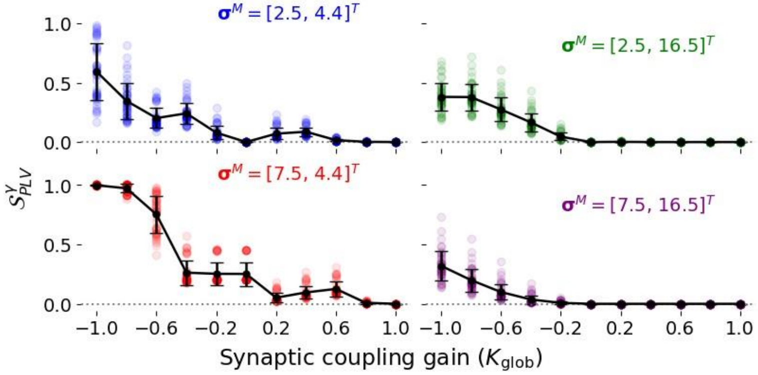
Spectral synchrony with differential adjustments in excitatory and inhibitory heterogeneities .

Lowest $D^M \approx 0$

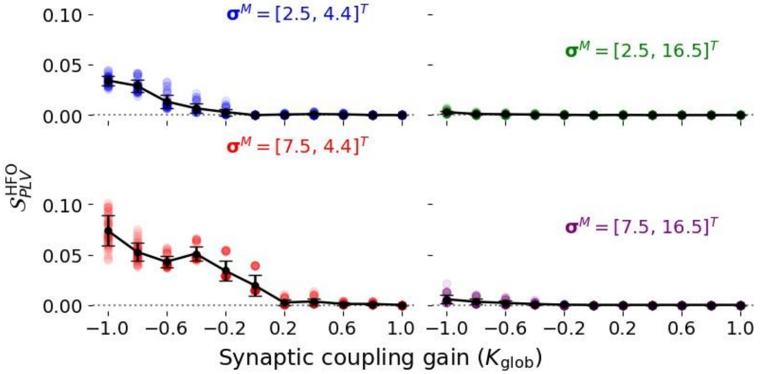
Highest $D^M \approx 1$

Gamma- frequency band

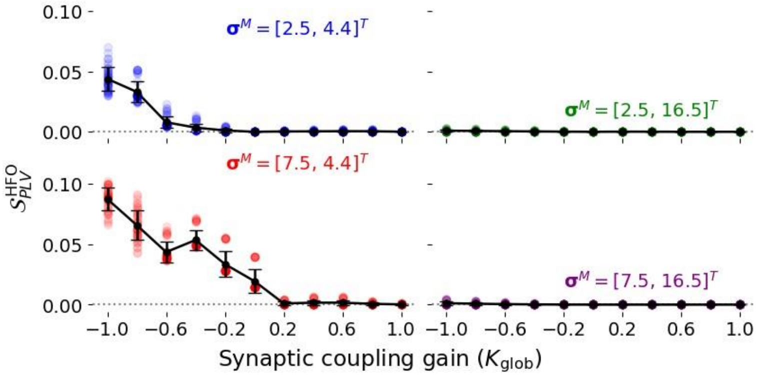




High frequency oscillations



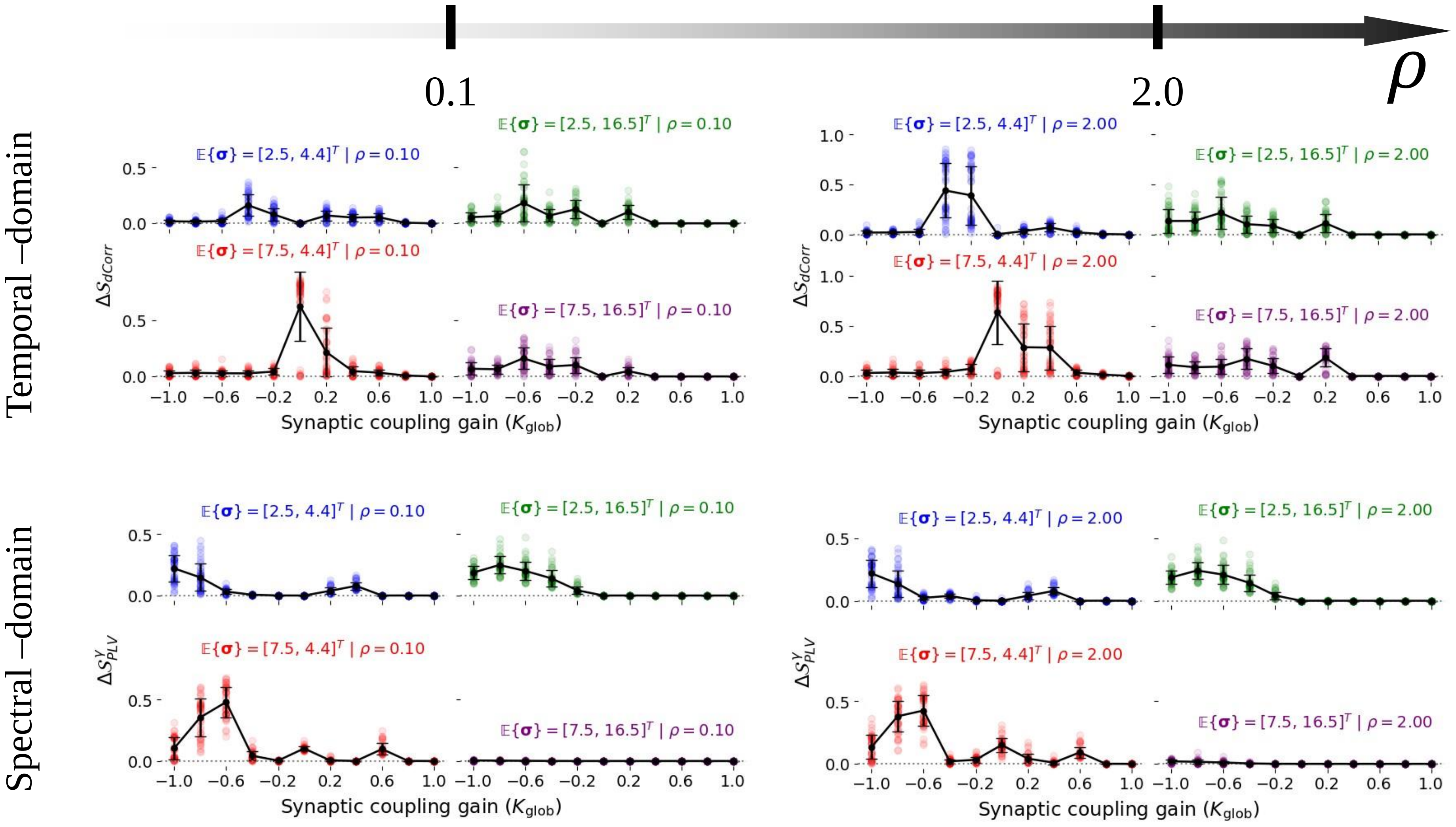
D^M increases



- Inhibitory heterogeneity with excitatory coupling **attains asynchrony** in high frequency bands.
- When D^M increases, so does the synchronization between the nodes.

Figure: Change in spectral synchronization with differential excitatory and inhibitory heterogeneities as D^M changes.

Change in the distribution of intrinsic excitability heterogeneity affects the synchrony.



- Significant synchrony changes for $K_{\text{glob}} < 0$ and/or reduced $\mathbb{E}\{\sigma_{\text{i}}\}$.

Figure: Change in the pair-wise synchrony measures as ρ changes

Outline

1. Motivation (Chapter 1 and 2)

2. Proposed mathematical brain model (Chapter 3)

3. Key results (Chapter 4 and 5)

4. Conclusions and future work (Chapter 6)

Key takeaways

- Intrinsic excitability heterogeneity and synaptic coupling collectively **reduce volatility** of a perturbed macro-scale network and **improves stability**.
- Intrinsic heterogeneity **trivializes the network** and **improves the resilience** against external perturbation.
- In both situations, **inhibitory heterogeneity is more important than the excitatory heterogeneity**. Also, synaptic coupling gain plays a critical role in them as well.
- At extreme heterogeneity and excitatory synaptic coupling, the network achieves asynchrony – irrespective of the measure.
- The weighted connectivity of the perturbed node imposes subtle changes on the dynamic properties.
- As the heterogeneity distribution of the unmodulated nodes are varied, the dynamical properties changes slightly, apart from the trivialization and improvement of resilience.

Overall, the intrinsic excitability heterogeneity of underlying finer-scale neurons improves stability and resilience of macro-scale dynamics against external modulations and achieve asynchronous states among the brain activities.

Limitations and Future Work

- Derive intrinsic excitability heterogeneity parameters from neuroimaging data to align the proposed framework with biophysical models.
  - MODEL VERIFICATION!!

- Excitability heterogeneity at each brain region was sampled using identical Gaussian distributions.
  - Use region-specific heterogeneity distributions (inter-regional diversity in local excitability heterogeneity).

- Vary structural connectivity to simulate different topologies.

Thank You!

Q & A

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Backup slides

Parameter manifolds within $(K_{\mathrm{glob}}, \sigma)$ separating stable and unstable states.

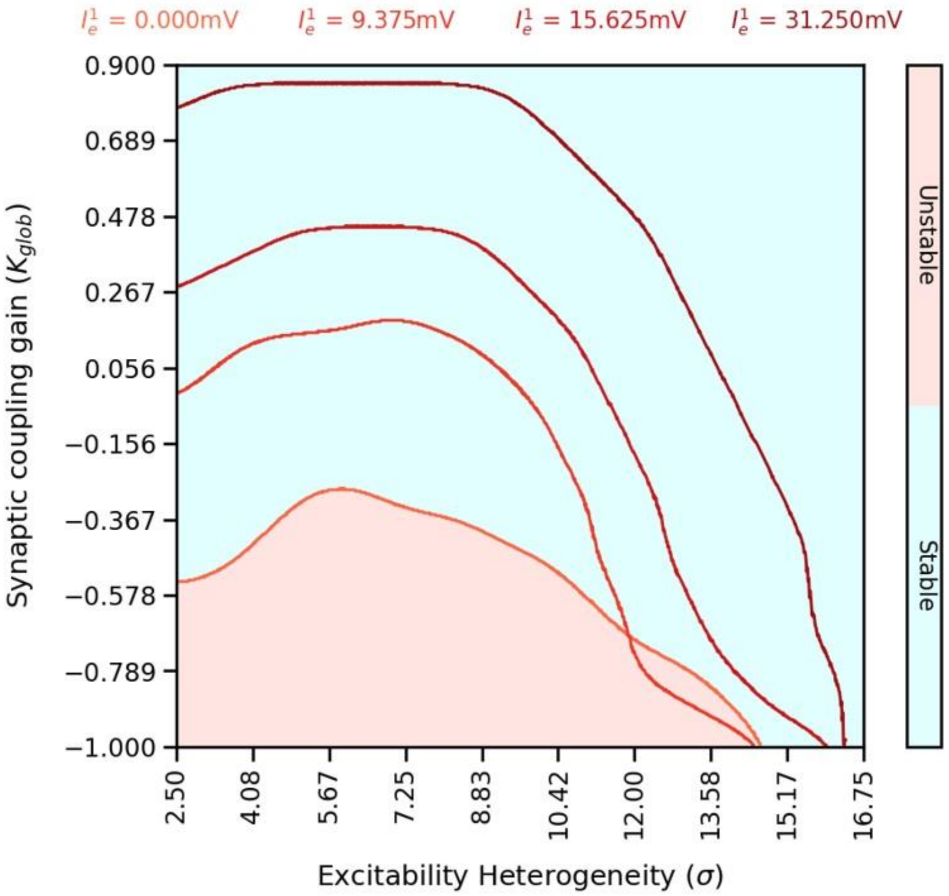


Figure: Parameter manifolds with different levels of external perturbation amplitudes.

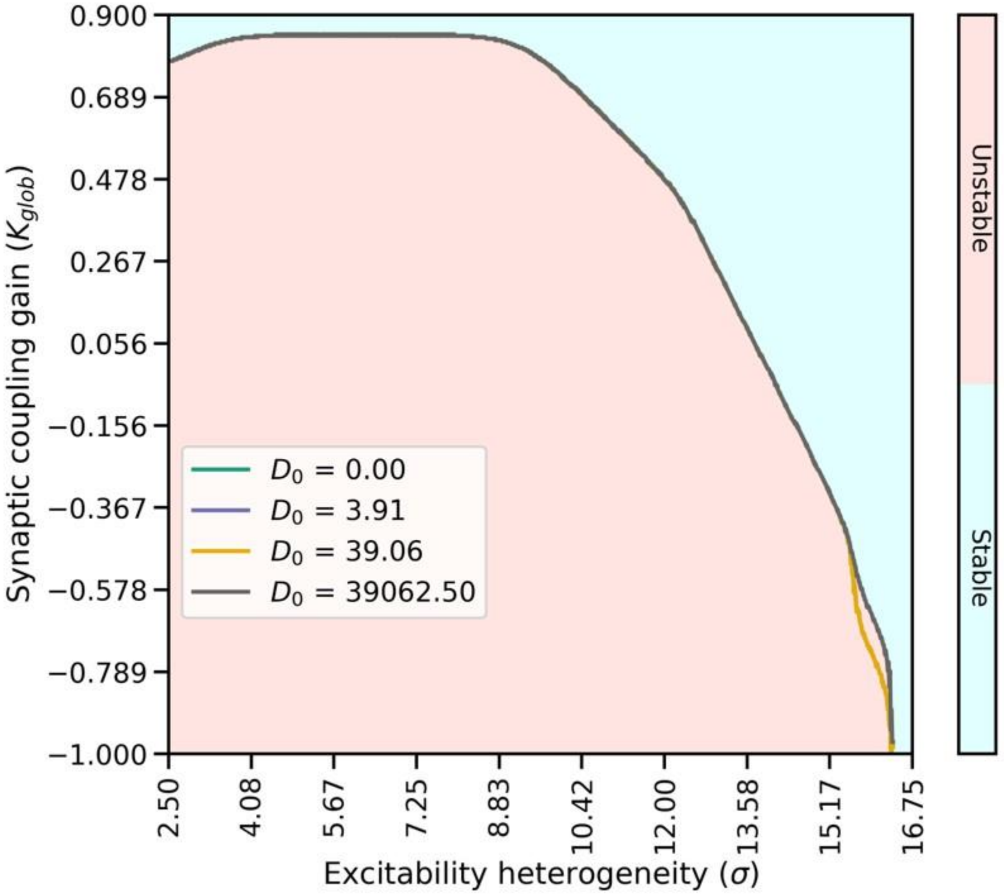


Figure: Parameter manifolds do not change with the addition of noise into the system dynamics. Results are for $I_e^1 = 31.25$mV.

Parameter manifolds within (σ_e, σ_i) separating stable and unstable (volatile) states.

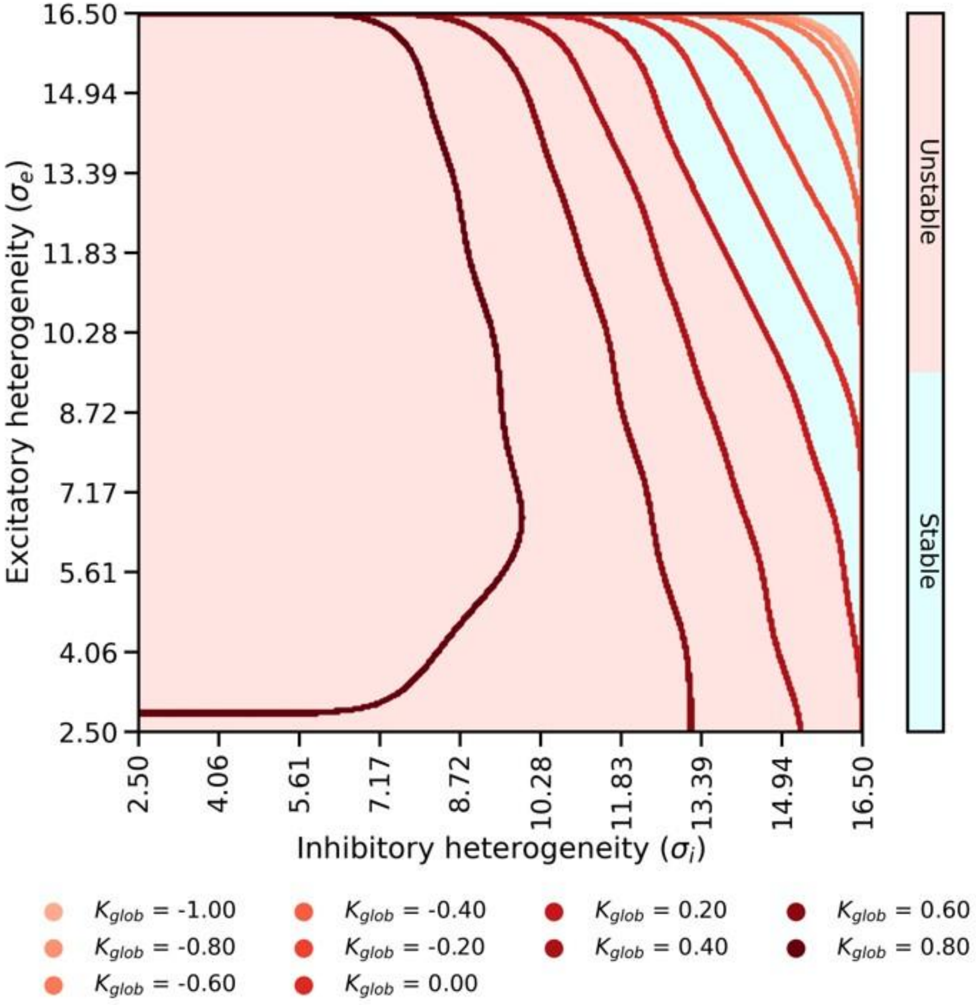


Figure: Parameter manifolds for Lyapunov stability with different synaptic coupling values for $I_e^1 = 31.25$mV.

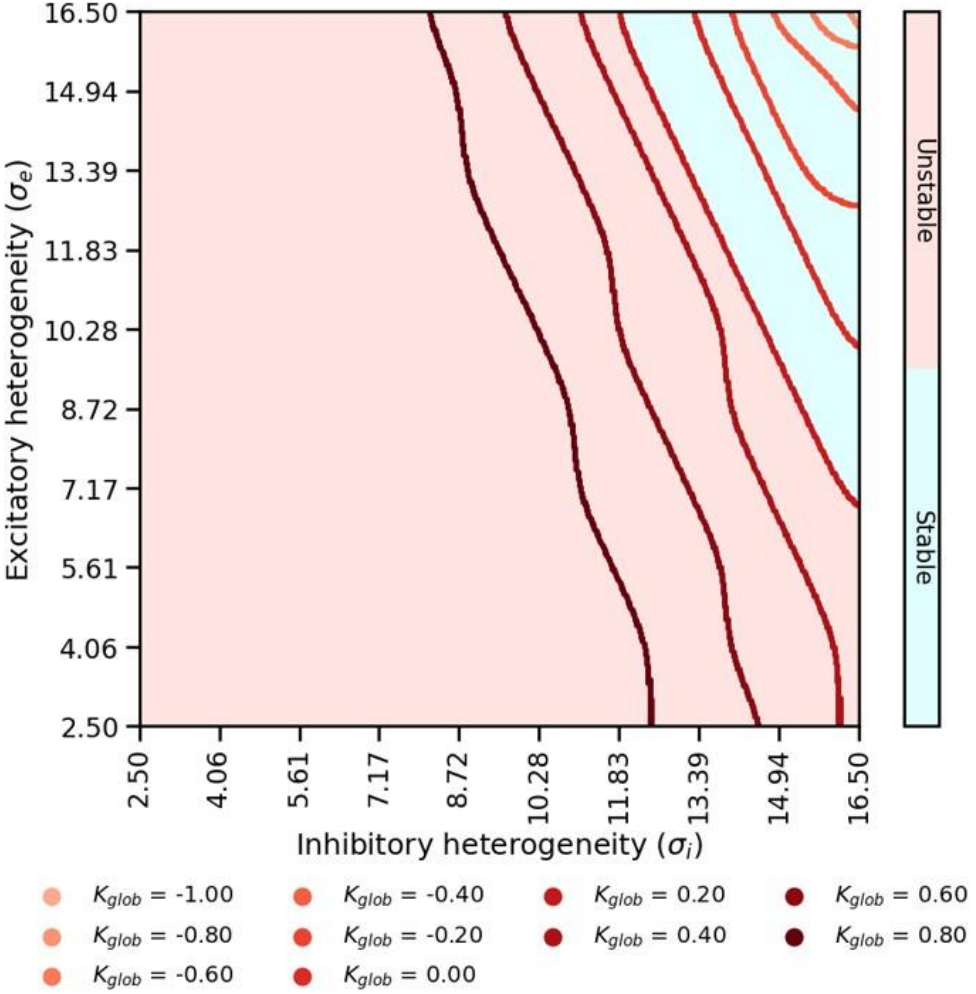
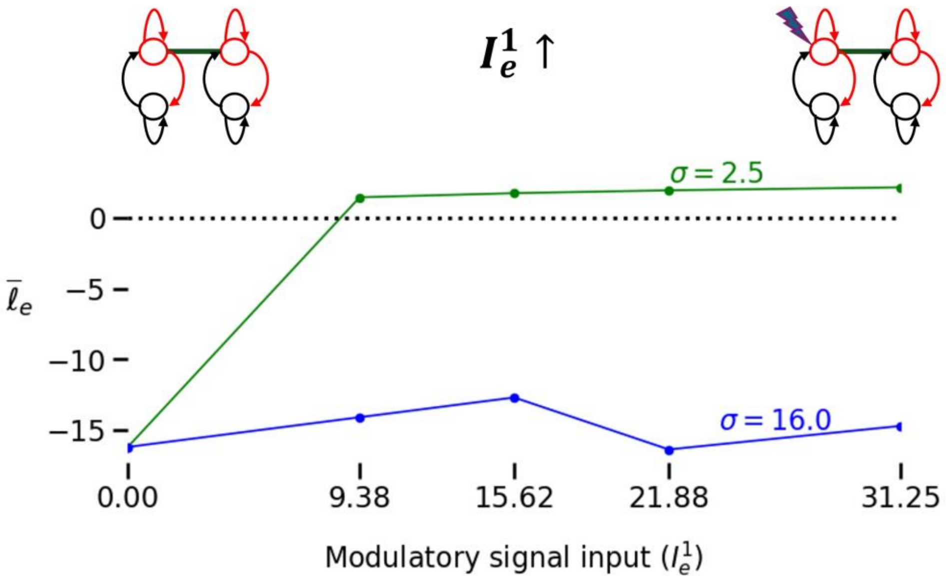


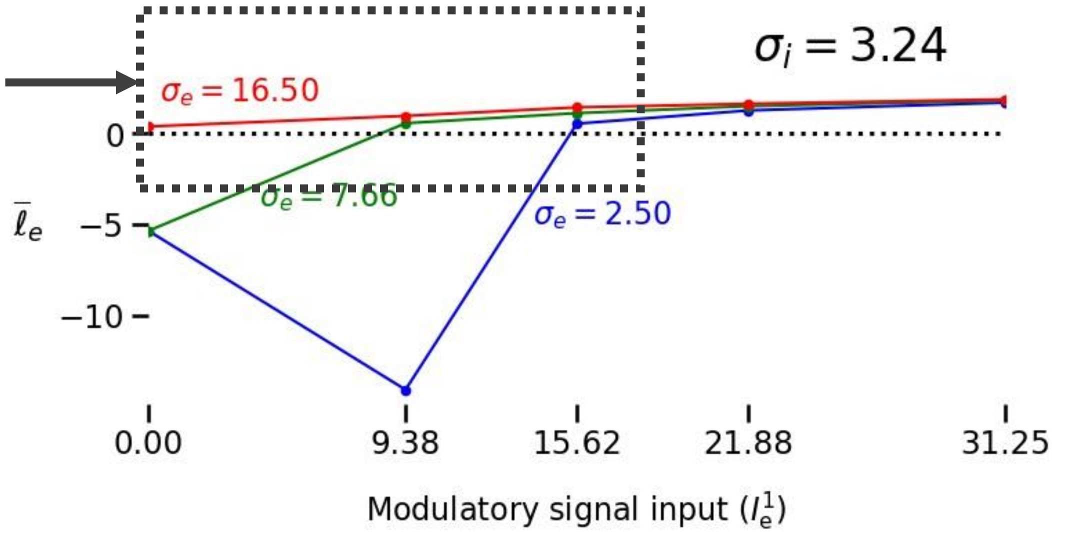
Figure: Parameter manifolds for network volatility with different synaptic coupling values for $I_e^1 = 31.25$mV.

Change of Lyapunov stability with different levels of perturbations.

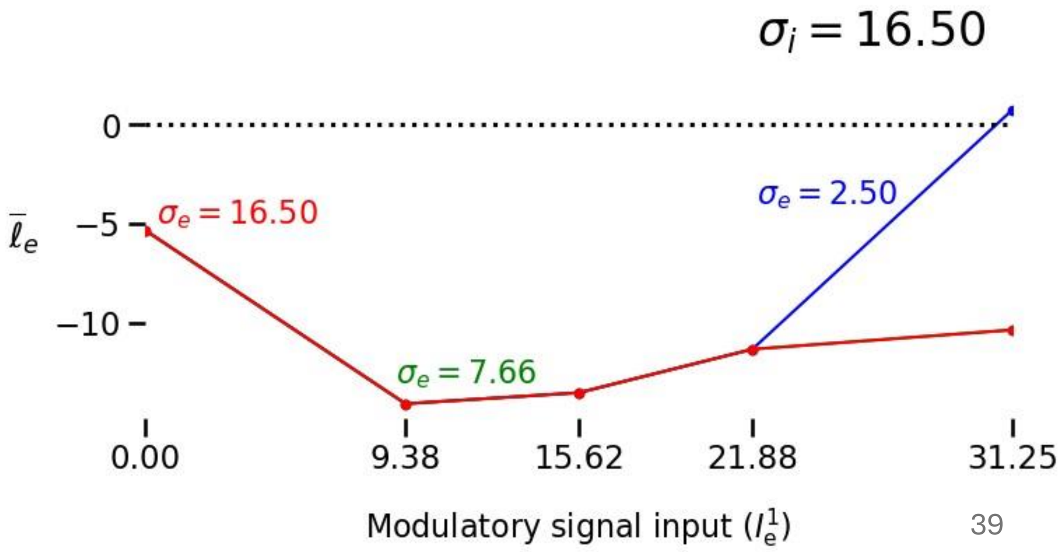
Increase in excitability heterogeneity renders perturbed node unstable for lower perturbations.



Homogeneous inhibitory sub-populations



Heterogeneous inhibitory sub-populations



Powell's Hybrid method to find roots of non-linear systems

Objective function:

$F(x) = \frac{1}{2}\|f(x)\|^2$, where $f: \mathbb{R}^n \rightarrow \mathbb{R}$

Optimization:

$$x^* = \operatorname{argmin}_x F(x)$$

Iterative implementation:

$x_k = x_{k-1} + \delta_k$ and $x_k \rightarrow x^*$ as k $\rightarrow K^*$

$$\delta_k = \begin{cases} \delta_{gn} & \text{if } \|\delta_{gn}\| \leq \Delta \\ \frac{\Delta}{\|\delta_{sd}\|}\delta_{sd} & \text{if } \|\delta_{gn}\| > \Delta, \|\delta_{sd}\| > \Delta \\ t\delta_{sd} + s(\delta_{gn} - t\delta_{sd}) & \text{if} \|\delta_{gn}\| > \Delta, \|\delta_{sd}\| \leq \Delta \end{cases}$$

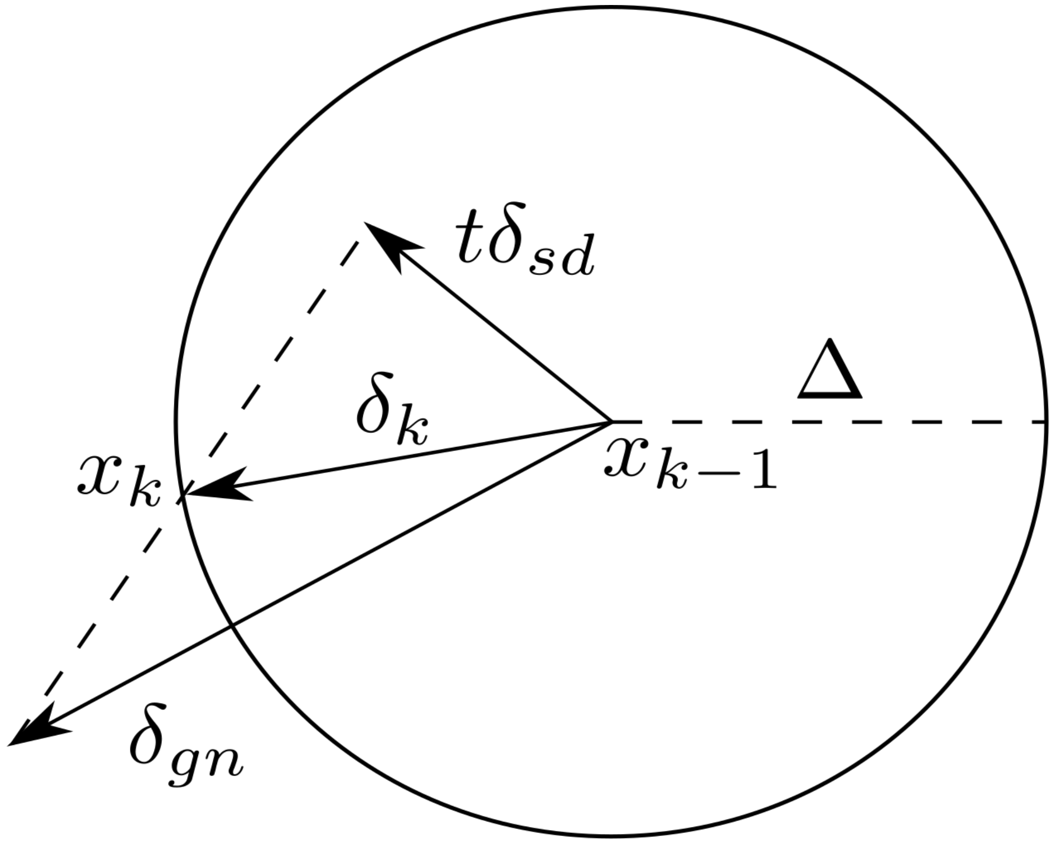


Figure: Selection of different δ_k in Powell's dog leg method

Categorizing the 2-node system fixed points

Table 3.2: Determining local system stability based on eigenvalues of the linearized system.

Stable state	**Stability criteria** $\Re\{\lambda^{l,k}\} = a^{l,k}$	$\Im\{\lambda^{l,k}\} = b^{l,k}$	**Color code**
Stable spiral	$a^{l,k} < 0 \ \forall \ k \in [2N]$	$\exists b^{l,k}$ s.t. $b^{l,k} \neq 0$	Brown
Stable node	$a^{l,k} < 0 \ \forall \ k \in [2N]$	$b^{l,k} = 0 \ \forall \ k \in [2N]$	Green
Centre	$a^{l,k} = 0 \ \forall \ k \in [2N]$	$b^{l,k} = 0 \ \forall \ k \in [2N]$	Black
Unstable spiral	$a^{l,k} > 0 \ \forall \ k \in [2N]$	$\exists b^{l,k}$ s.t. $b^{l,k} \neq 0$	Purple
Unstable node	$a^{l,k} > 0 \ \forall \ k \in [2N]$	$b^{l,k} = 0 \ \forall \ k \in [2N]$	Blue
Saddle Point	$\exists a^{l,k}$ s.t. $a^{l,k} > 0$	$b^{l,k} = 0 \ \forall \ k \in [2N]$	Red

Where $\lambda_i's$ are the eigen values of the Jacobian ($\mathbf{J}$) of 2-node coupled system evaluated at the equilibriums.

Using Linearization around the fixed point:

$$\frac{d}{dt}\Delta\mathbf{U_{eq}} = \mathbf{J}_{4\times4}\big|_{\mathbf{U}=\mathbf{U_{eq}}}\Delta\mathbf{U_{eq}}$$

Define:

$$\mathbf{J} = \begin{bmatrix} \frac{\partial G_1}{\partial U_e^1} & \frac{\partial G_1}{\partial U_i^1} & \frac{\partial G_1}{\partial U_e^2} & \frac{\partial G_1}{\partial U_i^2} \\ \frac{\partial G_2}{\partial U_e^1} & \frac{\partial G_2}{\partial U_i^1} & \frac{\partial G_2}{\partial U_e^2} & \frac{\partial G_2}{\partial U_i^2} \\ \frac{\partial G_3}{\partial U_e^1} & \frac{\partial G_3}{\partial U_i^1} & \frac{\partial G_3}{\partial U_e^2} & \frac{\partial G_3}{\partial U_i^2} \\ \frac{\partial G_4}{\partial U_e^1} & \frac{\partial G_4}{\partial U_i^1} & \frac{\partial G_4}{\partial U_e^2} & \frac{\partial G_4}{\partial U_i^2} \end{bmatrix} =$$

$$\begin{bmatrix} -\alpha_e + w_{ee}\alpha_e R_e^1 & w_{ie}\alpha_e R_i^1 & C\alpha_e a_{12} & 0 \\ w_{ei}\alpha_i R_e^1 & -\alpha_i + w_{ii}\alpha_i R_i^1 & 0 & 0 \\ C\alpha_e a_{21} & 0 & -\alpha_e + w_{ee}\alpha_e R_e^2 & w_{ie}\alpha_e R_i^2 \\ 0 & 0 & w_{ei}\alpha_i R_e^2 & -\alpha_i + w_{ii}\alpha_i R_i^2 \end{bmatrix}$$

where:

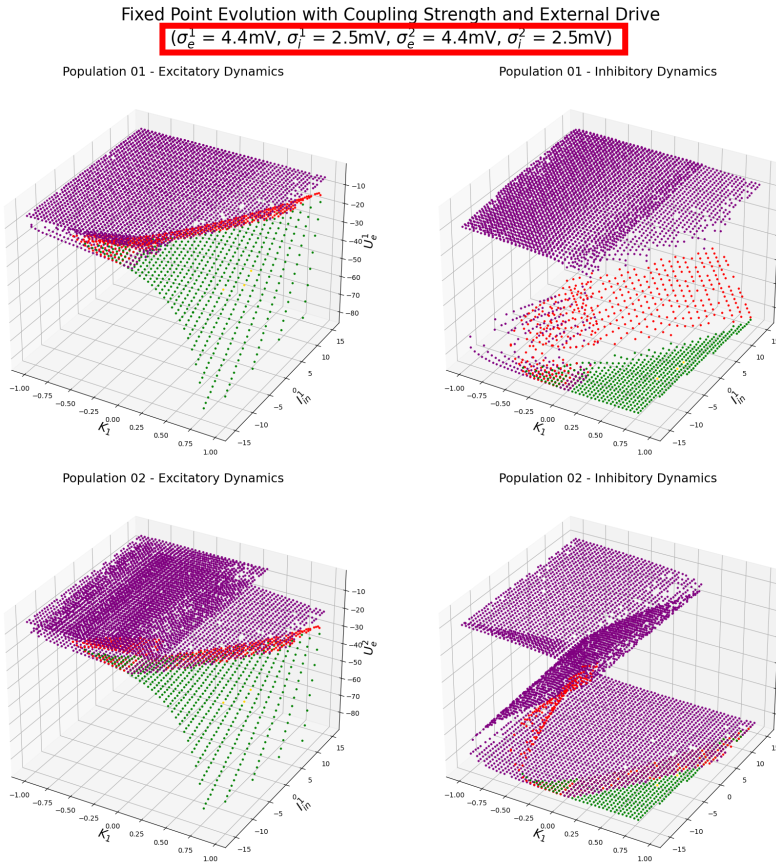
$$G_1 \equiv \alpha_e\left[-U_e^1 + w_{ee}F(U_e^1, \sigma_e^1) + Ca_{12}U_e^2 + w_{ie}F(U_i^1, \sigma_i^1) + I_e^o + I_e(t)\right]$$
$$G_2 \equiv \alpha_i\left[-U_i^1 + w_{ei}F(U_e^1, \sigma_e^1) + w_{ii}F(U_i^1, \sigma_i^1) + I_i^o\right]$$
$$G_3 \equiv \alpha_e\left[-U_e^2 + w_{ee}F(U_e^2, \sigma_e^2) + Ca_{21}U_e^1 + w_{ie}F(U_i^2, \sigma_i^2) + I_e^o\right]$$
$$G_4 \equiv \alpha_i\left[-U_i^2 + w_{ei}F(U_e^2, \sigma_e^2) + w_{ii}F(U_i^2, \sigma_i^2) + I_i^o\right]$$
$$R_{e,i}^k \equiv \int_{-\infty}^{+\infty} f'(U_{e,i}^k + v, 0)\rho^k(v)dv \text{ and } \rho^k = \mathcal{N}(0, \sigma_{e,i}^k)$$

Figure: Jacobian Derivation for 2-node coupled system.

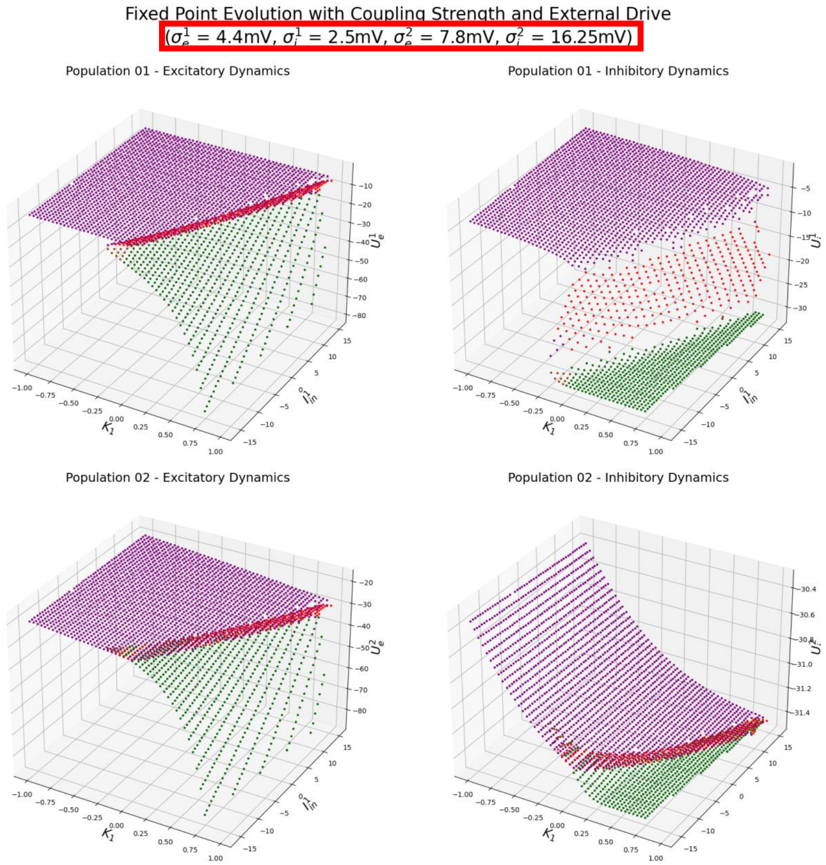
Note: $\alpha_{e,i} = \frac{1}{\tau_{e,i}}$ and $C = K_{\text{glob}}$

Local stability analysis of the system with both K_{glob} and $I_{in}^{1} = I_{\text{e}}^{1} + I_{\text{e},0}$

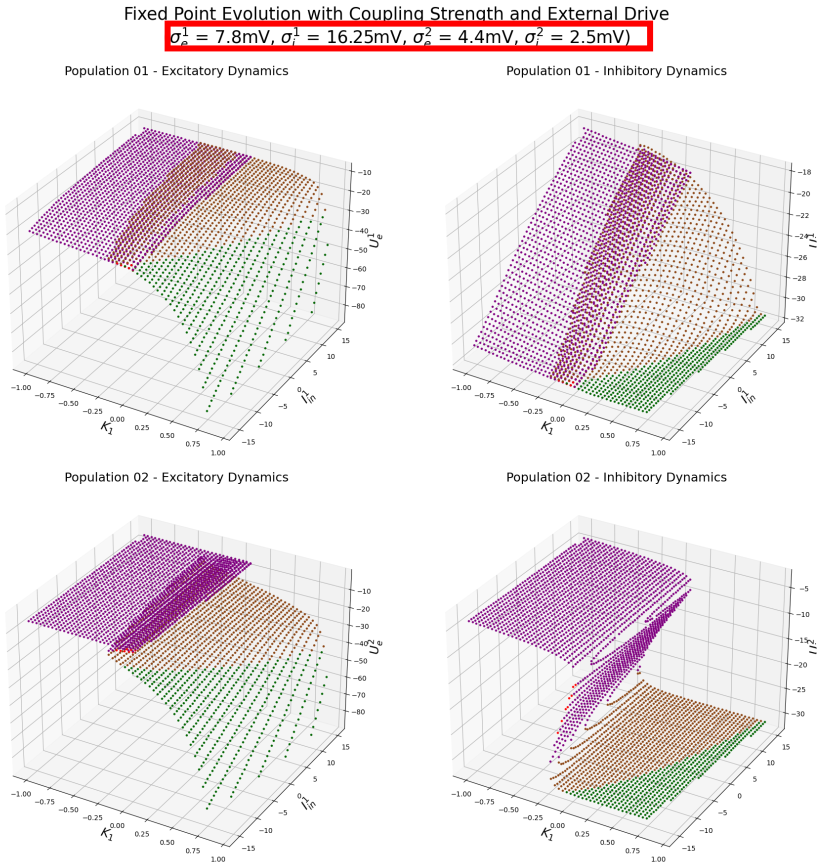
- The results are obtained by assigning different values to $(\sigma_e^1, \sigma_i^1, \sigma_e^2, \sigma_i^2)$.
- Interpretation and presentation of the results is difficult.



(A) In this analysis, both nodes are highly homogenous.



(B) In this analysis, node 1 is homogenous and node 2 is relatively heterogeneous.

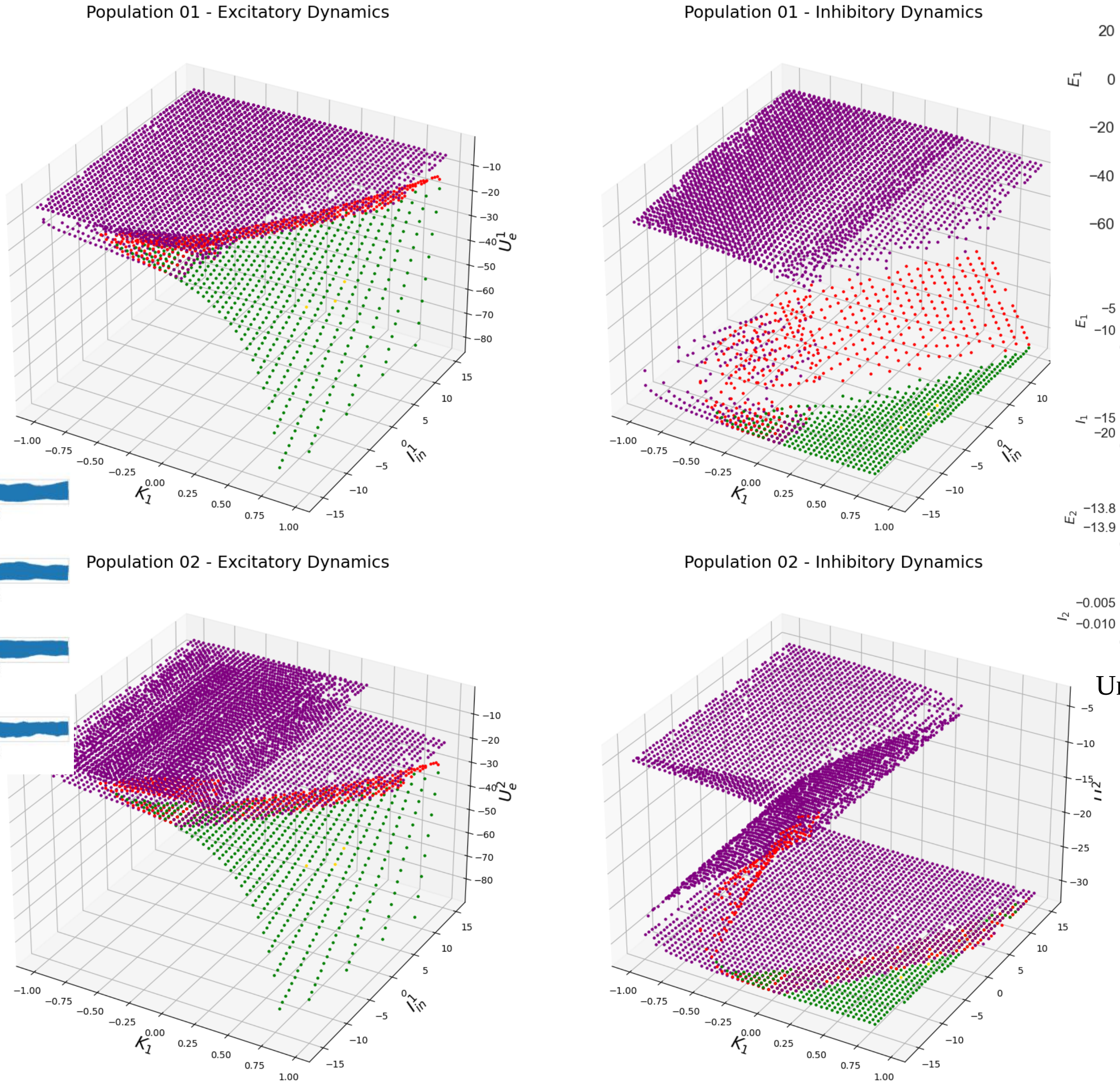


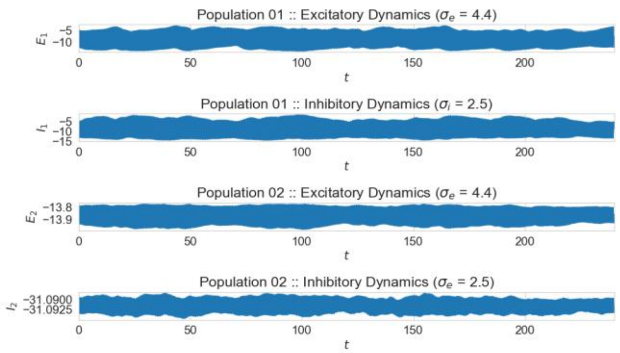
(C) In this analysis, node 1 is heterogenous and node 2 is relatively homogenous.

Slide 42
Figure (A)

Fixed Point Evolution with Coupling Strength and External Drive

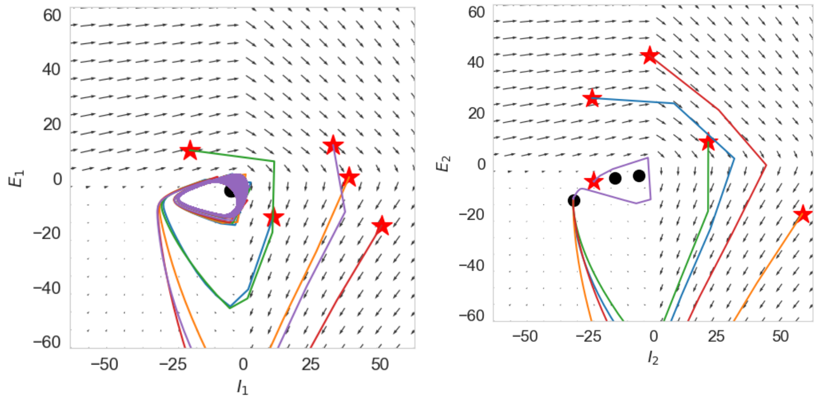
$(\sigma_e^1 = 4.4\text{mV}, \sigma_i^1 = 2.5\text{mV}, \sigma_e^2 = 4.4\text{mV}, \sigma_i^2 = 2.5\text{mV})$

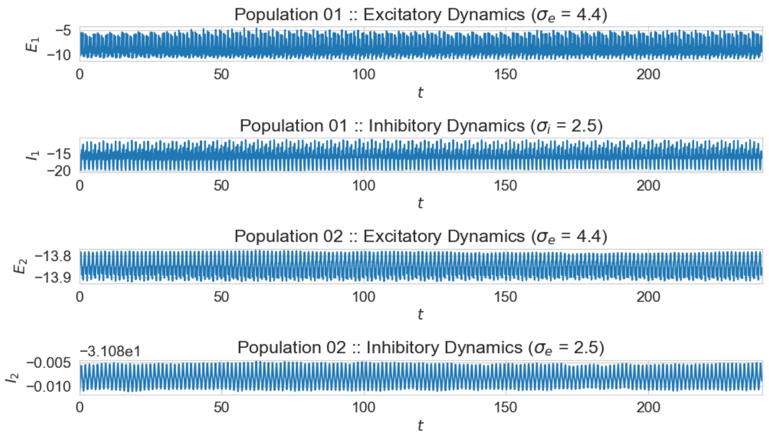




Unstable spiral (purple

$K_{12} = -0.2, I_{in}^1 = 12.5$





Unstable spiral (purple), unstable saddle node (red) and stable node (green) (Multi-stability)

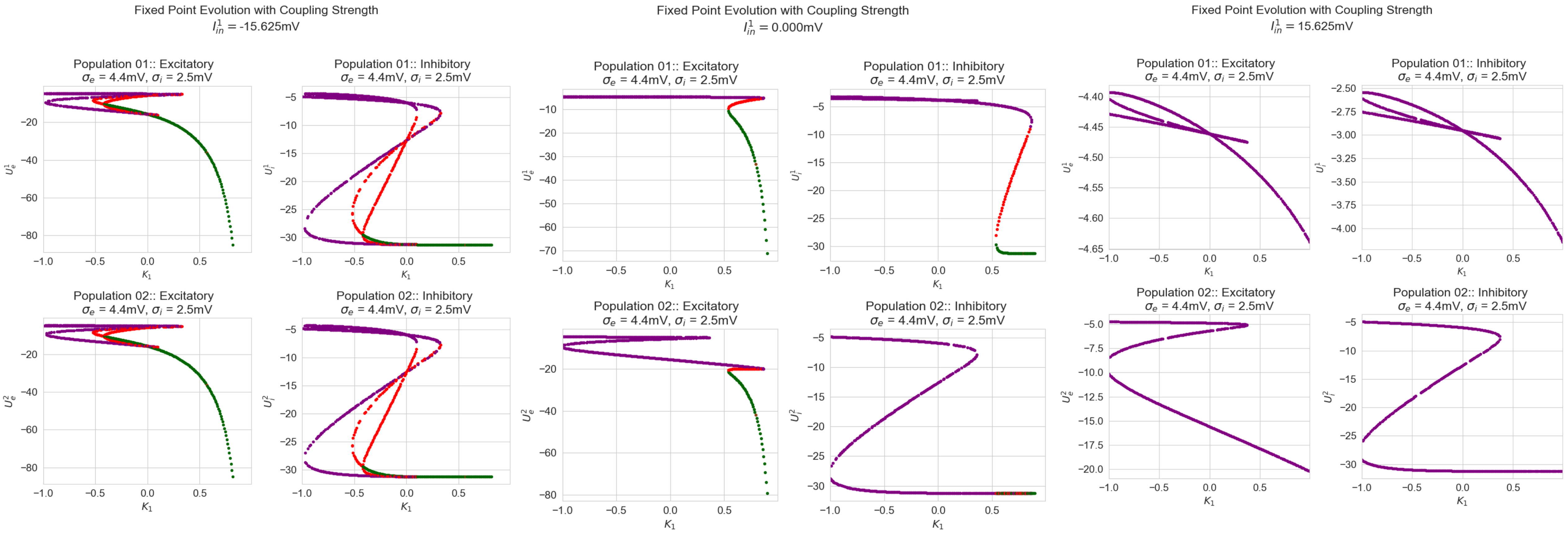
$K_{\text{glob}} = -0.2, I_{in}^1 = -6.25$

Conclusion:

Either both nodes have unstable spiral (purple) or a multi-stable state (i.e., node 1 showing an unstable spiral and node 2 a stable node).

Figure 5: Fixed Point Analysis - K_{glob} vs. I_{in}^1 (Node 1 – Homogenous | Node 2 – Homogenous)

Projections from Figure (A) of Slide 42 onto $I_{in}^{1} = \gamma$: Multi-stability region shifts to right rapidly as γ increases when both nodes are homogenous.



(a) Fixed points show 3 different types of behaviors. Unstable spiral (purple), Unstable saddle point (red), and Stable node (green).

(b) Fixed points show 3 different types of behaviors. Unstable spiral (purple), Unstable saddle point (red), Stable node (green). Multi stability region moves towards right with higher external input. Operating space is dominated by the unstable/multi-stable states.

(c) For any level of coupling, the system only is in an unstable spiral state.

Figure: Fixed Point Analysis with K_{glob} for a given I_{in}^{1} (Node 1 – Homogenous | Node 2 – Homogenous)

Analysis of multi-stable regions when both nodes are homogenous

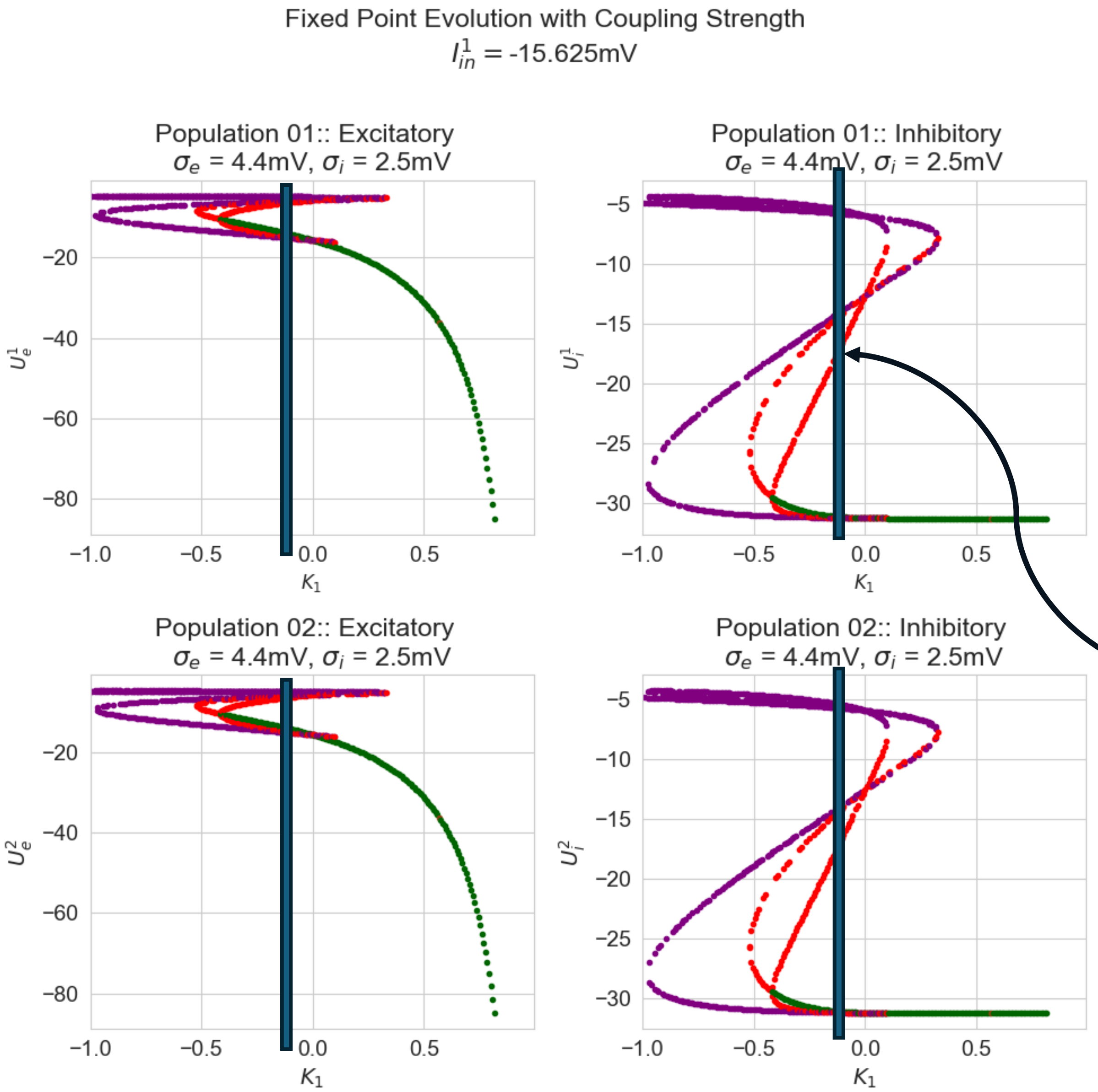
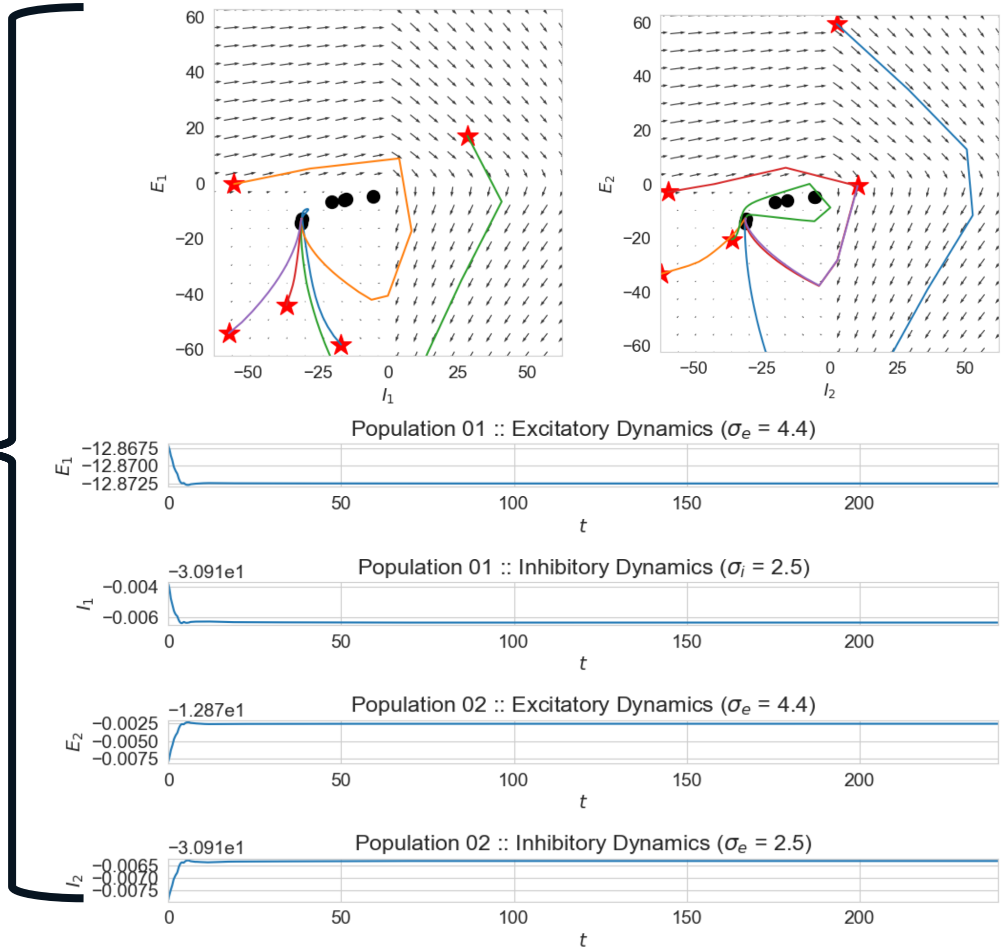
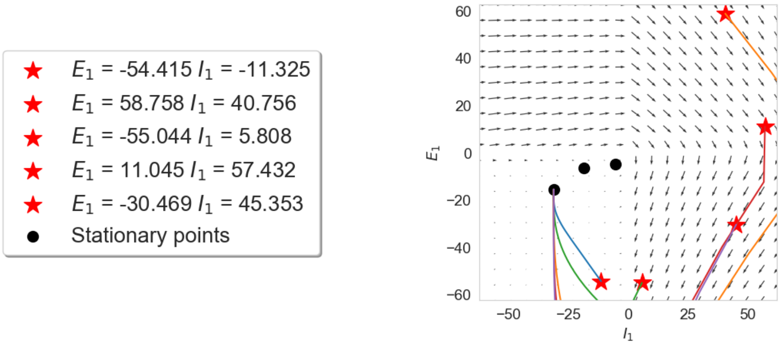


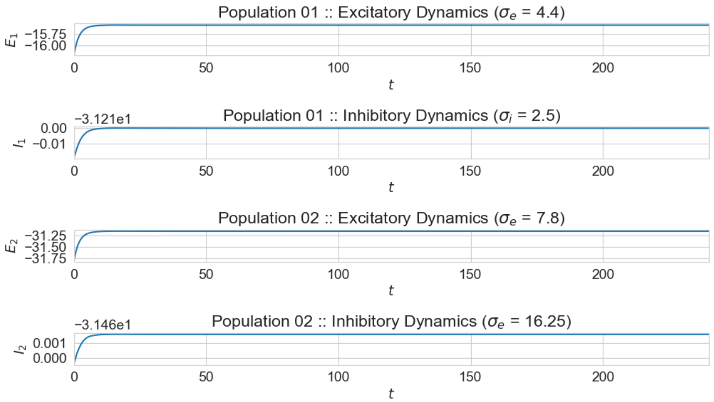
Figure: Fixed Point Analysis for specific conditions (Node 1 – Homogenous | Node 2 – Homogenous)



Multiple fixed points are present (7) with one stable node (green)
$K = -0.2$, $I_{in}^1 = -15.625$mV.

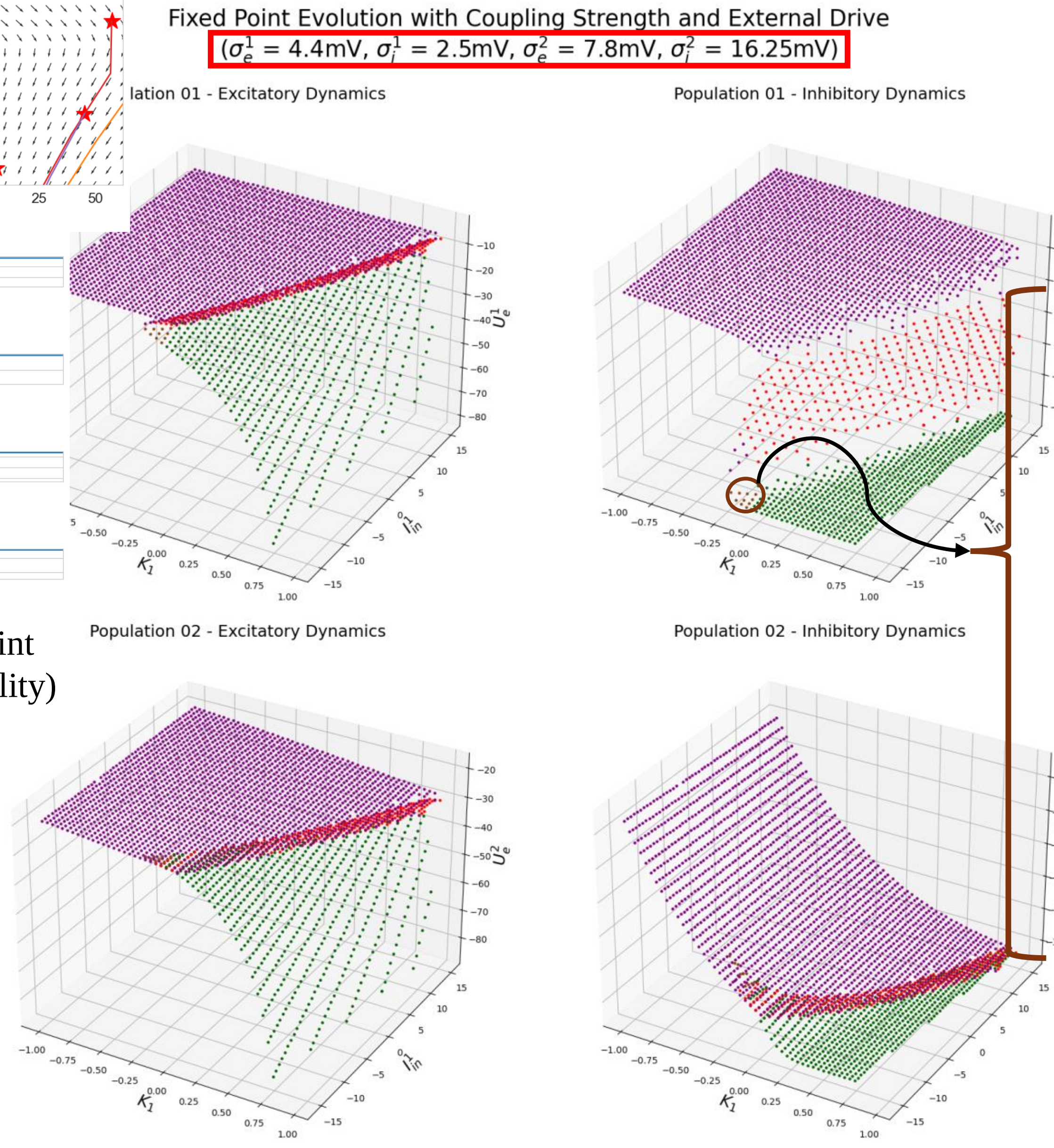
Slide 42 Figure (B)

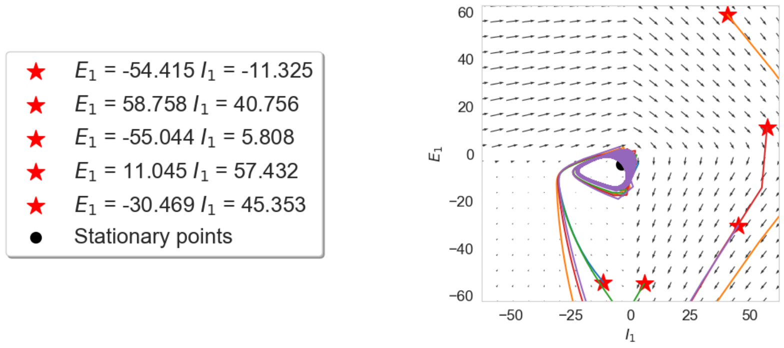


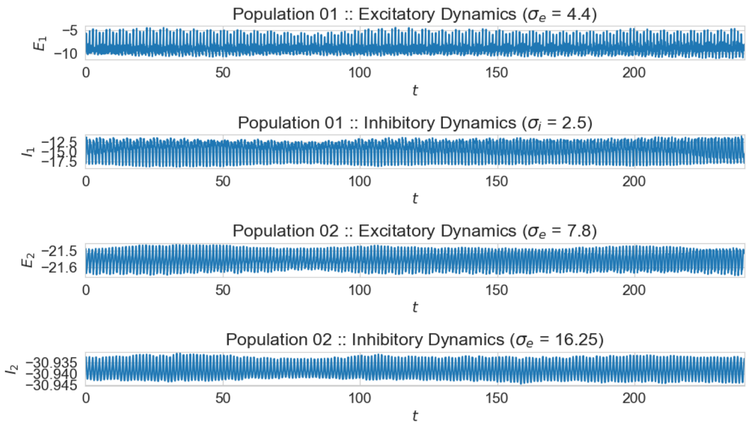


Stable node (green), Unstable saddle point (red), Unstable spiral (purple) (Multi stability)

$K_{12} = 0.5, I_{in}^1 = 0.0$







Stable Spiral (brown) (Mono stability)

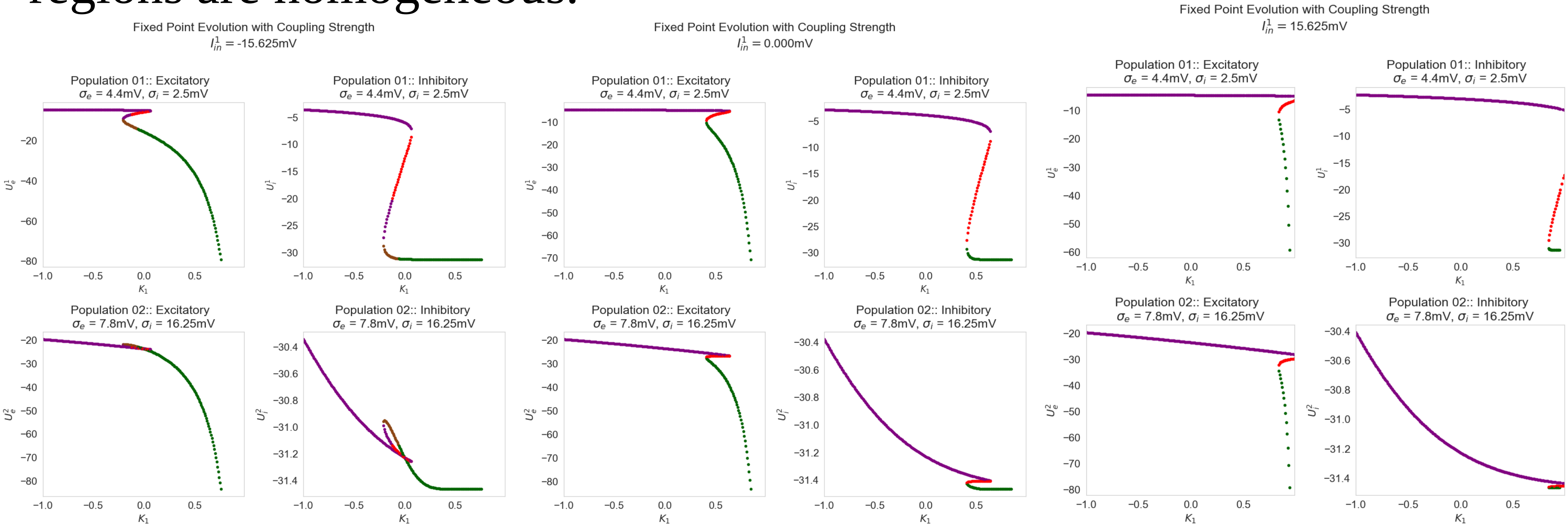
$K_{\text{glob}} = -0.2, I_{in}^1 = -6.25$

Conclusion:

When the heterogeneity is added to the coupled system, multi-stable regions are created such that for certain coupling values and external inputs system is stable.

Figure 6: Fixed Point Analysis - K_{glob} vs. I_{in}^1 (Node 1 – Homogenous | Node 2 – Heterogenous)

Projections from Figure (B) of Slide 42 onto $I_{in}^{1} = \gamma$: Multi-stability region shifts towards right as γ increases at a lower rate than when both regions are homogeneous.



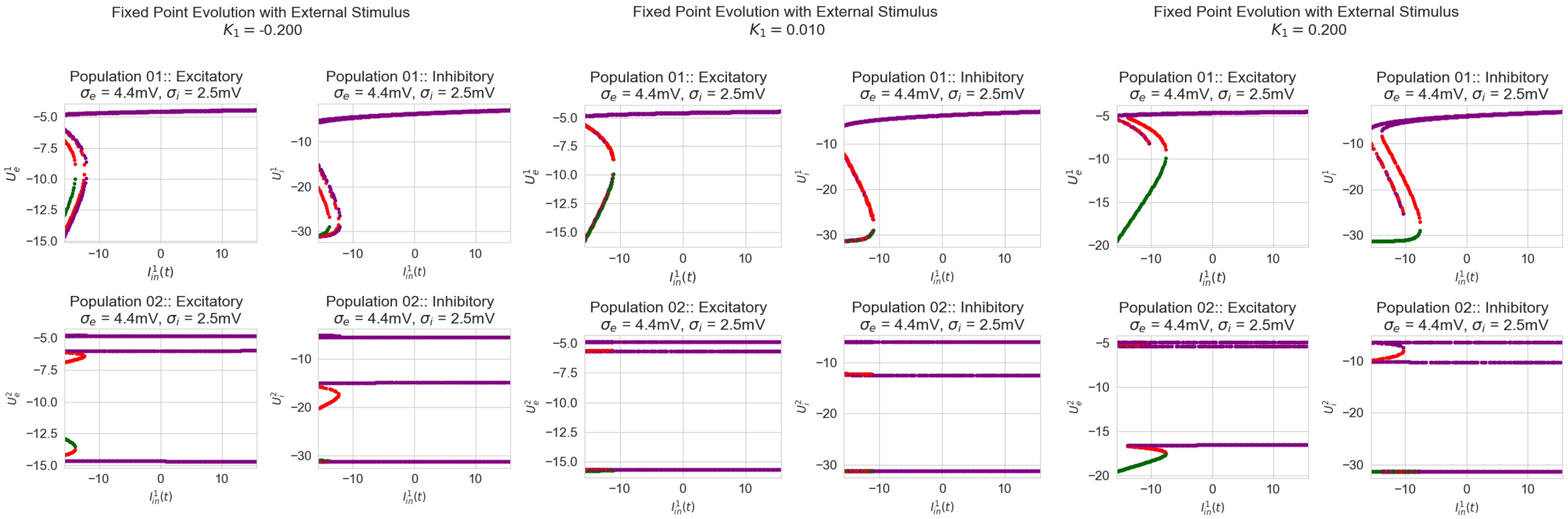
(a) Fixed points show 4 different types of behaviors. Unstable spiral (purple), Unstable saddle point (red), Stable Spiral (brown) and Stable node (green)

(b) Fixed points show 3 different types of behaviors. Unstable spiral (purple), Unstable saddle point (red), Stable node (green).
Multi stability region moves towards right with higher external input.

(c) Fixed points show same types of behaviors as of (b). Multi stability region moves even further towards right with higher external input.

Figure: Fixed Point Analysis with K_{glob} for a given I_{in}^{1} (Node 1 – Homogenous | Node 2 – Heterogenous)

Projections from Figure (B) of Slide 42 onto $K_{12} = \gamma$: Multi-stability region shifts to right at a very slow rate as γ increases.



(a) Fixed points show 3 behaviors at the far-left: most of the operative space has a unstable spiral (purple).

(b) Fixed points now show 3 different types of behaviors. Unstable spiral (purple), Unstable saddle point (red), Stable node (green).
Multi stability region seems to have states mingled together and moves towards right. Still unstable spiraling dominates.

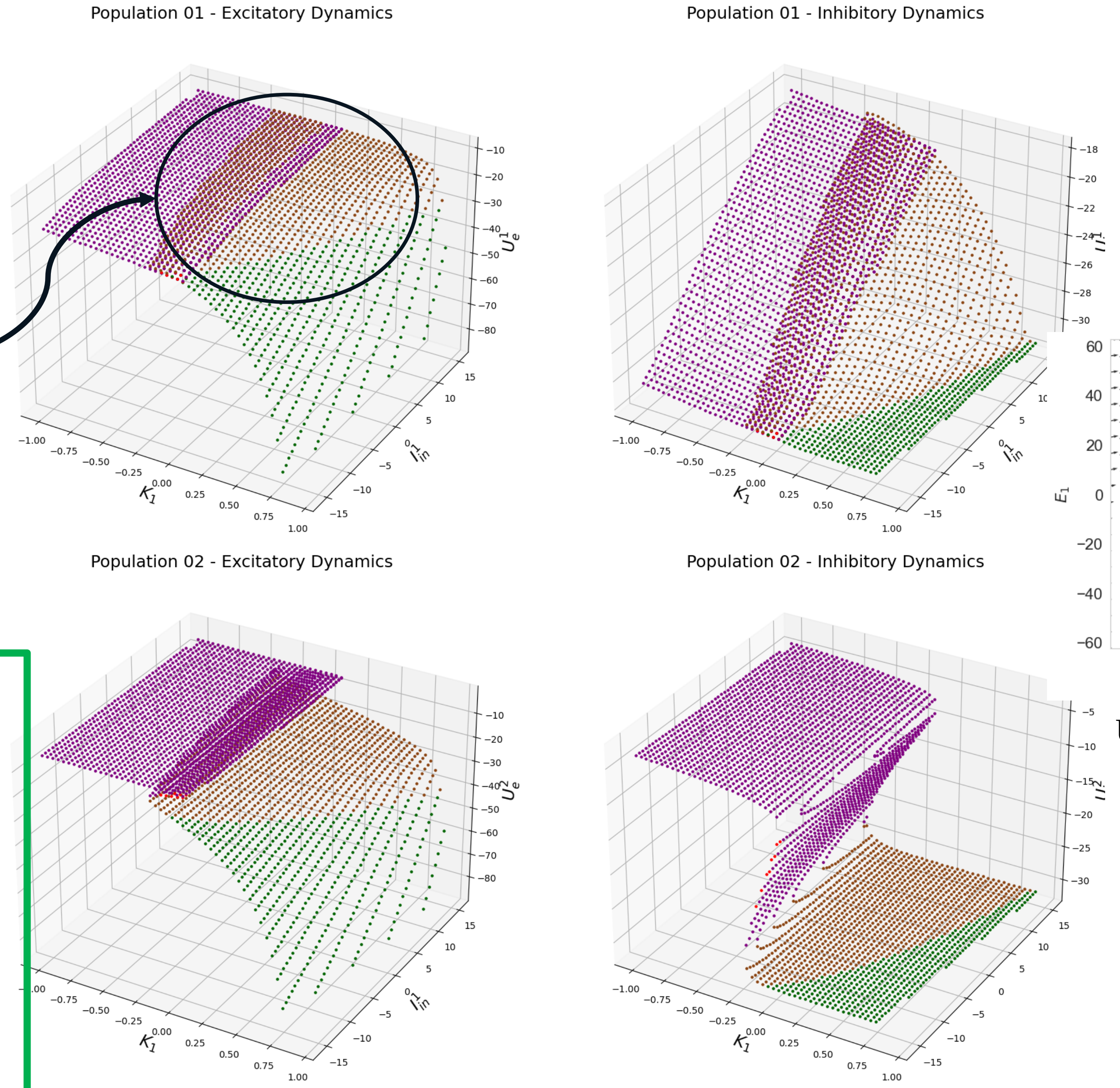
(c) Fixed points show same types of behaviors as of (b). Multi stability region moves slightly towards.

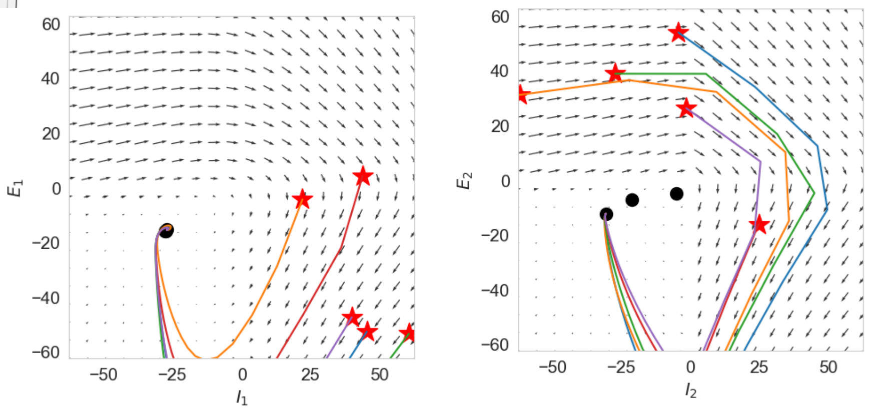
Figure: Fixed Point Analysis with I_{in}^1 for a given K_{glob} (Node 1 – Homogenous | Node 2 – Homogenous)

Slide 42
Figure (C)

Fixed Point Evolution with Coupling Strength and External Drive
(σ_e^1 = 7.8mV, σ_i^1 = 16.25mV, σ_e^2 = 4.4mV, σ_i^2 = 2.5mV)

Majority of the multi-stable region is dominated by stable spirals as opposed to the unstable spirals when node 1 is homogenous.



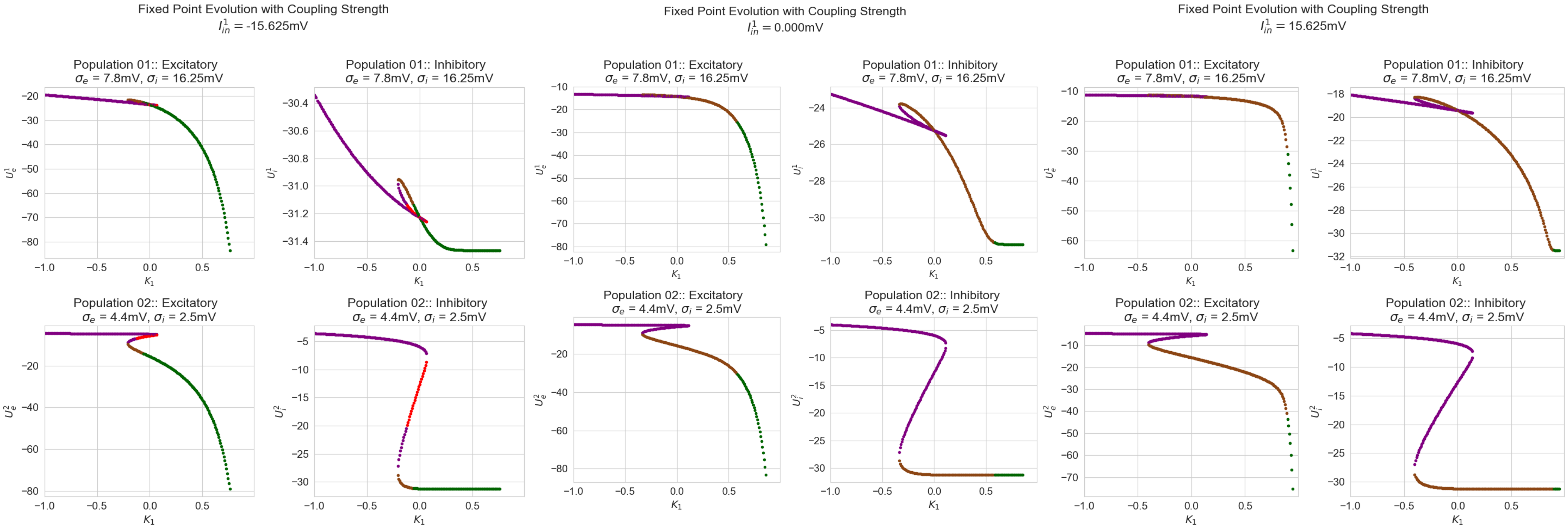


Unstable spiral (purple), Stable spiral (brown) and Stable node (green)
Multi-stability is dominated by the higher homogenous node's dynamics.
$K_{\text{glob}} = -0.2$, $I_{in}^1 = -6.25\text{mV}$

Conclusion:
When the heterogeneity is added to the coupled system, multi-stable regions are created such that for certain coupling values, the system become resilient to higher values of inputs.

Figure 7: Fixed Point Analysis - K_{glob} vs. I_{in}^1 (Node 1 – Heterogenous | Node 2 – Homogenous)

Projections from Figure (C) in Slide 42 onto $I_{in}^{1} = \gamma$: Multi-stability region remains almost the same with significant portion having stable dynamics as γ increases.



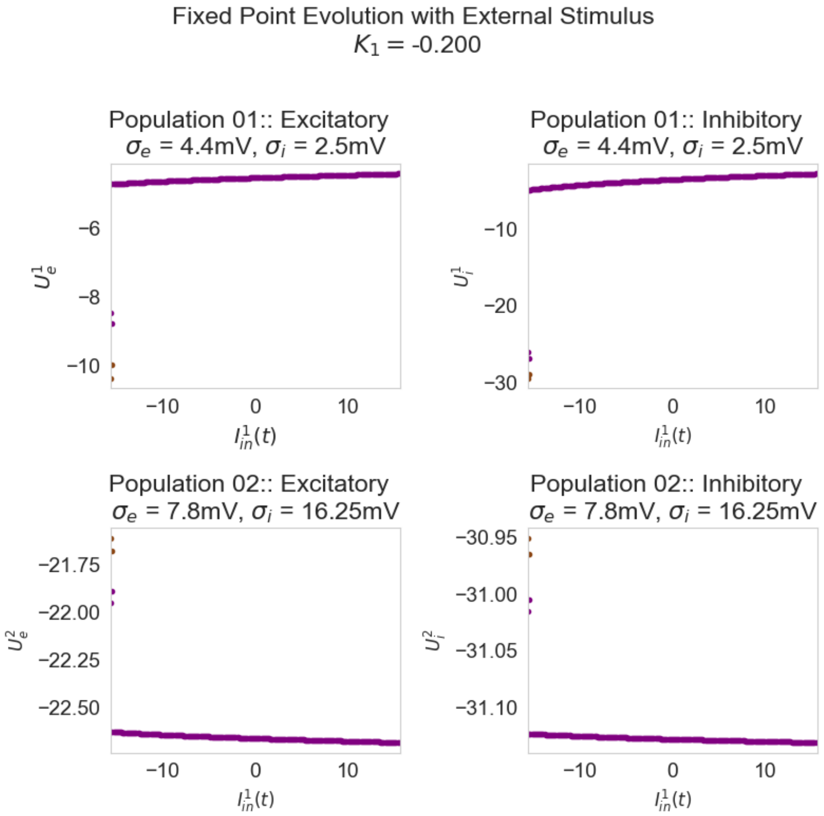
(a) Fixed points show 4 different types of behaviors. Unstable spiral (purple), Unstable saddle point (red), Stable spiral (brown) and Stable node (green). Most of the excitatory coupling shows stable node.

(b) Fixed points show 3 different types of behaviors. Unstable spiral (purple), Stable node (green) and Stable spiral (brown).

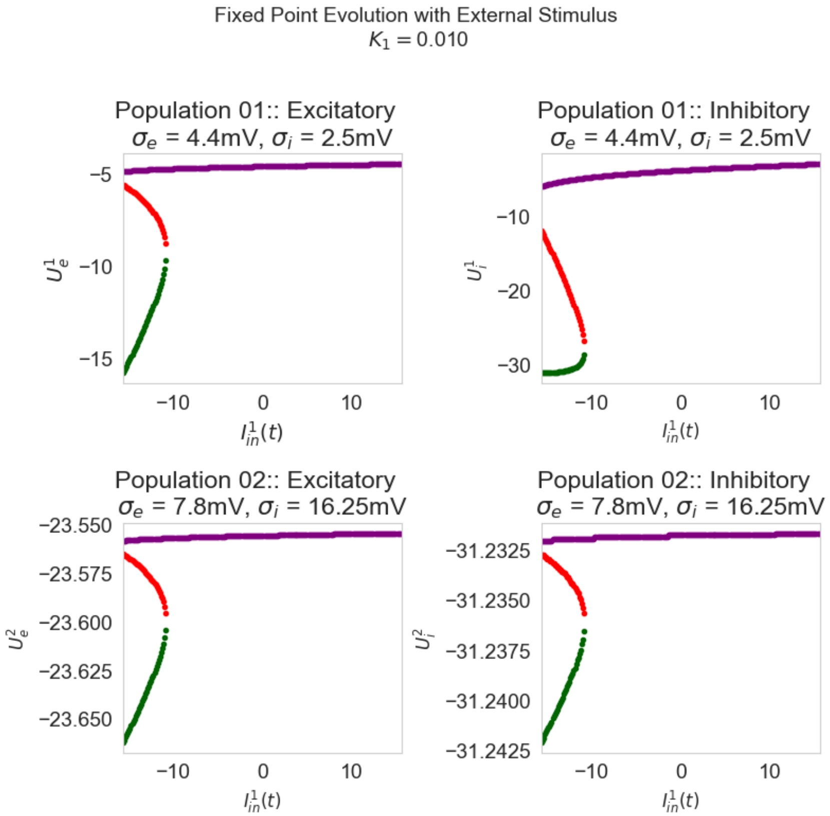
(c) For inhibitory coupling values, the system shows stable spiral activities. For all excitatory coupling, either multi-stable state or stable states can be observed. Multistability region remains unchanged from (b).

Figure: Fixed Point Analysis with K_{glob} for a given I_{in}^{1} (Node 1 – Heterogenous | Node 2 – Homogenous)

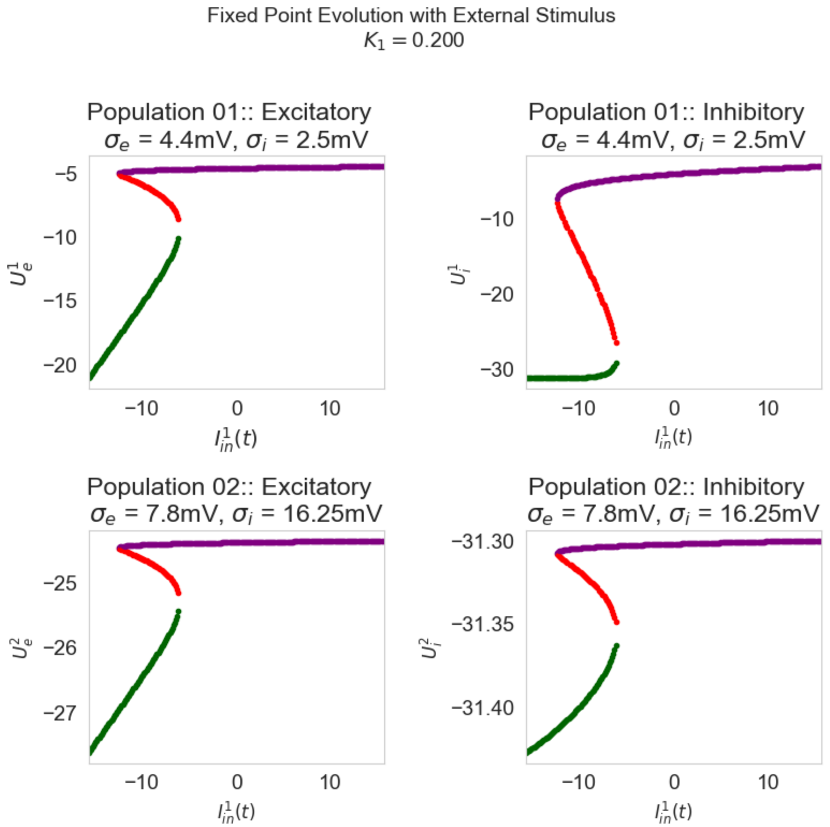
Projections from Figure (C) of Slide 42 onto $K_{\text{glob}} = \gamma$: Multi-stability region shifts to right at a higher rate with homogeneous perturbed region.



(a) Fixed points show only 1behaviors: unstable spiral (purple). Multi-stability point has been moved further to the left when coupling is more inhibitory.



(b) Fixed points now show 3 different types of behaviors. Unstable spiral (purple), Unstable saddle point (red), Stable node (green).
Multi stability region moves towards right with higher coupling value.

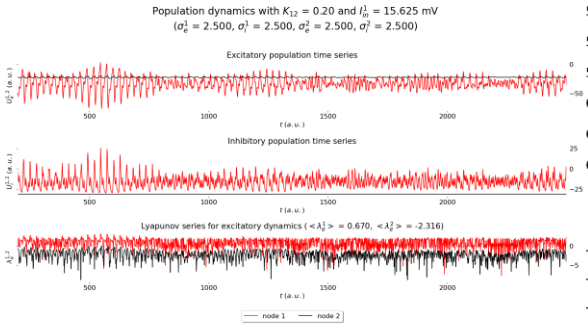


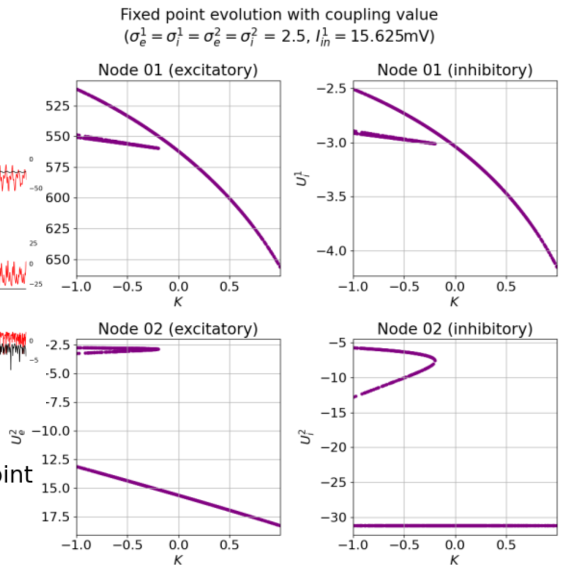
(c) Fixed points show same types of behaviors as of (b). Multi stability region moves even further towards right and fixed point has been increased in excitatory populations.

Figure: Fixed Point Analysis with I_{in}^1 for a given K_{glob} (Node 1 – Homogenous | Node 2 – Heterogenous)

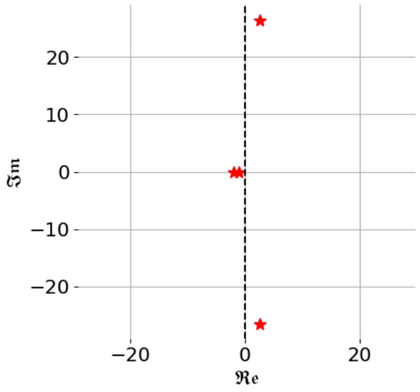
Projection on to $I_{in}^{1} = 15.625\text{mV}$ & $\sigma_{e,i}^{1,2} = \sigma$

Summary: When σ increases, for higher amplitude of external perturbation, the system converges to a stable state for any K.

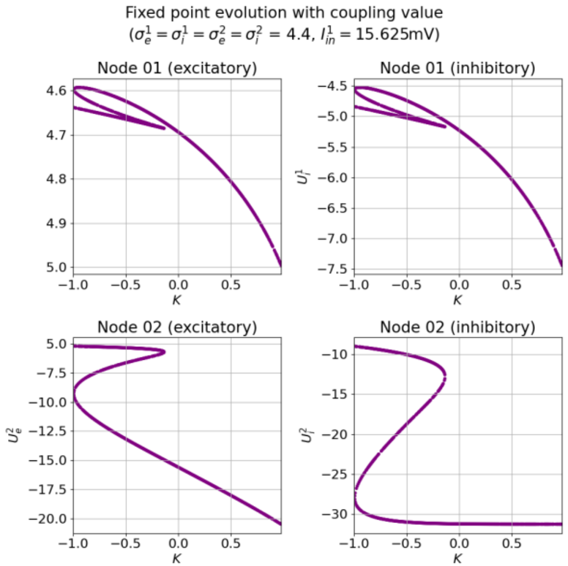




Eigenvalues of Jacobian evaluated at fixed point
(σ = 2.50, I_{in}^1 = 15.625mV, K = 0.2)

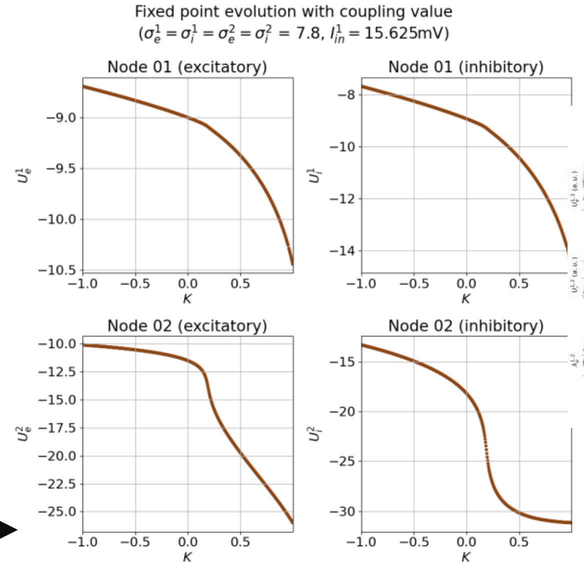


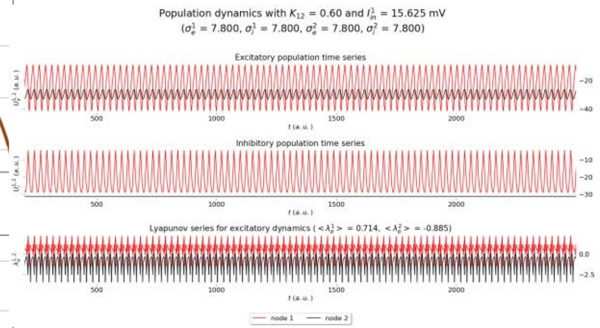
Unstable Spiral



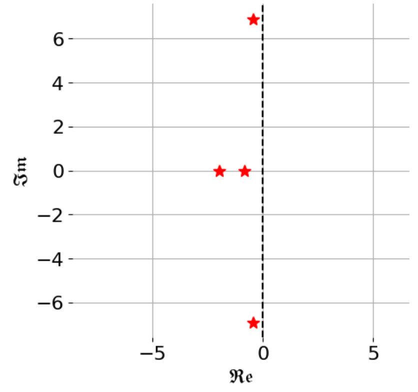
σ

Increase in heterogeneity

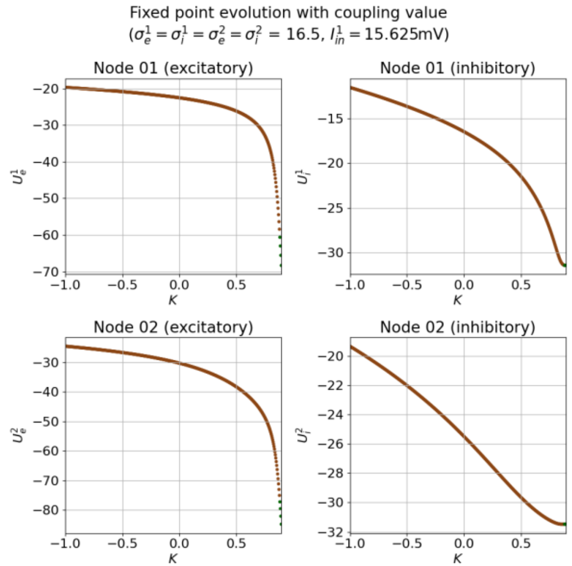




Eigenvalues of Jacobian evaluated at fixed point
(σ = 7.80, I_{in}^1 = 15.625mV, K = 0.6)

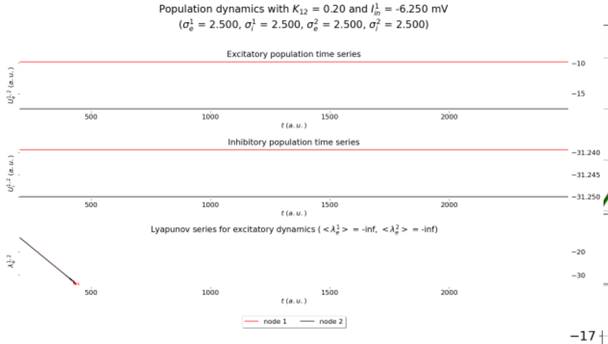


Stable Oscillations

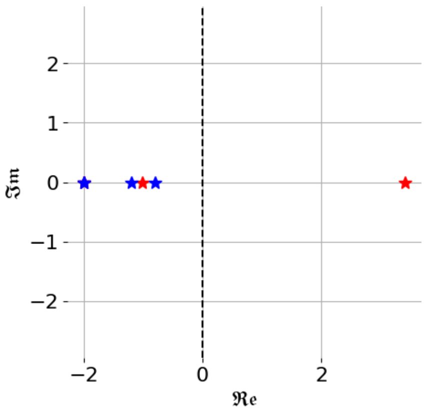


Projections onto $K = 0.2$ & $\sigma_{\mathrm{e,i}}^{1,2} = \sigma$

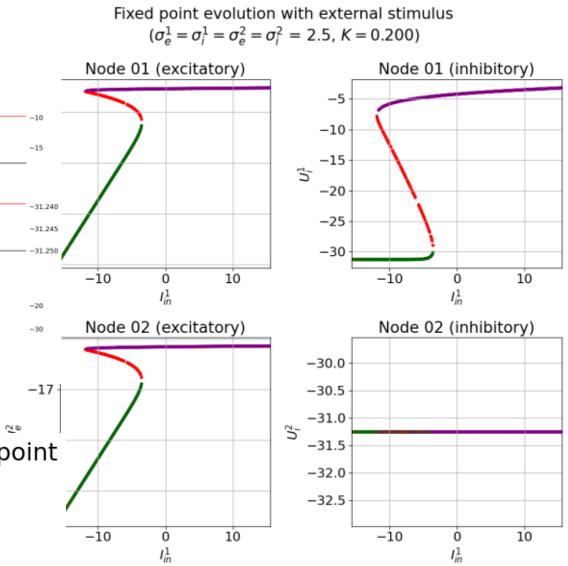
Summary: As <u>σ increases</u> the multi-stability vanishes and system <u>converges to a stable state.</u>

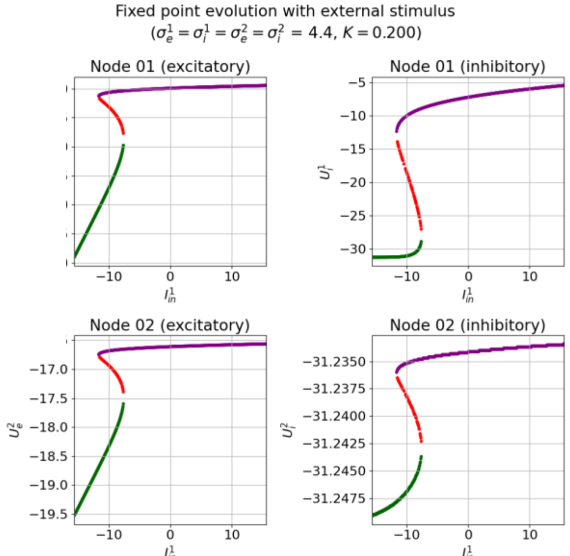


Eigenvalues of Jacobian evaluated at fixed point
(σ = 2.50, I^1_{in} = -6.250mV, K = 0.2)



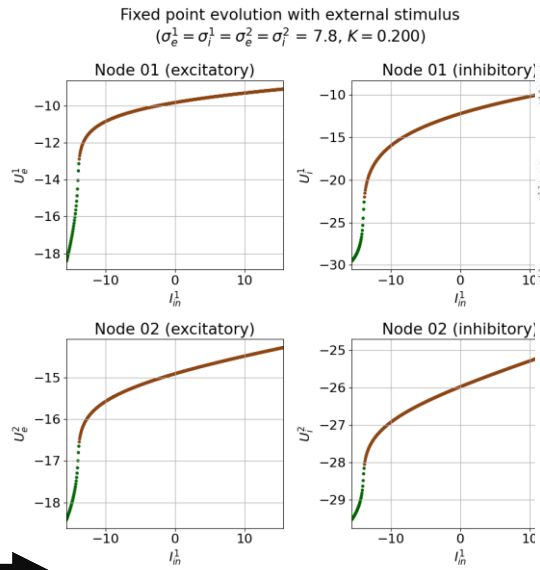
Existence of multi-stable region at negative inputs (high input levels destabilize the system)

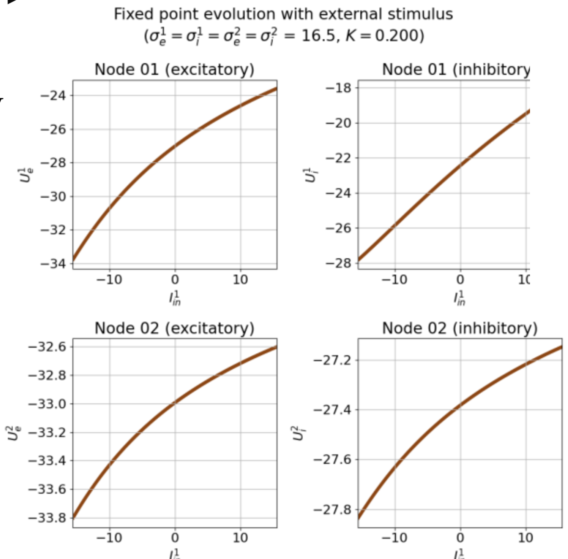


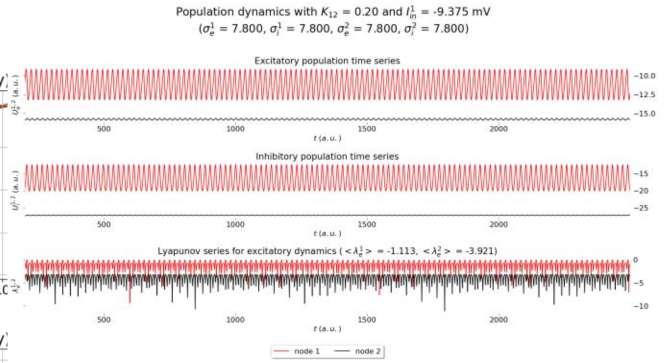


σ

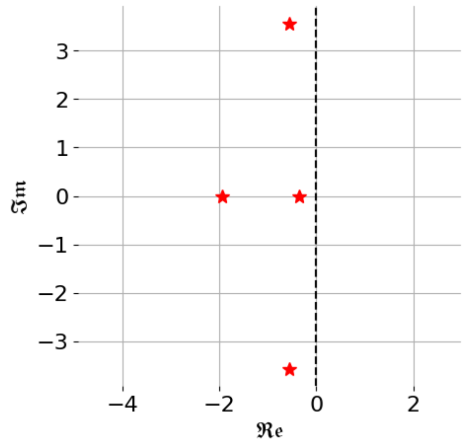
Increase in heterogeneity







Eigenvalues of Jacobian evaluated at fixed point
(σ = 7.80, I^1_{in} = -9.375mV, K = 0.2)



Multi-stability vanishes and stable (low-ampl.) oscillations persists

Trivialization analysis with simplified parameter space.

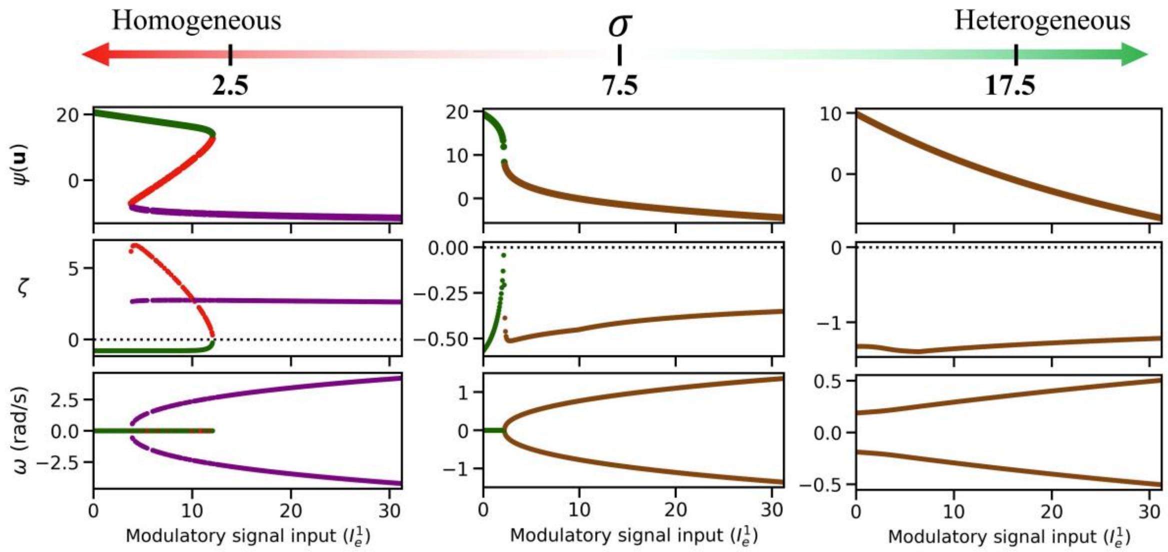


Figure: Excitability heterogeneity reduces the emergence of multiple equilibria (trivialize) at the macro-scale and achieves resilience against external perturbations. Results are for $K_{\text{glob}} = 0.2$.

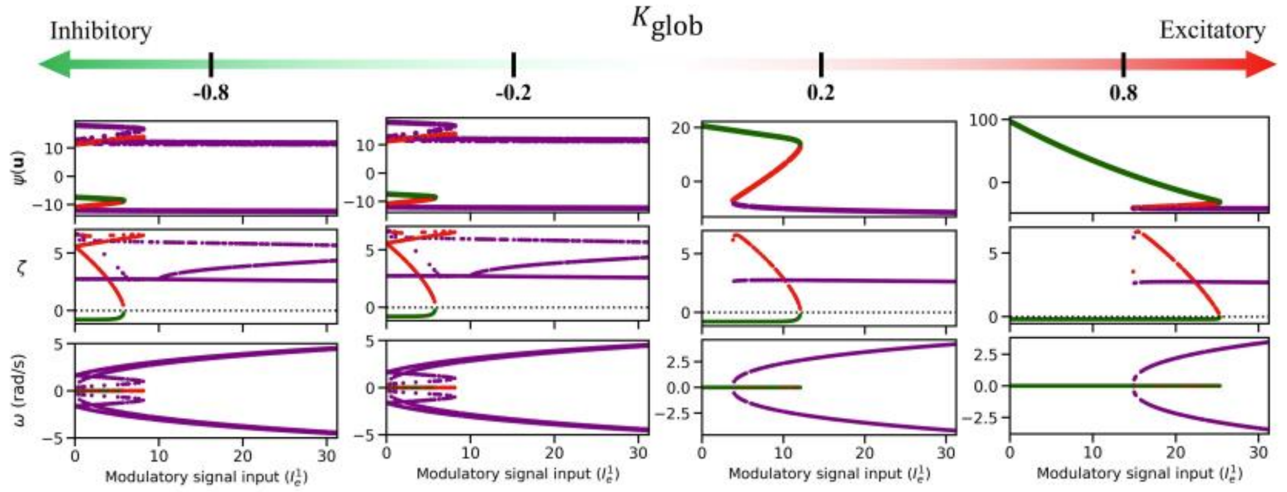


Figure: Global coupling parameter's effect on trivializing the two-node system is less significant compared to heterogeneity. Results are for $\sigma = 2.5$.

Excitatory and inhibitory heterogeneity unevenly impact network trivialization and resilience.

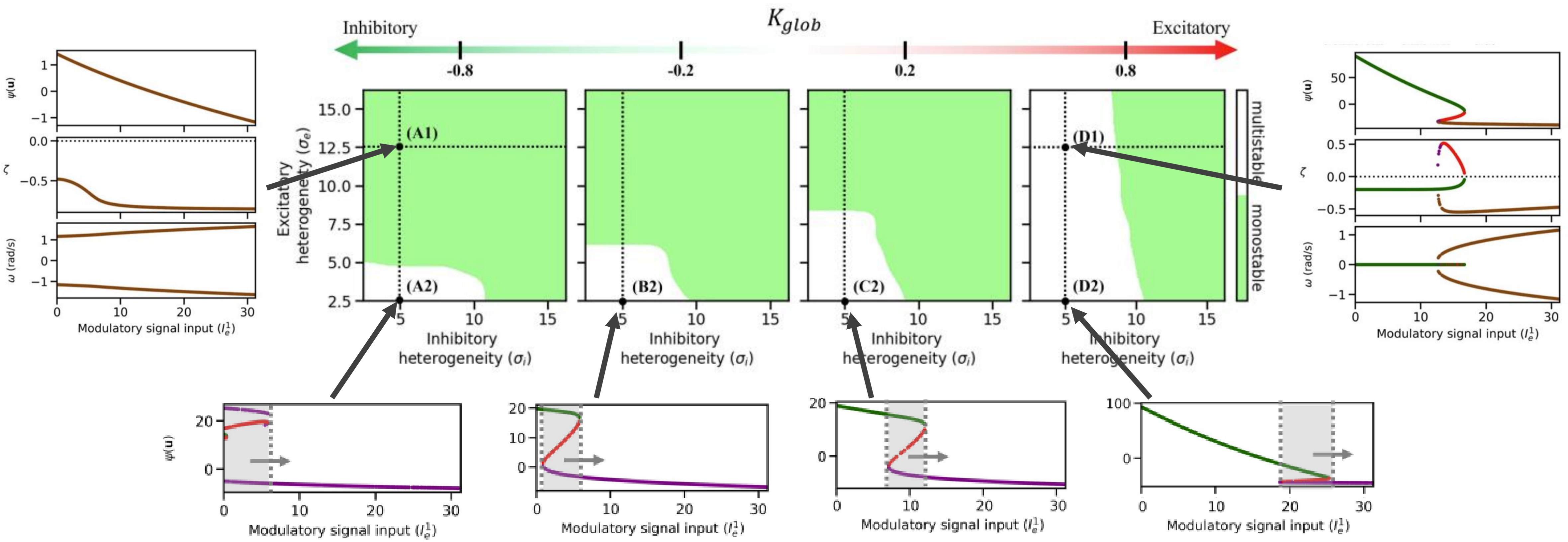


Figure: Excitatory and inhibitory heterogeneity differentially affect the trivialization as K_{glob} changes.

As K_{glob} increases, inhibitory heterogeneity becomes the most significant property for trivialization.

Analyzing spectral radius and eigenvalue distribution – Preliminary Results

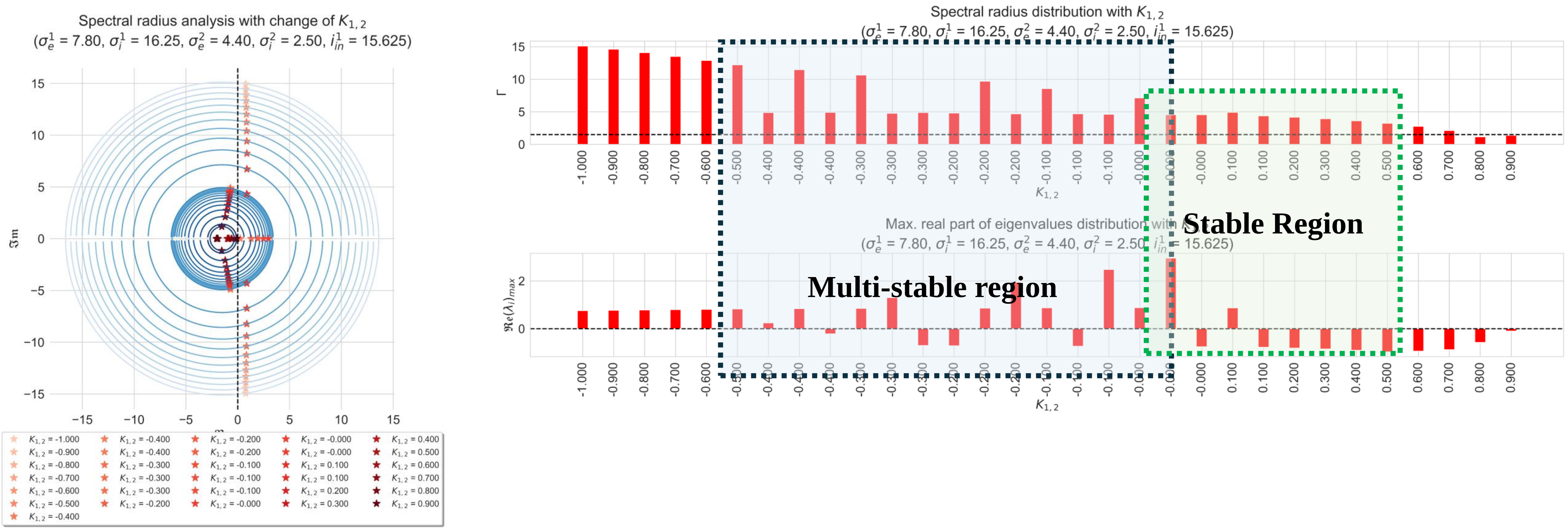


Figure: Eigen spectrum of the system with reduced heterogeneity in Node 2

****These were created without the system simplifications. Systematic analysis based on the suggested simplifications is required.**

Temporal synchronization analysis with 2-node macro-scale brain network.

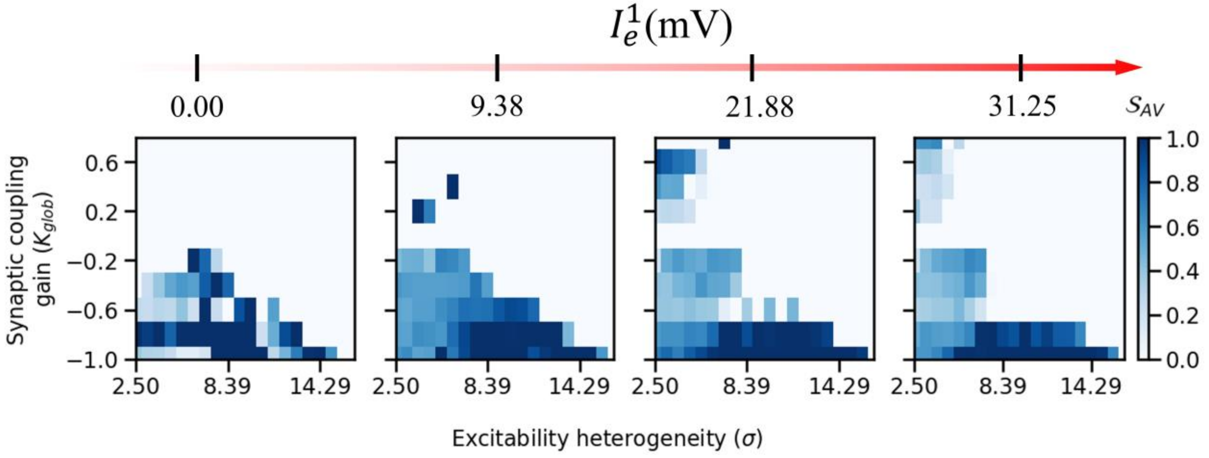


Figure: Change in avalanche event synchrony within $(K_{\mathrm{glob}}, \sigma)$ parameter space as I_{e}^1 increases.

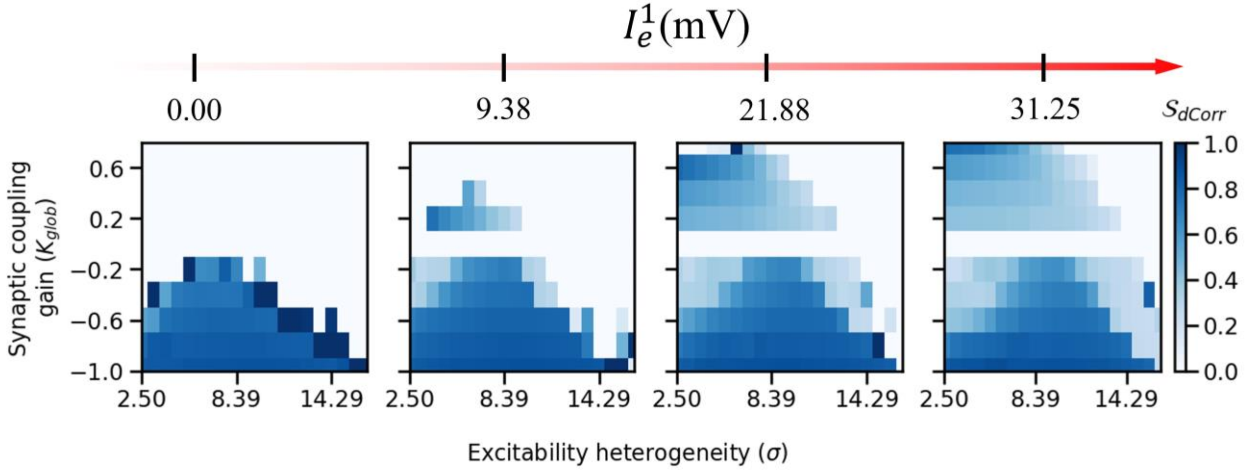


Figure: Change in distance correlation within $(K_{\mathrm{glob}}, \sigma)$ parameter space as I_{e}^1 increases.

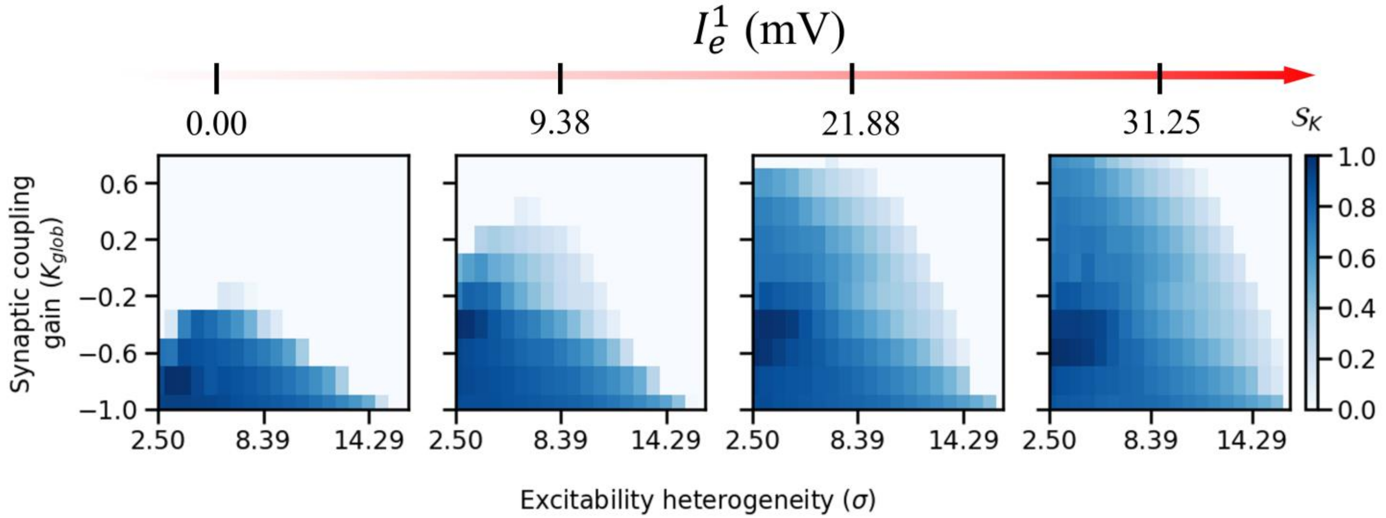


Figure: Change in Kuramoto order parameter within $(K_{\mathrm{glob}}, \sigma)$ parameter space as I_{e}^1 increases.

Spectral synchronization analysis with 2-node macro-scale brain network.

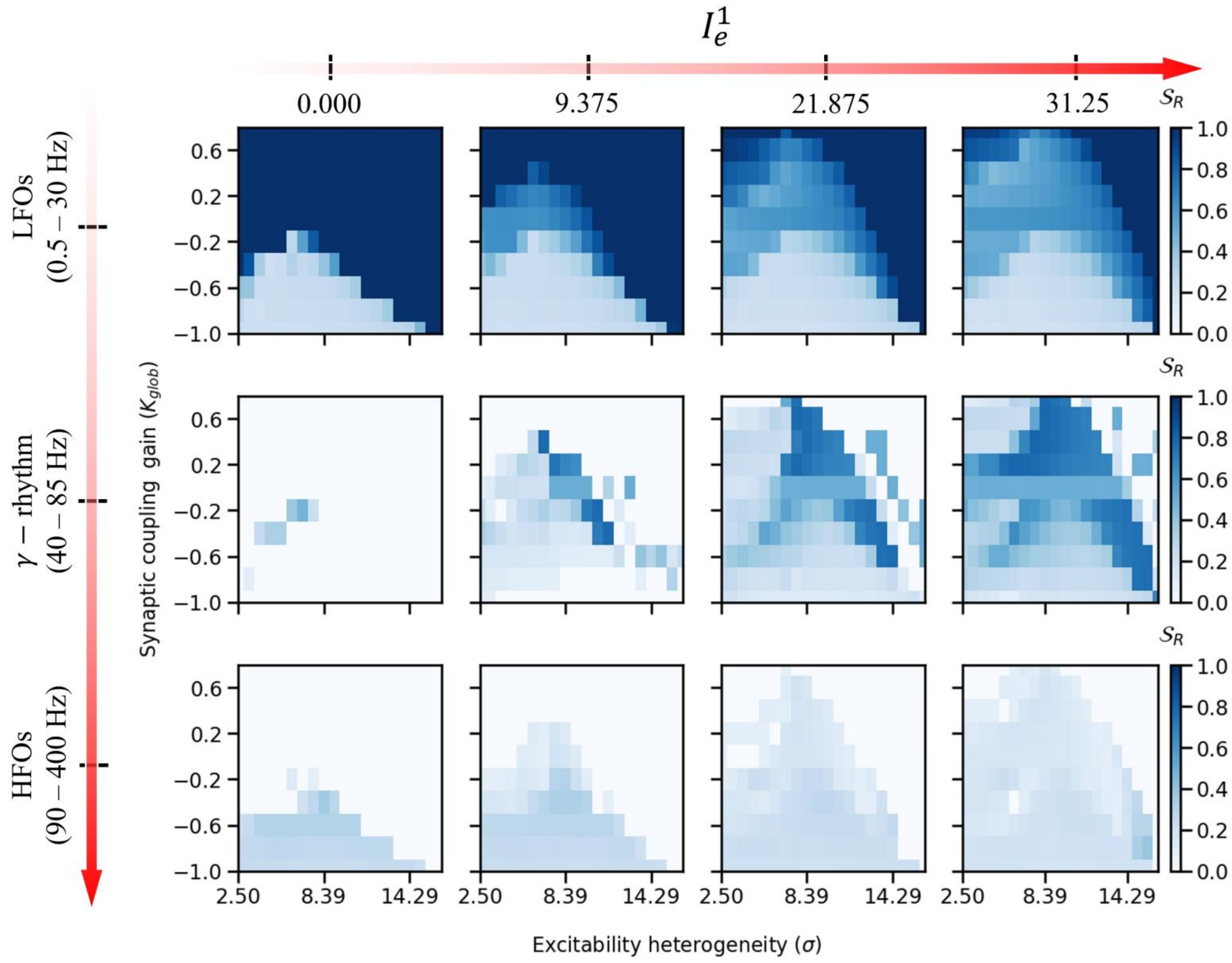


Figure: Change in global phase synchrony within $(K_{\mathrm{glob}}, \sigma)$ parameter space as I_{e}^1 increases for different frequency bands.

Takes the relative spectral power into consideration.

$$\widetilde{\mathrm{R}}^w_{f_b}(t)\widetilde{\Phi}^w_{f_b}(t) = \frac{1}{N}\sum_{n=1}^{N} \mathrm{a}^n_{f_b}(t)\, e^{j\phi^n_{f_b}(t)}$$

$$\mathcal{S}_R^{f_b} = \frac{1}{N_w}\sum_{w=1}^{N_w}\mathbb{E}_t[\widetilde{\mathrm{R}}^w_{f_b}(t)] = \frac{1}{N_w}\sum_{w=1}^{N_w}\left\{\frac{1}{l_w}\sum_{t=t_o^w}^{t_f^w}\widetilde{\mathrm{R}}_{f_b}(t)\right\}$$

Spectral synchronization analysis with 2-node macro-scale brain network ctd.

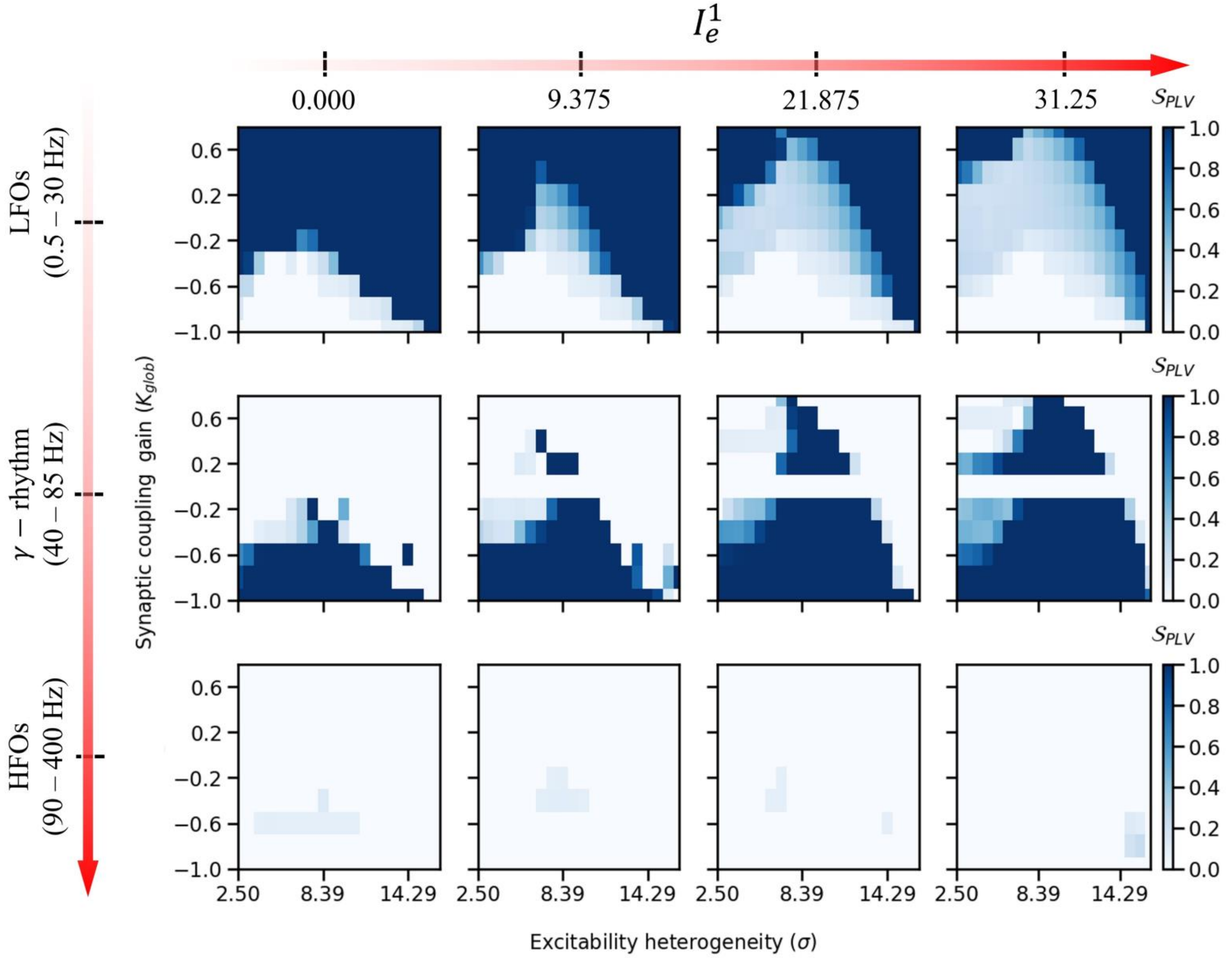


Figure: Change in Phase Locked Value (PLV) within $(K_{\mathrm{glob}}, \sigma)$ parameter space as I_e^1 increases for different frequency bands.

Takes the relative spectral power into consideration.

$$\left(\tilde{\eta}^{f_b}_{\mathrm{PLV}}\right)^{w}_{nm} = \left|\mathbb{E}_t\left[\langle \mathbf{a}^{m}_{f_b} e^{j\boldsymbol{\phi}^{m}_{f_b}}, \mathbf{a}^{n}_{f_b} e^{j\boldsymbol{\phi}^{n}_{f_b}}\rangle\right]\right| \in [0,1]$$

$$\mathcal{S}^{f_b}_{PLV} = \frac{1}{N_w}\sum_{w=1}^{N_w}\left(\tilde{\eta}^{\mathrm{PLV}}_{f_b}\right)^{w}_{nm}$$

Spectral synchronization analysis with 2-node macro-scale brain network ctd.

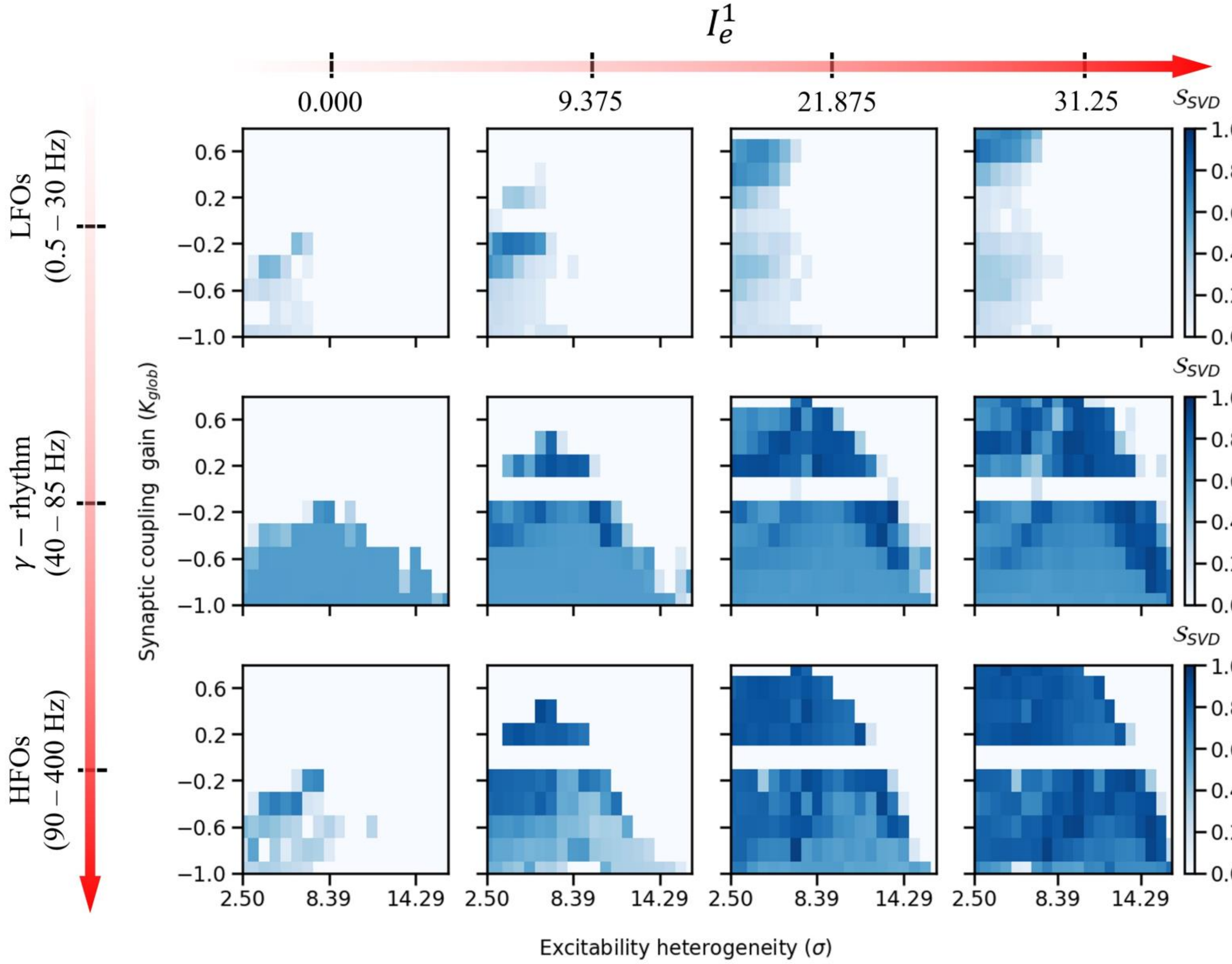


Figure: Change in SVD-based synchrony within $(K_{\mathrm{glob}}, \sigma)$ parameter space as I_{e}^1 increases for different frequency bands.

Relative contributions from node 1 and node 2

$$\left(\eta^{f_b}_{\mathrm{SVD}}\right)^w = \min\left(\left|\frac{\tilde{v}_1^w}{\tilde{v}_2^w}\right|, \left|\frac{\tilde{v}_2^w}{\tilde{v}_1^w}\right|\right) \in [0,1]$$

$$\mathcal{S}^{f_b}_{SVD} = \frac{1}{N_w}\sum_{w=1}^{N_w} \left(\eta^{\mathrm{SVD}}_{f_b}\right)^w$$

Multi-node network – connectivity statistics

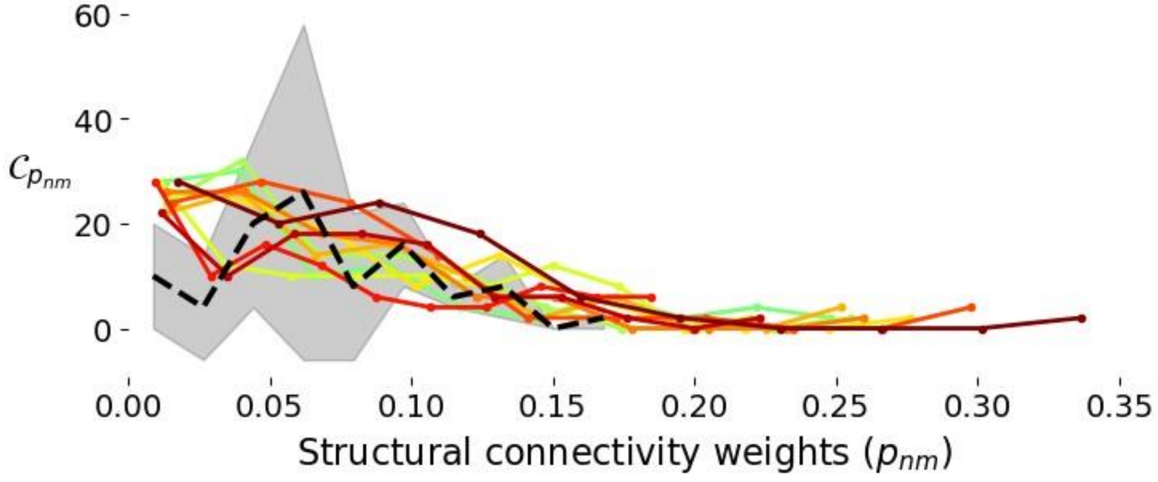


Figure: Distributions of weight-counts of statistically realized structural connectivity matrices.

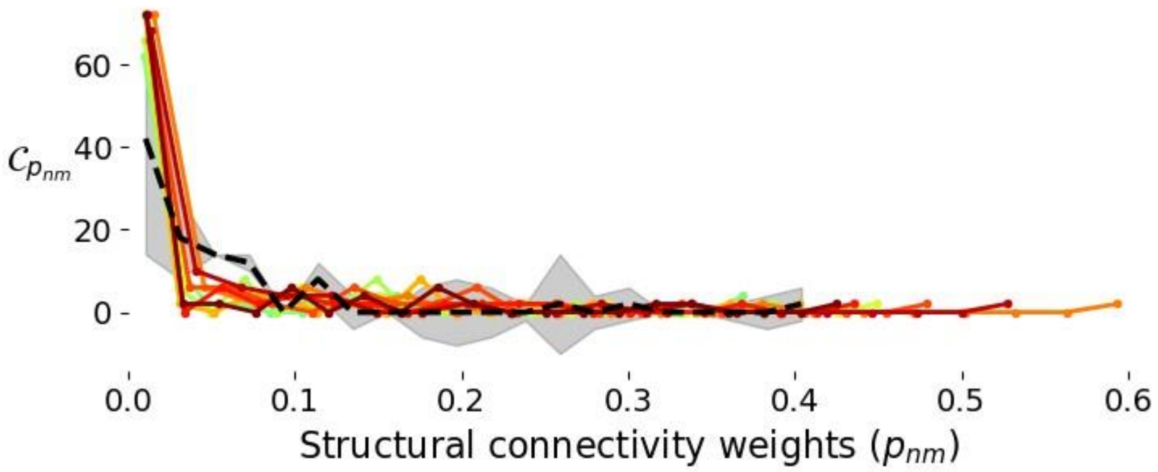


Figure: Distributions of weight-counts of sub-sampled empirical structural connectivity matrices.

Multi-node network – connectivity statistics ctd.

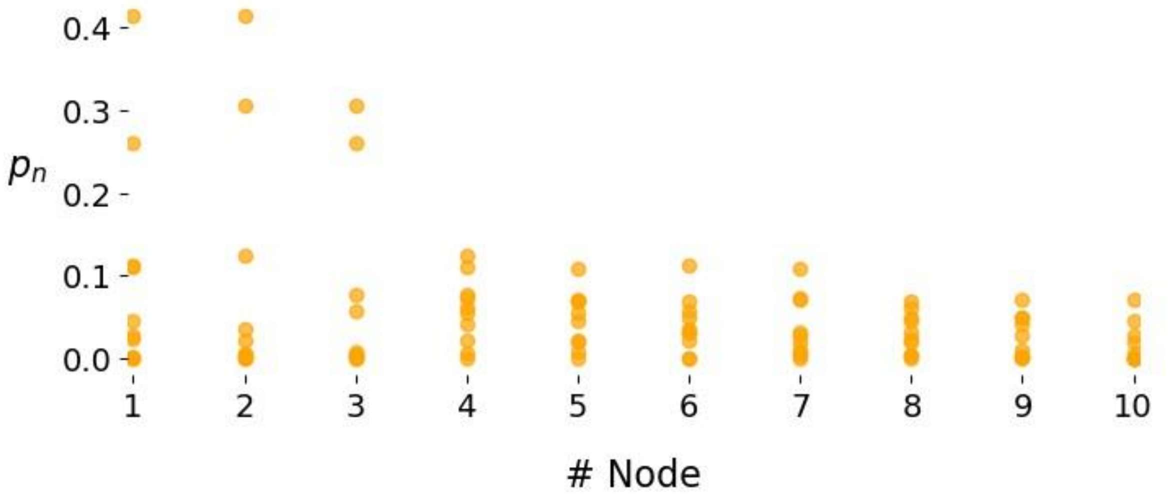


Figure: Distributions of connectivity weights of statistically realized structural connectivity matrices for each node .

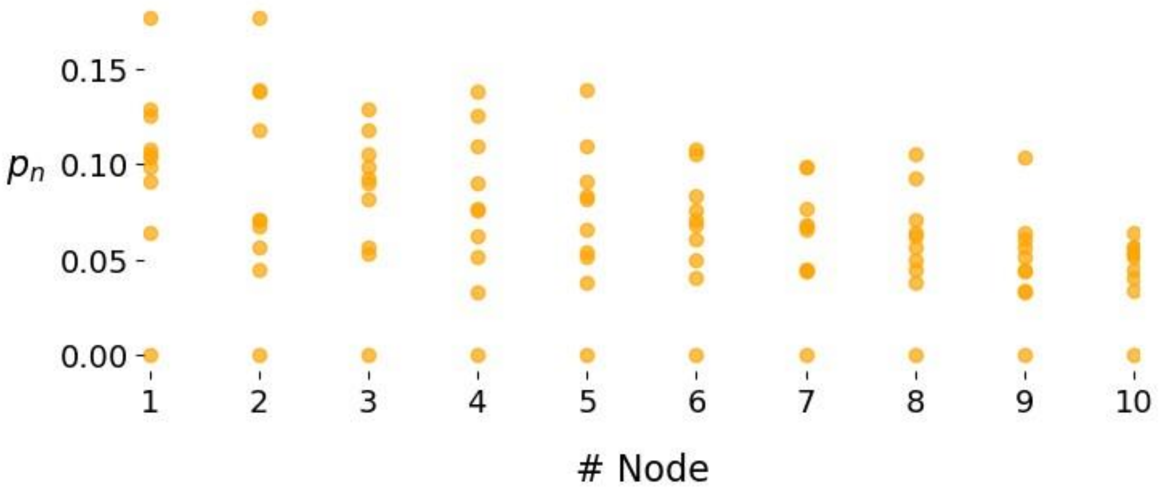


Figure: Distributions of connectivity weights of sub-sampled empirical structural connectivity matrices for each node .

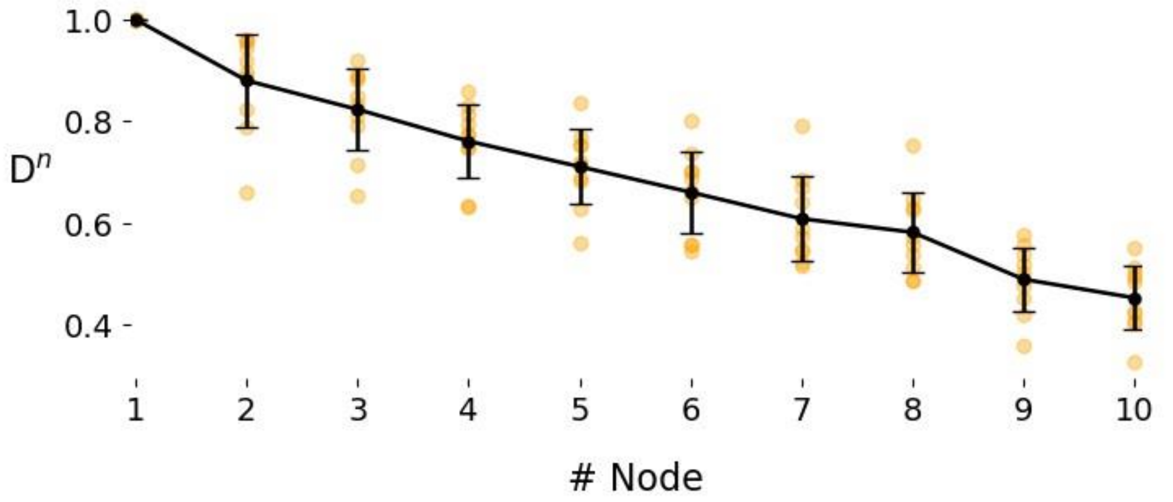


Figure: Degree distribution of each node with statistically realized structural connectivity matrices.

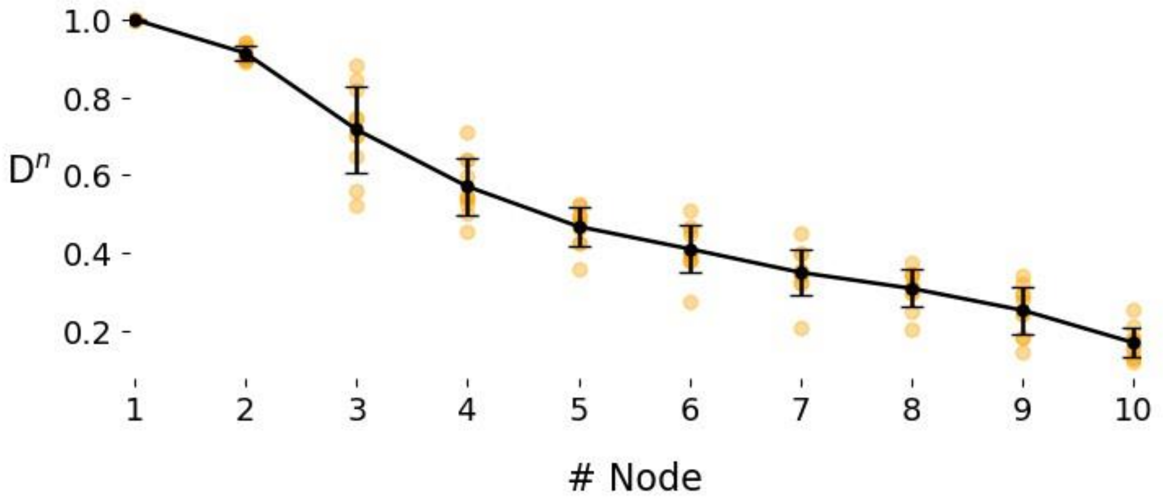


Figure: Degree distribution of each node with sub-sampled empirical structural connectivity matrices.

Inter-regional local excitability heterogeneity distributions

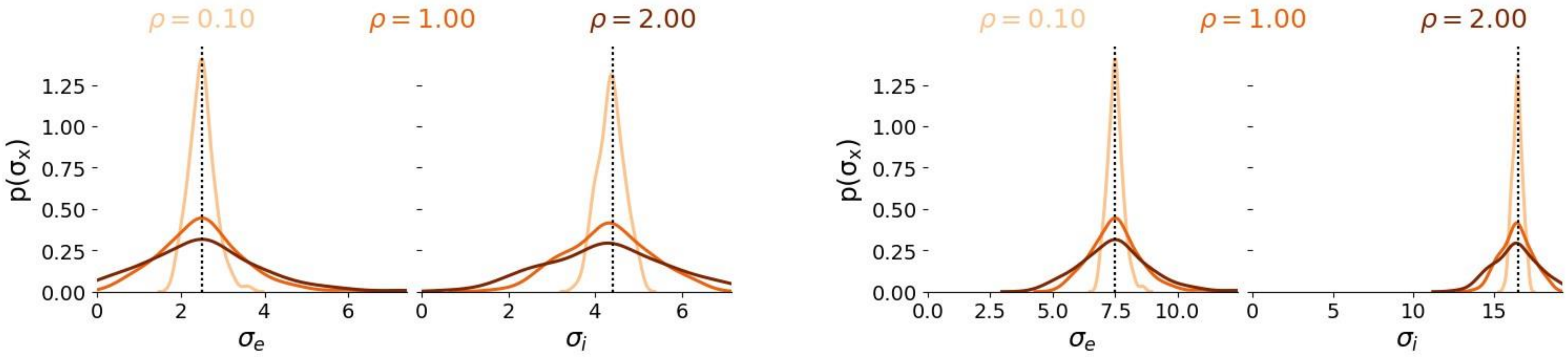


Figure: The spreads of the heterogeneity parameters in the unmodulated nodes as ρ changes